HEREDITARY COLOR HIERARCHY FOR RESTRICTED ARITHMETIC PROGRESSIONS: QUASI-CENTRAL HOSTS, NILRETURN WINDOWS, AND FINITE REDUCTIONS

RESEARCH MEMORANDUM PREPARED FOR SOHAIL FARHANGI

Abstract. The hereditary color-hierarchy conjecture remains unresolved for arbitrary large hosts. We prove the full strict hierarchy in every additive quasi-central host and in its positive image under any nonconstant integer polynomial p with $p(0) = 0$ and positive leading coefficient. The subsets have the all-nil-window property P_k and a $(k+1)$-color blocker at length $400 \cdot 2^k$. For arbitrary $(k+1)$-large D we prove a simultaneous localization: every k-coloring of a nilreturn window contains arbitrarily long monochromatic progressions whose difference lies in D and in any prescribed finite-order nil-Bohr neighborhood of zero. We also give finite-extraction equivalences, a uniform one-extra-color theorem for nil-Bohr complements, binary-blockable seeds in IP_0 hosts, a finite van der Corput transfer for binary triples, and two explicit unresolved host benchmarks. The proofs and separately stated inputs are identified throughout. This memorandum records internal research audits.

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1. The question and the status of this report

Throughout, $\mathbb{N} = \{1, 2, \ldots\}$, and a progression has positive common difference. Its initial point belongs to $\mathbb{N}$ unless another ambient group is specified. An r-coloring means a coloring using at most r colors. Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, let $\{x\}$ denote the fractional part of a real number, and let $\|x\|_{\mathbb{T}}$ denote its distance to the nearest integer.

Definition 1.1. For integers $r \geq 1$ and $h \geq 2$, a set $D \subseteq \mathbb{N}$ is (r, h)-*forcing* if every r-coloring of $\mathbb{N}$ contains a monochromatic progression

$$a, a + d, \ldots, a + (h - 1)d, \qquad a \in \mathbb{N}, \quad d \in D.$$

An (r, h)-*blocker* for D is a coloring witnessing failure of this property. A set is r-*large* if it is (r, h)-forcing for every $h \geq 2$, and *large* if it is r-large for every finite r. These are restrictions on the allowed *differences*; the terms of the progression are not required to lie in D.

Conjecture 1.2 (Hereditary color hierarchy)**.** *For every large $A \subseteq \mathbb{N}$ and every integer $k \geq 1$, there is $D \subseteq A$ which is k-large and is not $(k + 1)$-large.*

The $k = 1$ case is elementary. A large set is nonempty, so choose $d \in A$. The singleton $\{d\}$ is 1-large. The binary coloring

$$n \longmapsto \left\lfloor \frac{n - 1}{d} \right\rfloor \pmod 2$$

changes color between n and $n + d$, so $\{d\}$ is not 2-large. For $k \geq 2$, the conjecture remains open in the present work. No counterexample host has been constructed either.

For arbitrary large hosts, the central reduction is to a finite problem with an exact uniformity requirement. Fix $k \geq 2$ and put $q = k + 1$. The required finite packets must force unbounded k-color progression lengths while admitting q-color blockers at *one fixed length* H. The value of H may depend on a large subhost; it cannot grow with the target forcing length. Section 4 proves the equivalence carefully, including the passage from local bounds to a common bound on a smaller large host.

A *counterhost at level k* will mean a large set A satisfying

$$\forall D \subseteq A \qquad \big[D \text{ is } k\text{-large} \Longrightarrow D \text{ is } (k + 1)\text{-large}\big]. \tag{C_k}$$

Thus the existence of a counterhost is exactly a negative answer to Conjecture 1.2 at that level. Some localization results below only assume the implication in (C_k), without assuming that A itself is large; their applications use large counterhosts.

The strongest positive result proved here is Theorem 5.1: every additive quasi-central host contains the full strict hierarchy, with all-nil-window lower bounds and a uniform obstruction length $400 \cdot 2^k$. Theorem 6.1 extends it to zero-constant polynomial images of such hosts. Quasi-centrality means membership in an additive idempotent ultrafilter all of whose members are piecewise syndetic; its precise definition and the full proofs appear below. Boundary returns in arbitrary minimal systems, intersected with a thick set and deprived of a non-piecewise-syndetic set, are covered by a centrality lemma. Every IP_0 host also contains a pointwise all-nil recurrent seed with a binary blocker of length 6400 (Theorem 7.1); combined with piecewise syndeticity this gives two-color separation. These classes and conclusions do not settle arbitrary large hosts.

The relative gates, stabilizer arguments, palette induction, compact extraction, and assembly are proved in this memorandum. Their external inputs are identified explicitly, principally the polynomial return-time structure theorem and standard nilsystem and ultrafilter structure. The polynomial marker uses quantitative Weyl equidistribution; the three-step consequence uses the Huang–Shao–Ye characterization. One older theorem about general intersective polynomial images is recorded as a separate unpublished input. Compilation and page inspection check the artifact, not mathematical correctness, and internal proof audits do not amount to external peer review.

2. Finite witnesses, large-set permanence, and dilation

Lemma 2.1 (Compactness of finite forcing)**.** *If D is (r,h)-forcing, it has a finite (r,h)-forcing subset. Moreover there is an N such that every r-coloring of $[1,N]$ already contains a monochromatic h-term progression with difference in that finite subset.*

Proof. For every permitted progression, the event that its sites have one color is a clopen cylinder in the compact product space $\{1,\dots,r\}^{\mathbb{N}}$. Forcing means that these events cover the space. A finite subcover uses finitely many differences and finitely many sites. Take N larger than all of those sites. Any coloring of $[1,N]$ extends to $\mathbb{N}$, and a member of the finite subcover then provides the asserted progression inside $[1,N]$. □

Lemma 2.2 (Passing between one and two sides)**.** *A set $D \subseteq \mathbb{N}$ has an (r,h)-blocker on $\mathbb{N}$ if and only if it has one on $\mathbb{Z}$.*

Proof. Restriction proves one direction. For the other, translate a blocker of $\mathbb{N}$ to color $[-j,\infty)$, extend arbitrarily to the remaining integers, and choose a pointwise convergent subsequence in the compact space $\{1,\dots,r\}^{\mathbb{Z}}$. Every fixed forbidden progression is eventually in the correctly translated domain. It therefore remains nonmonochromatic in the limit. □

Proposition 2.3 (Large sets form a coideal)**.** *The nonlarge subsets of $\mathbb{N}$ form an ideal containing all finite sets. Consequently the family of large sets is upward closed, finitely partition regular, and invariant under finite changes.*

Proof. Subsets of a blocked set inherit its blocker. If D_i has an (r_i,h_i)-blocker c_i, the product coloring (c_1,c_2) blocks $D_1 \cup D_2$ at length $\max(h_1,h_2)$ using at most $r_1 r_2$ colors. This proves closure under finite unions.

For a finite nonempty F, put $M = \max F$. Color n by $\lfloor (n-1)/M \rfloor \bmod 2$. Along a progression with $1 \le d \le M$, successive block indices increase by zero or one. Over $2M+1$ sites their total increase is at least two. At least one increment is one, so the progression is not monochromatic. Thus every finite set is nonlarge. The three consequences follow from the ideal property and the definitions. □

Lemma 2.4 (Exact dilation invariance)**.** *For $\lambda \in \mathbb{N}$, D is (r,h)-forcing if and only if λD is (r,h)-forcing. Equivalently, blockability at each fixed palette and length is invariant under dilation.*

Proof. If D is forcing, apply its property to $n \mapsto c(\lambda n)$. Conversely, if c blocks D, write each positive integer uniquely as $n = \lambda t + j$, with $t \ge 0$ and $1 \le j \le \lambda$, and color it by $c(t+1)$. A step λd stays in one residue class and advances its quotient by d, so this colors λD with the same blocking parameters. □

Proposition 2.5 (Divisibility cores)**.** *If A is large and $\lambda \ge 1$, then*

$$A \cap \lambda\mathbb{N}, \qquad A/\lambda = \{d \in \mathbb{N} : \lambda d \in A\}, \qquad \lambda A$$

are all large.

Proof. Refine an arbitrary finite coloring by residues modulo λ, and use largeness of A. A monochromatic progression of length at least two has difference in $A \cap \lambda\mathbb{N}$. This proves largeness of that intersection. The other assertions follow from Lemma 2.4, since $A \cap \lambda\mathbb{N} = \lambda(A/\lambda)$. □

The residue refinement uses more colors. It does not prove that a fixed r-large set has r-large divisibility cores. A one-color-loss version of that assertion will instead follow from the marker lemma.

Proposition 2.6 (Large hosts need not contain progressions)**.** *For any sequence (b_i) of positive integers,*

$$\mathrm{FS}(b_i) = \left\{ \sum_{i \in I} b_i : \varnothing \neq I \subseteq \mathbb{N} \text{ finite} \right\}$$

is large. If $b_j > 2\sum_{i<j} b_i$ for all $j > 1$, it contains no nonconstant three-term arithmetic progression.

Proof. For a target (r, h), let m be a Hales–Jewett number for r colors and alphabet $\{0, \dots, h-1\}$ [12]. Color the word w by the color of $1 + \sum_{i=1}^{m} w_i b_i$. A monochromatic combinatorial line has a nonempty variable coordinate set I and gives a progression of difference $\sum_{i \in I} b_i$.

Under the growth assumption, suppose $a + c = 2b$ in $\mathrm{FS}(b_i)$. The coefficient of each b_i in $a + c - 2b$ belongs to $\{-2, -1, 0, 1, 2\}$. A highest nonzero coefficient cannot be canceled by all lower coefficients, since $b_j > 2\sum_{i<j} b_i$. Hence every coefficient is zero. At a binary coordinate this says $a_i + c_i = 2b_i$, forcing $a_i = b_i = c_i$ and therefore $a = b = c$. □

3. Uniform finite addition with the same palette

The next lemma allows a bound depending on the actual finite set being adjoined, but not on the old blocked set. That distinction is enough for the diagonal argument in the next section.

Theorem 3.1. *Let $q, H \geq 2$, and let $E \subseteq \mathbb{N}$ admit a q-coloring with no monochromatic H-term progression whose difference belongs to E. Let $F = \{f_0, \dots, f_{s-1}\}$ be a nonempty finite subset of $\mathbb{N}$, and put $M = \max F$. Choose a prime*

$$p > \max\{2sM,\ s(H+1)\}.$$

Then $E \cup F$ admits a q-color blocker of length $2p$. In particular the bound is uniform in E and in its original blocking coloring.

Proof. We may start with a blocking coloring $c : \mathbb{Z} \to \mathbb{Z}/q\mathbb{Z}$. Indeed, translate a coloring of $\mathbb{N}$ farther and farther to the left and pass to a pointwise convergent subsequence in the compact product $(\mathbb{Z}/q\mathbb{Z})^{\mathbb{Z}}$. Every fixed forbidden configuration lies in the translated positive domain eventually, so the limiting coloring is still a blocker.

Mark the s residue classes

$$r_j = 2Mj \pmod{p} \qquad (0 \leq j < s).$$

They are distinct, and $r_j - f_j$ is not marked: for $j \geq 1$ it lies strictly between r_{j-1} and r_j, while for $j = 0$ its representative $p - f_0$ lies above the last marked residue. Define

$$c'(n) = \begin{cases} c(n - f_j) + 1 \pmod{q}, & n \equiv r_j \pmod{p}, \\ c(n), & n \text{ belongs to an unmarked residue class.} \end{cases}$$

Fix $d \in E$. If $p \mid d$, a progression of step d stays in one residue class. Along the whole progression c' is either c or one fixed translate of c followed by addition of one. It therefore cannot be monochromatic on H sites. If $p \nmid d$, the successive indices of the progression cycle through all p residue classes. Among this cyclic list there is a run of at least H unmarked classes, since there are $p - s > sH$ unmarked classes distributed among s gaps. Any $2p$ consecutive sites contain a full such run, where $c' = c$. Monochromaticity again contradicts the blocker c.

Now fix $f_j \in F$. In any $p + 1$ sites of step f_j there is a marked site $n \equiv r_j \pmod{p}$ which is not the first site: $p > M \geq f_j$ ensures that f_j is invertible modulo p. Its immediately preceding site $n - f_j$ is unmarked. Hence

$$c'(n) = c'(n - f_j) + 1 \pmod{q},$$

so the progression is not monochromatic. Restricting c' to $\mathbb{N}$ proves the theorem. □

For a singleton $F = \{f\}$, the simpler choice $p > \max\{H, f\}$ suffices: mark just residue zero and use $c'(n) = c(n - f) + 1$ there. The same proof gives length $2p$. For $F = \varnothing$, no change to the original blocker is needed. Bertrand's postulate also turns the displayed prime choice into a bound of order $s(H + M)$, with an absolute constant.

Corollary 3.2 (Uniform removal from finite forcing)**.** *For fixed q, H, F, there is $L = L(H, F)$ such that whenever $D \subseteq \mathbb{N}$ is (q, L)-forcing, $D \setminus F$ is (q, H)-forcing.*

Proof. Otherwise $D \setminus F$ has a q, H blocker; the theorem would give a q, L blocker for its union with F, and hence for D. □

4. Approximate extraction and uniform counterhost compression

Fix $k \geq 2$ and $q = k + 1$ throughout this section. For $H \geq 2$, put

$$T_H(A): \quad \forall h \geq 2\ \exists E \subseteq A \text{ finite} \quad \begin{cases} E \text{ is } (k, h)\text{-forcing,} \\ E \text{ has a } (q, H)\text{-blocker.} \end{cases} \tag{1}$$

The same H must work for all h. The individual finite sets and their blocking colorings may vary with h.

Lemma 4.1 (Finite-change invariance)**.** *For each H, the property T_H is upward closed and invariant under finite changes of the host. It is increasing in H.*

Proof. Upward closure and monotonicity in H are immediate. To remove a fixed finite F, fix a target h and take the forcing length $L = L(h, F)$ from the finite-removal corollary in Section 3, applied to the palette k. Obtain a finite (k, L)-forcing, (q, H)-blockable packet E from $T_H(A)$. Then $E \setminus F$ is (k, h)-forcing and inherits the same (q, H)-blocker. This proves $T_H(A \setminus F)$. Adding finitely many elements follows by upward closure, so finite changes preserve the property. □

Lemma 4.2 (Uniformity on a large subhost)**.** *Suppose every large $C \subseteq A$ satisfies $T_H(C)$ for some $H \geq 2$. Then every large $C \subseteq A$ contains a large B and has an H such that $T_H(D)$ holds for every large $D \subseteq B$.*

Proof. If this failed for some C, recursively choose decreasing large sets $C_j \subseteq C$, for $j \geq 2$, such that $T_j(C_j)$ fails. Indeed, failure says that inside every large current subhost and at every prescribed length bound there is a further large subhost where that bound fails.

Choose a finite (j, j)-forcing $F_j \subseteq C_j$, using Lemma 2.1, and let $B = \bigcup_{j \geq 2} F_j$. This set is large: for any (r, h) use $j \geq \max(r, h, 2)$. Also $B \setminus C_j$ is finite for every j. By the hypothesis, $T_{H_0}(B)$ holds for some H_0. Lemma 4.1 gives $T_{H_0}(B \cap C_{H_0})$, and upward closure gives $T_{H_0}(C_{H_0})$, a contradiction. □

Lemma 4.3 (Mixed-radix assembly)**.** *Suppose finite nonempty sets $G_j \subseteq \mathbb{N}$ have (q, H)-blockers c_j, and positive integer scales satisfy $\lambda_{j+1} = \lambda_j b_j$ with $b_j > 2H \max G_j$. Then $\bigcup_j \lambda_j G_j$ has a $(q, 2H)$-blocker.*

Proof. Read the colors of each c_j in $\mathbb{Z}/q\mathbb{Z}$. Every $n \geq 0$ has a unique expansion

$$n = r(n) + \sum_{j \geq 1} \lambda_j z_j(n), \qquad 0 \leq r(n) < \lambda_1, \quad 0 \leq z_j(n) < b_j,$$

with finitely many nonzero digits. Define

$$C(n) = \sum_{j \geq 1} (c_j(1 + z_j(n)) - c_j(1)) \pmod q.$$

The normalization makes every zero digit contribute zero, so the sum is finite. For a step $d = \lambda_j g$, $g \in G_j$, all lower digits and the remainder modulo λ_1 are fixed. Over $2H$ sites the quotient by λ_{j+1} changes by at most one, because $(2H - 1)g < b_j$. Thus at least H consecutive sites have

the same higher prefix. On them all digits except the jth are fixed, and the jth digit advances by g. If C were constant, c_j would have a monochromatic H-progression of step g, contrary to its definition. □

Theorem 4.4 (Exact approximate-extraction equivalence)**.** *For a large A, the following are equivalent:*

(1) *Every large $C \subseteq A$ contains a k-large subset which is not $(k+1)$-large.*
(2) *Every large $C \subseteq A$ satisfies $T_H(C)$ for some $H \geq 2$.*

Proof. For the first implication, take a separating subset of C and fix one of its (q, H)-blockers. Its k-largeness, together with Lemma 2.1, gives the required finite packets.

For the converse, apply Lemma 4.2 inside any given large $C \subseteq A$. Obtain a large $B \subseteq C$ and a fixed H for which T_H holds on every large subhost of B. Start with $\lambda_1 = 1$. At stage j, the set $B \cap \lambda_j \mathbb{N}$ is large by Proposition 2.5. Choose in it a finite $(k, j+2)$-forcing, (q, H)-blockable packet E_j. Let $G_j = E_j/\lambda_j$, choose $b_j > 2H \max G_j$, and set $\lambda_{j+1} = \lambda_j b_j$. Exact dilation invariance gives a (q, H)-blocker of G_j.

The union $D = \bigcup_j E_j \subseteq C$ is k-large because it contains forcing packets for arbitrarily large lengths. By Lemma 4.3, it has a $(q, 2H)$-blocker. It is therefore not q-large, as required. □

The theorem is hereditary in its hypotheses: it quantifies over *every* large subhost of A. It does not assert that $T_H(A)$ at one isolated host, without the subhost condition, already gives a separating subset of that same host.

Theorem 4.5 (Uniform compression in every counterhost)**.** *If a large A satisfies* (C_k)*, it contains a large B and integers $h(H) \geq 2$, for $H \geq 2$, such that*

$$\forall H \geq 2 \; \forall D \subseteq B \qquad [D \text{ has a } (k+1, H)\text{-blocker}] \Longrightarrow [D \text{ has a } (k, h(H))\text{-blocker}].$$

Proof. By Theorem 4.4, failure of hereditary separation in A gives a large $B \subseteq A$ for which $T_H(B)$ fails for every H. For each H, choose $h(H)$ witnessing this failure. No finite $(k+1, H)$-blockable subset of B is then $(k, h(H))$-forcing. If any possibly infinite such D were forcing, Lemma 2.1 would give a finite forcing subset, which inherits the blocker and contradicts the choice of $h(H)$. □

No contradiction to these simultaneous compression bounds is known from the arguments in this report. Constructing them in an actual large host would also be a substantial additional step.

5. The hierarchy in every additive quasi-central host

5.1. **The nilwindow property and the structural input.** For a nilsystem (Y, T), a point y, and an open neighborhood V of y, let $N_T(y, V) = \{n \geq 1 : T^n y \in V\}$. A set $D \subseteq \mathbb{N}$ has property P_r if every r-coloring of every such return window has arbitrarily long monochromatic progressions contained in the window and having difference in D. Nilrotations are translations on compact nilmanifolds. Disconnected nilmanifolds and nonminimal translations are allowed; each point's orbit closure is a minimal nilsystem. The one-point test shows that P_r implies r-largeness.

We shall use the following precise consequence of the polynomial return-time structure theorem [9, Theorem A]. For a minimal invertible compact metric system (X, T) with maximal infinite-step pronilfactor $\pi : X \to Z_\infty$, let

$$R_{\mathbf{p}}(U_1, \ldots, U_t) = \{n \in \mathbb{Z} : \bigcap_{i=1}^{t} T^{-p_i(n)} U_i \neq \varnothing\}.$$

If all U_i are nonempty and open, and every $p_i - p_j$ with $i \neq j$ is nonconstant, then

$$R_{\mathbf{p}}((\pi U_1)^\circ, \ldots, (\pi U_t)^\circ) \setminus R_{\mathbf{p}}(U_1, \ldots, U_t)$$

is not piecewise syndetic. The interiors are part of the theorem. In our applications the relevant image is already open. The factor Z_∞ is an inverse limit of finite-dimensional minimal nilsystems and contains every nilsystem factor of X.

We use the standard facts that nilrotations are distal and have minimal nilsystem orbit closures, that lattice-adapted Malcev coordinates have rational structure constants, and that connected component returns have unipotent affine presentations. These facts underlie the explicit tower and countable-test arguments below; see [18, 21].

A set is *thick* if it contains intervals of every finite length, *syndetic* if it has bounded gaps, and *piecewise syndetic* if finitely many of its shifts have thick union. Piecewise-syndetic sets form an upward-closed partition-regular family. Finite unions of non-piecewise-syndetic sets remain non-piecewise-syndetic.

We use the additive Stone–Čech semigroup $\beta\mathbb{N}$, with composition convention $(p+q)x = p(qx)$ in every compact dynamical system. A set A is *central* if $A \in u$ for a minimal idempotent $u \in \beta\mathbb{N}$. Minimal idempotents belong to the smallest ideal $K(\beta\mathbb{N})$, the union of the minimal left ideals. A set belongs to some member of $K(\beta\mathbb{N})$ exactly when it is piecewise syndetic. We use these standard characterizations and the Ellis idempotent lemma; see [7, 14]. A set A is *quasi-central* if $A \in u$ for an idempotent $u \in \overline{K(\beta\mathbb{N})}$ [13]. Equivalently, A belongs to an idempotent all of whose members are piecewise syndetic: $u \in \overline{K}$ exactly when every basic neighborhood $\overline{B}$, $B \in u$, meets K. Every central set is quasi-central.

Theorem 5.1. *For every quasi-central $A \subseteq \mathbb{N}$ and every $k \geq 1$ there is $D_{k,A} \subseteq A$ with P_k and a coloring in $k+1$ colors blocking $(D_{k,A}, H_k)$, where*

$$H_k = 400 \cdot 2^k.$$

Consequently $D_{k,A}$ is k-large and not $(k+1)$-large. The assertion also applies after deleting any non-piecewise-syndetic set from A. In particular the conclusion holds in every central set.

5.2. **Semigroup facts, including division of the host.**

Lemma 5.2. *If A is quasi-central and $\lambda \geq 1$, then $A/\lambda = \{n \geq 1 : \lambda n \in A\}$ is quasi-central. If E is not piecewise syndetic, $A \setminus E$ is quasi-central. Both statements also preserve centrality.*

Proof. Choose an idempotent $u \in \overline{K(\beta\mathbb{N})}$ with $A \in u$. Its image under reduction modulo λ is the idempotent 0, so $u \in S = \overline{\lambda\mathbb{N}}$. The map $n \mapsto \lambda n$ extends to a semigroup isomorphism $\beta\mathbb{N} \to S$. Its inverse image v of u is idempotent. If $B \in v$, then $\lambda B \in u$ is piecewise syndetic, which implies B is piecewise syndetic, by restricting bounded gaps on arbitrarily long intervals to 0 (mod λ) and dividing indices by λ. Thus $v \in \overline{K(\beta\mathbb{N})}$ and $A/\lambda \in v$.

For the additional centrality claim, suppose u is minimal and put $L = \beta\mathbb{N} + u$. The intersection $L_0 = L \cap S$ is a nonempty compact left ideal of S. If $p, q \in L_0$, minimality of L gives $q = v + p$ for some $v \in \beta\mathbb{N}$. Reduction modulo λ forces $v \in S$. Thus $S + p = L_0$ for every $p \in L_0$, and L_0 is a minimal left ideal of S. The inverse image of u is therefore a minimal idempotent containing A/λ.

No member of $u \in \overline{K(\beta\mathbb{N})}$ can be non-piecewise-syndetic. Thus $E \notin u$ and $A \setminus E \in u$. □

Lemma 5.3. *If every member of $u \in \beta\mathbb{N}$ is piecewise syndetic, then the same is true of every $p \in \beta\mathbb{N} + u$.*

Proof. Write $p = v + u$. If $B \in p$, then $\{n \geq 1 : -n + B \in u\} \in v$. Choose one such n. The set $-n + B$ is piecewise syndetic, and hence so is B. □

Lemma 5.4 (Generic frequencies on prescribed return sets)**.** *For countably many infinite subsets $S_j \subseteq \mathbb{N}$ and every $M \geq 1$, almost every $\alpha \in \mathbb{T}^M$ has $S_j\alpha$ dense in $\mathbb{T}^M$ for every j. One can also require rational independence of $1, \alpha_1, \ldots, \alpha_M$.*

Proof. Enumerate each S_j without repetitions as $(n_{j,t})_{t\ge1}$. For every nonzero $m \in \mathbb{Z}^M$, orthogonality gives

$$\int_{\mathbb{T}^M} \left| \frac{1}{N} \sum_{t=1}^{N} e(n_{j,t} m \cdot \alpha) \right|^2 d\alpha = \frac{1}{N}, \qquad e(t) = \exp(2\pi i t).$$

Along $N = \ell^2$, Chebyshev's inequality and Borel–Cantelli give convergence to zero almost everywhere. The ratio $(\ell+1)^2/\ell^2 \to 1$ fills in the intermediate N. Intersect the full-measure sets over j, m and use Weyl's criterion. Rational dependence also excludes only a null set. □

For a distal compact system (X, T), its Ellis semigroup $E(X,T) = \overline{\{T^n : n \ge 1\}} \subseteq X^X$ is a compact right-topological algebraic group of bijections. Right multiplication is continuous; joint continuity is not assumed.

Lemma 5.5. *A nonempty compact subsemigroup of an Ellis group is a subgroup.*

Proof. Let P be the subsemigroup. A minimal left ideal J of P contains an idempotent. The ambient group has only one idempotent, its identity 1, so $1 \in J$ and $P = P1 \subseteq J$. Thus $J = P$. For $p \in P$, the compact left ideal Pp equals P, so $qp = 1$ for some $q \in P$. Hence $q = p^{-1}$. □

5.3. **The stabilizer dimension bound.** We spell out the finite-dimensional feature needed from nilsystems. First restrict each tested nilsystem, including a product test, to the cycle of connected components containing the tested point. This changes neither its orbit nor the relevant dimension bound. Such a system admits a finite tower over a finite system whose successive fibers are compact tori acting freely and commuting with the transformation. The sum of the torus dimensions is the manifold dimension. For a connected affine presentation N/Λ with transformation $z \mapsto bA(z)$, A is a lattice-preserving unipotent automorphism. The center of $\mathrm{Lie}(N)$ is a nonzero rational A-invariant subspace whenever $\dim N > 0$. Since A is unipotent, its restriction has a nonzero rational fixed subspace there. Exponentiating that subspace gives a central torus in N/Λ fixed by A; its action commutes with bA. Quotient and repeat. For finitely many cyclically permuted components, do this for the return map on a component and transport the torus action to the other components by the transformation. Commutation with the return map makes the transported action well defined. The zero-dimensional remainder is finite. This also covers powers, products, and the affine presentations used in compact extraction.

Lemma 5.6. *Let P be a compact subgroup of a distal Ellis group with an induced action on a finite-dimensional nilsystem W, and let $\tau : P \to \mathbb{T}^M$ be any continuous group homomorphism. For $w \in W$ let $P_w = \{g \in P : gw = w\}$. Then*

$$\dim \tau(P) - \dim \tau(P_w) \le \dim W.$$

Proof. Use the finite central-torus tower of W, restricting to the component cycle containing w. A finite base costs no dimension because its stabilizers have finite index. At a top torus fiber K, let P_0 fix the projected basepoint and write $gw = \rho(g)w$ for $g \in P_0$. Central fiber translations commute with every Ellis map, so $\rho : P_0 \to K$ is a continuous homomorphism with kernel P_w. The map

$$\rho(P_0) \longrightarrow \tau(P_0)/\tau(P_w), \qquad \rho(g) \longmapsto \tau(g) + \tau(P_w)$$

is well defined, continuous, and onto. Its source is a compact Lie subgroup of K, so the dimension drop is at most $\dim K$. Induct through the tower. This proof requires only that τ is a continuous homomorphism, not that it comes from a rotation factor of the original system. □

5.4. **The finite torus gate.** For a blocking palette $q \geq 2$, put

$$Q = 100q, \qquad a = Q^{-1}, \qquad \eta = (10^6 Q^2)^{-1}.$$

Let $O_M \subseteq \mathbb{T}^M$ consist of vectors whose centered coordinates satisfy, for some i,

$$|\theta_i + a| < \eta, \quad |\theta_j| < \eta \ (j \neq i), \quad \sum_{j \neq i} |\theta_j| < 2a.$$

This is the finite-dimensional version of Alweiss's root window [3]; the constants and palette mechanism follow [1].

Lemma 5.7. *Given b, q, every connected subtorus of $\mathbb{T}^M$ of codimension at most b meets O_M when M is sufficiently large in terms of b, q.*

Proof. Write its tangent space as the kernel of a matrix with columns $v_1, \dots, v_M$ spanning dimension at most b. Give this span the norm whose unit ball is $\operatorname{conv}\{\pm v_j\}$. The unit ball is covered by at most $(1 + 2/\epsilon)^b$ balls of radius ϵ. For large M, one such ball contains v_i and L further indexed columns. Their average $\bar{v}$ satisfies $v_i - \bar{v} = \sum c_j v_j$, with $\sum |c_j| \leq 2\epsilon$. Move $c_i v_i$ to the left to obtain $v_i = \sum_{j \neq i} t_j v_j$, where

$$\sum_{j \neq i} |t_j| \leq \frac{1 + 2\epsilon}{1 - 2\epsilon}, \qquad \max_{j \neq i} |t_j| \leq \frac{L^{-1} + 2\epsilon}{1 - 2\epsilon}.$$

Choose ϵ small and L large so these are strictly below 2 and η/a. The tangent vector with coordinate $-a$ at i and at_j elsewhere reduces to a point of O_M. The zero-dimensional column span is immediate. □

Proposition 5.8 (Quasi-central-host gate)**.** *Let A be quasi-central. For every $q \geq 2, h \geq 2, C \geq 0$, there exist $M = M(q, hC)$ and $\alpha \in \mathbb{T}^M$ such that $1, \alpha_1, \dots, \alpha_M$ are rationally independent and, simultaneously for every nilsystem (Z, T) of dimension at most C, every $z \in Z$ and every open $V \ni z$, the set*

$$A \cap \{d \geq 1 : d\alpha \in O_M\} \cap \{d \geq 1 : T^{jd} z \in V \ (1 \leq j \leq h)\} \tag{2}$$

is piecewise syndetic. In particular the frequencies are chosen before the test system, window, and any coloring.

Proof. Choose an idempotent $u \in \overline{K(\beta\mathbb{N})}$ with $A \in u$. Let $x \in \{0,1\}^{\mathbb{Z}}$ be the indicator of A on the positive coordinates, extended arbitrarily elsewhere. Let S be the left shift and set $y = ux$. Then $y_0 = 1$ and $uy = y$. Choose a countable neighborhood basis (V_j) at y. Each return set $N_S(y, V_j)$ belongs to u and is infinite, since u is a free idempotent. Choose M by Lemma 5.7 with $b = hC$, and choose α by Lemma 5.4 for all these return sets. Then

$$\{y\} \times \mathbb{T}^M \subseteq \overline{\{(S^n y, n\alpha) : n \geq 1\}}. \tag{3}$$

Indeed every product neighborhood of (y, θ) is met. No minimality assertion about the orbit closure of y is used.

Put $L = \beta\mathbb{N} + u$ and

$$K = \{p \in L : px = y\}.$$

This is nonempty and compact. For every $p \in L$, $p + u = p$; hence if $p \in K$, then $py = p(ux) = px = y$. Consequently K is a subsemigroup: for $p, q \in K$, $(p+q)x = py = y$, and $p+q \in L$ because L is a left ideal.

Apply these ultrafilters to the distal system

$$(Z^h \times \mathbb{T}^M, T \times T^2 \times \cdots \times T^h \times (+\alpha)).$$

The image P of K in its Ellis group is a compact subgroup by Lemma 5.5. It has full auxiliary image $\tau(P) = \mathbb{T}^M$. Indeed, for any $\theta \in \mathbb{T}^M$, (3) provides $v \in \beta\mathbb{N}$ with $vy = y$ and $v0 = \theta$ on the

torus. Then $v+u\in K$, while u acts as the identity on the torus because its image is idempotent. Thus $\tau(v+u)=\theta$.

Lemma 5.6 at $(z,\dots,z)$ gives a connected subtorus of dimension at least $M-hC$ inside the auxiliary image of its stabilizer in P. By Lemma 5.7, choose an element of that stabilizer with auxiliary displacement in O_M, and choose a preimage $p\in K$. Along p the host indicator converges to y with $y_0=1$, all nil coordinates converge to z, and the torus coordinate converges into O_M. Therefore the set in (2) belongs to p. Since $p\in L=\beta\mathbb{N}+u$, Lemma 5.3 says that set is piecewise syndetic, as required. □

5.5. **One palette for a relative stage.** For these frequencies define

$$F(n)=\sum_{j=1}^{M}\operatorname{Re}(e(n\alpha_j)-1),\qquad J(n)=\lfloor qF(n)\rfloor,\qquad e(t)=\exp(2\pi it).$$

On every $2Q$-term progression with $d\alpha\in O_M$, J visits all residues modulo q. Here is the uniform estimate. A step changes qF by at most $2\pi q(3/Q+\eta)<1$. Choose $0\le t<Q$ for which $\|n\alpha_i-t/Q\|\le 1/(2Q)$, and put $m=Q/2$. At the progression indices $t,t+m,t+2m$, the second difference of F has distinguished contribution

$$\operatorname{Re}\left(e(n\alpha_i+t\theta_i)(e(m\theta_i)-1)^2\right)>3.96-4\pi Q\eta-\pi^2Q^2\eta^2.$$

The other coordinates contribute error less than $2\pi^2Q\eta$, using $\sum_{j\neq i}\theta_j^2<2a\eta$. Thus the second difference exceeds 3.9, and the range of F has width greater than 1.95. The floors of qF move by at most one and visit at least q consecutive integers.

For input labels $i\in\{0,\dots,q-2\}$, define a chart ι_J by choosing the unique integer $t\in[J-(q-1),J-1]$ congruent to i modulo $q-1$ and putting $\iota_J(i)=t-1 \pmod q$. Its missing output is $J-1 \pmod q$. Adjacent charts change exactly one input output, moving it to the previously missing output. Consequently

$$|J-J'|\le 1,\quad \iota_J(i)=\iota_{J'}(i')\quad\Longrightarrow\quad i=i'.$$

Suppose $B\subseteq A$ has a $(q-1)$-color blocker f at length $H\ge 2$. Put

$$B_M=\left\{d\in B:\sum_{j=1}^{M}\|d\alpha_j\|<(4\pi qH)^{-1}\right\},\qquad S_{A,M}=(A\cap G_M)\cup B_M,\quad G_M=\{d\ge 1:d\alpha\in O_M\}.$$

Then $n\mapsto\iota_{J(n)}(f(n))$ is a q-color blocker for $S_{A,M}$ at length

$$L=\max(200q,H).\tag{4}$$

For G_M directions all possible outputs are omitted at some site. On the first H sites in a B_M direction, the values of qF differ by less than $1/2$, so the charts take at most two consecutive values. A constant output then forces constant input, contrary to f. In particular L is independent of M, the host, and the target nilwindow.

5.6. **Forcing one more color inside the host.**

Lemma 5.9. *Assume $B\subseteq A$ has P_s and an $(s+1)$-color blocker of length H, and A is quasi-central. Put $q=s+2$. For every h,C, choose M,α from Proposition 5.8. Then $S_{A,M}$ forces a monochromatic h-term progression for every $(s+1)$-coloring of $N_T(z,V)$, for every nilmanifold of dimension at most C and every open $V\ni z$. For $s=0$, take $B=\varnothing$, $H=2$ and constant input f; the assertion remains valid.*

Proof. Extend the coloring to $\mathbb{Z}$ and take a minimal subsystem X in the forward orbit limit set of the colored sequence joined to z. It projects onto the minimal nil-orbit closure Z_0 of z. Factor this projection through the maximal infinite-step pronilfactor:

$$X\xrightarrow{\pi}Z_\infty\xrightarrow{\rho}Z_0.$$

Put $V_0 = V \cap Z_0$, $W = \rho^{-1}(V_0)$, and let C_1 be the first-color cylinder in X.

If $\pi(C_1) \supseteq W$, the open set $U = C_1 \cap \pi^{-1}(W)$ has $\pi(U) = W$. For an open set O, let $\mathrm{Ret}_h(O)$ be the positive d for which $\bigcap_{t=0}^{h-1} T^{-td}O \neq \varnothing$. Surjectivity gives $\mathrm{Ret}_h(W) = \mathrm{Ret}_h(V_0)$. By [9, Theorem A],

$$\mathrm{Ret}_h(W) \setminus \mathrm{Ret}_h(U)$$

is not piecewise syndetic. The theorem's forms $n, 2n, \dots, hn$ give the same return differences as $0, n, \dots, (h-1)n$ by invertibility. Proposition 5.8, applied directly to the original target (Z_0, T, z, V_0), supplies a piecewise syndetic subset of $A \cap G_M \cap \mathrm{Ret}_h(V_0)$. It cannot be contained in the exceptional set, so some such difference belongs to $\mathrm{Ret}_h(U)$. This gives an h-term first-color progression in the required window. Crucially, M depends on the original target dimension, not on the symbolic extension or a factor chosen after seeing the coloring.

Otherwise $W \setminus \pi(C_1)$ is nonempty and open. The inverse-limit description gives a nonempty open set V' in a finite nilfactor Z' whose full inverse image lies in this complement. Above V' only the other s colors occur, and all points project into V_0. This is impossible when $s = 0$. Pick a point over $z' \in V'$. Along its orbit color visits to V', and restrict further to $n\alpha \in O$ for a small open neighborhood O of 0 with

$$\sum_j \|v_j\| < (4\pi q H)^{-1} \quad (v \in O - O).$$

This is a nilwindow for $(z', 0)$ in $Z' \times \mathbb{T}^M$. Apply P_s of B. The first two sites of the resulting monochromatic progression show that its difference lies in B_M. The configuration therefore again has its difference in $S_{A,M}$ and its sites in the original target window.

Finally, the finite color and open-window conditions obtained in X occur arbitrarily far forward in the original joined orbit, because X was chosen in its forward limit set. Thus the original coloring has the required progression with positive starting index. □

5.7. **Compact extraction and mixed-radix assembly.** We prove Theorem 5.1 simultaneously over all quasi-central hosts. Start with the vacuous level $s = 0$ and $H_0 = 2$. Assume level s has been constructed for every quasi-central host. Fix a quasi-central A and put $q = s + 2$, $L = \max(200q, H_s)$.

Here are the compact test parameters used in the extraction. There are countably many rational Malcev presentations (N, Λ) and countably many lattice-preserving unipotent automorphisms R. Write $T_b(z) = bR(z)$. Exhaust $b \in N$ by compact subsets, and use countably many window pairs $K \subset V$ with K closed and V open, chosen so every point in every open window belongs to one such K with V inside that window. Include a positive integer m recording a component-return period. These tests cover every nilwindow: on a connected component of a nilmanifold, a return power of a niltranslation has this affine form. Indeed, writing the return translating element as $b\gamma$ with $b \in G^0$ and $\gamma \in \Gamma$ gives $b\,\mathrm{Ad}(\gamma)$; on the simply connected cover the induced lattice automorphism is unipotent because sufficiently long commutators vanish. Countability follows from rational structure constants and the finite rational data for a lattice and its automorphisms.

Set $\lambda_1 = 1$. At stage j, take the first j test families, retaining those with $m \mid \lambda_j$; include the one-point test at every stage. Let $h_j = j + 2$ and let C_j be the maximum target dimension. By Lemma 5.2, $A_j = A/\lambda_j$ is quasi-central. Use its already constructed set $B_j = D_{s,A_j}$ and apply Lemma 5.9 with h_j, C_j. The resulting direction set $S_j \subseteq A_j$ has a q-color blocker of length L.

For each retained test, use the fixed transformation $T_b^{\lambda_j/m}$ and the compact parameter space

$$\{b \text{ in its compact set}\} \times K \times \{1, \dots, s+1\}^{\mathbb{Z}}.$$

For each positive start a_0 and $d \in S_j$, the conditions of monochromaticity and

$$(T_b^{\lambda_j/m})^{a_0+td} z \in V \quad (0 \le t < h_j)$$

are open in this parameter space. Lemma 5.9 says these open sets cover it. A finite subcover, for each of the finitely many tests, provides a finite nonempty $F_j \subseteq S_j \subseteq A_j$ which works for all their parameters and colorings. No uniform time in a symbolic extension is assumed: finite extraction occurs only after the pointwise forcing statement.

Choose an integer $b_j > 2L \max F_j$ such that $(j+1)! \mid \lambda_j b_j$, put $\lambda_{j+1} = \lambda_j b_j$, and define

$$D_{s+1,A} = \bigcup_{j \geq 1} \lambda_j F_j.$$

Containment in A follows from $F_j \subseteq A/\lambda_j$. To verify P_{s+1}, take a nilwindow and a coloring of its visits, and choose its component-return test with period m. For a sufficiently large j, that test is among the first j, $m \mid j! \mid \lambda_j$, and h_j exceeds the desired length. Read the coloring at multiples of λ_j and apply the packet property. Multiplying its indices by λ_j gives the required original-window progression with difference in $\lambda_j F_j$.

For the obstruction, let $c_j : \mathbb{N} \to \mathbb{Z}/q\mathbb{Z}$ block F_j at length L, and let $z_j(n)$ be the mixed-radix digit of n at place λ_j, with $0 \leq z_j(n) < b_j$. Define

$$c(n) = \sum_{j \geq 1} \big(c_j(1 + z_j(n)) - c_j(1)\big) \pmod q.$$

The sum is finite. On $2L$ sites of step $\lambda_j f$, $f \in F_j$, the lower digits are fixed. The higher prefix $\lfloor n/\lambda_{j+1} \rfloor$ changes at most once since $(2L-1)f < b_j$. At least L consecutive sites have constant lower and higher digits; their jth digit increases by f. A constant output would contradict c_j. Thus

$$H_{s+1} = 2 \max(200(s+2), H_s), \qquad H_0 = 2,$$

which yields $H_k = 400 \cdot 2^k$ for $k \geq 1$. This completes the simultaneous induction and the proof of Theorem 5.1.

5.8. **Scope and nearby host classes.** The hypothesis is additive quasi-centrality. It cannot be replaced in this proof merely by the existence of some additive idempotent containing the host: every ultrafilter in its generated left ideal must have only piecewise-syndetic members, so that the structural exceptional set can be removed. An arbitrary large set need not be quasi-central or even piecewise syndetic. No conclusion for all large hosts follows here.

Lemma 5.10 (Boundary return hosts are central)**.** *Let (Y, S) be any minimal compact metric system, let U be nonempty and open, and suppose $y \in \overline{U}$. If $T \subseteq \mathbb{N}$ is thick and $E \subseteq \mathbb{N}$ is not piecewise syndetic, then*

$$(N_S(y, U) \cap T) \setminus E$$

is central. No nilsystem or distality hypothesis is needed.

Proof. Put $R = N_S(y, U)$. The closure of the thick set T contains a minimal left ideal L of $\beta\mathbb{N}$. We claim that

$$P = L \cap \overline{R} \cap \{p : py = y\}$$

is nonempty. For every neighborhood V of y, $U \cap V$ is nonempty and open. Thus $R \cap N_S(y, V) = N_S(y, U \cap V)$ is syndetic. Every syndetic set has closure meeting every minimal left ideal: finitely many of its translates cover $\mathbb{N}$, so translating any member of the ideal by one of these integers gives a member containing that set. Applying this to finite intersections of the neighborhoods V, compactness gives $P \neq \varnothing$.

The set P is compact and closed under addition. For $p, q \in P$, $p + q \in L$ and $(p+q)y = y$. If $n \in R$, then $S^n y \in U$; since $qy = y$ and U is open, $-n + R \in q$. Hence $R \subseteq \{n : -n + R \in q\}$, which belongs to p, so $R \in p + q$. Ellis's lemma supplies an idempotent $u \in P$. It is minimal because $u \in L$, and $R \cap T \in u$. Finally $E \notin u$, proving the assertion. □

Theorem 5.1 therefore covers every superset of a boundary minimal-return host as in Lemma 5.10, including the earlier boundary nilsystem hosts. This observation does not say that an arbitrary central set can be intersected with an arbitrary thick set and remain central.

Corollary 5.11 (The absolute hierarchy)**.** *For every $k \geq 1$ there is $D_k \subseteq \mathbb{N}$ with P_k and a $(k+1, 400 \cdot 2^k)$-blocker.*

Proof. The full set $\mathbb{N}$ belongs to every minimal idempotent and is therefore central. Apply Theorem 5.1. □

Corollary 5.12 (A pointwise all-nil recurrence seed)**.** *Every quasi-central A contains a pointwise nil-recurrent set S with a binary blocker of length 800. Here pointwise nil recurrence means that for every nilrotation T, every point z, and every open $V \ni z$, some $d \in S$ has $T^d z \in V$. In fact the same d may be required to satisfy $T^{jd} z \in V$ for any prescribed finite range $1 \leq j \leq h$.*

Proof. At each stage use Proposition 5.8 with $q = 2$ and with the finite list of pointwise tests from the compact extraction above. The stage set $A/\lambda \cap G_M$ has a binary blocker at length $200q = 400$: use the palette construction with the empty old direction set and a constant one-color input. The gate gives actual pointwise returns, so compactness of the translation and point parameters extracts a finite packet returning every tested point at all required multiples. No coloring parameter is needed for this extraction. Use factorially divisible scales exactly as in the preceding assembly, and apply Lemma 4.3 with $H = 400$. The resulting subset has a binary blocker at length 800. The exhausting pointwise tests give the asserted simultaneous returns. This conclusion uses the gate itself; it is stronger than the statement P_1 alone. □

6. Polynomial images of quasi-central sources

The source variable and the eventual progression difference must be distinguished in this extension. Piecewise syndeticity will hold in the source parameter n, while the difference is $p(n)$ and may run through a sparse set. The hypothesis $p(0) = 0$ is used both by the clock and by the exact division formula at assembly.

Theorem 6.1. *Let $A \subseteq \mathbb{N}$ be quasi-central, and let $p \in \mathbb{Z}[n]$ be nonconstant, have positive leading coefficient, and satisfy $p(0) = 0$. Put $p_+(A) = \{p(n) : n \in A,\ p(n) > 0\}$. For every $k \geq 1$ there is $D_{k,p,A} \subseteq p_+(A)$ with P_k and a $(k+1)$-coloring of $\mathbb{N}$ having no monochromatic $D_{k,p,A}$-progression of length*

$$H_k = 400 \cdot 2^k.$$

Thus every such polynomial image contains the strict hierarchy.

We use the elementary torus gate, chart coloring, compact nilwindow tests, and mixed-radix argument proved in Section 5; the precise uses are restated below. The external inputs are the polynomial return-time structure theorem [9, Theorem A] and polynomial nil-orbit linearization [18, Proposition 3.14]. The latter's finite-generator construction gives the uniformity explained in Lemma 6.3. The new step is that the top coordinate of a unipotent torus clock supplies a continuous homomorphism on the subgroup fixing its lower coordinates. It is not a rotation factor of the whole clock.

6.1. Two uniform polynomial-clock facts.

Lemma 6.2. *Fix $d, M \geq 1$ and countably many infinite $S_j \subseteq \mathbb{N}$. For almost every $\alpha \in \mathbb{T}^M$, each sequence of points*

$$\left(\binom{n}{1}\alpha, \binom{n}{2}\alpha, \ldots, \binom{n}{d}\alpha \right), \quad n \in S_j,$$

is uniformly distributed in $\mathbb{T}^{dM}$ when S_j is enumerated without repetition. In particular its image is dense.

Proof. For a nonzero character $(m_1, \dots, m_d) \in (\mathbb{Z}^M)^d$, the vector-valued polynomial $Q(n) = \sum_{r=1}^{d} \binom{n}{r} m_r$ is nonconstant, of degree at most d, and takes integer vector values. Every value of Q has at most d preimages: use any nonconstant coordinate. Consequently, for distinct $n_1, \dots, n_N$,

$$\int_{\mathbb{T}^M} \left| \frac{1}{N} \sum_{t=1}^{N} e(Q(n_t) \cdot \alpha) \right|^2 d\alpha \le \frac{d}{N}, \qquad e(t) = \exp(2\pi i t).$$

Chebyshev and Borel–Cantelli on $N = \ell^2$ give almost-everywhere convergence to zero; consecutive squares have ratio tending to one, so all N follow. Intersect the countably many full-measure sets over characters and j, then apply Weyl's criterion. □

Lemma 6.3 (Uniform polynomial nil-orbit linearization)**.** *Fix finitely many nilmanifold presentations, a degree bound d, and $h \ge 1$. There is a finite integer C_*, depending on these data, with the following property. For every translation T in one of these presentations, every point z, and every $p \in \mathbb{Z}[n]$ of degree at most d with $p(0) = 0$, there are a nilsystem (W, R) of dimension at most C_*, a point w, and a continuous map $\psi : W \to Z^h$ such that*

$$\psi(R^n w) = (T^{p(n)} z, T^{2p(n)} z, \dots, T^{hp(n)} z) \quad (n \in \mathbb{Z}), \qquad \psi(w) = (z, \dots, z).$$

For the compact extraction, it is enough that the bound is uniform on each fixed finite list of presentations; no universal bound solely in terms of $\dim Z$ *is asserted.*

Proof. Represent the polynomial orbit in the product nilpotent group G^h, absorbing a representative of z into the constant term of the polynomial sequence. In fixed Malcev coordinates, the coordinate degrees of this sequence are bounded in terms of the presentation, d, and h. The bounds do not depend on the coefficients of p, the translating element, or the representative of z: the coordinate laws for products and powers are fixed polynomials.

Apply the construction in [18, Proposition 3.14]. For each coordinate polynomial it introduces a chain of generators of length one more than its degree in a free nilpotent Lie group of fixed nilpotency class. The resulting group, lattice, and unipotent automorphism have bounded dimension in terms of the number of coordinate polynomials, their degree bounds, and the original nilpotency class. Their coefficients determine the starting element. The projection of its linear orbit is the prescribed polynomial orbit. Realize the unipotent automorphism as a niltranslation in the semidirect-product presentation, as in [9, proof of Lemma 4.7]; the extra discrete generator does not increase manifold dimension. This proves the claimed bound. At $n = 0$ the desired projection equals $(z, \dots, z)$ because $p(0) = 0$.

Connected affine presentations $T_b = bB$ with fixed unipotent lattice automorphism B are included by using the fixed nilpotent group $N \rtimes_B \mathbb{Z}$. Its nilpotency class and coordinate laws are fixed while b varies. Powers fixed at a particular extraction stage also have uniform degree bounds. □

6.2. **A character on the top-clock stabilizer.** We use the general continuous-homomorphism form of Lemma 5.6.

Define the affine unipotent clock on $\mathbb{T}^{dM}$ by

$$R_\alpha(x_1, \dots, x_d) = (x_1 + \alpha, x_2 + x_1, \dots, x_d + x_{d-1}).$$

Its orbit of 0 is the vector in Lemma 6.2. It is a nilsystem in the semidirect-product presentation for an affine unipotent torus map, and in particular is distal. Translations of the top coordinate x_d commute with R_α and hence with all its Ellis maps. If a subgroup fixes the lower-clock basepoint $(x_1, \dots, x_{d-1}) = 0$, evaluation of its displacement in x_d is therefore a continuous homomorphism to $\mathbb{T}^M$. For $d = 1$, the lower clock is the one-point system.

6.3. **The polynomial quasi-central gate.** For $q \geq 2$ put $Q = 100q$, $a = Q^{-1}$ and $\eta = (10^6 Q^2)^{-1}$. Let $O_M \subseteq \mathbb{T}^M$ consist of vectors with centered coordinates satisfying, for some i,

$$|\theta_i + a| < \eta, \qquad |\theta_j| < \eta \ (j \neq i), \qquad \sum_{j \neq i} |\theta_j| < 2a.$$

The torus gate lemma says that every connected subtorus of codimension at most b meets O_M for sufficiently large $M = M(b, q)$; its convex-column proof is in Section 5.

Proposition 6.4. *Fix a quasi-central source A, a polynomial p as in Theorem 6.1, q, h, and finitely many nilpresentations for target systems. There are M and $\alpha \in \mathbb{T}^M$, chosen before any target translation, point, window, or coloring, such that for every one of these targets and every open $V \ni z$, the parameter set*

$$\{n \in A : p(n) > 0,\ p(n)\alpha \in O_M,\ T^{jp(n)} z \in V \ (1 \leq j \leq h)\} \tag{5}$$

is piecewise syndetic in the variable n.

Proof. Let $d = \deg p$ and write $p(n) = \sum_{r=1}^{d} c_r \binom{n}{r}$ with $c_r \in \mathbb{Z}$, $c_d \neq 0$. Choose C_* from Lemma 6.3 and $M = M(C_*, q)$ from the torus gate lemma. Choose an idempotent $u \in \overline{K(\beta\mathbb{N})}$ containing A, let x be its bilateral indicator, and put $y = ux$ under the shift. Then $y_0 = 1$ and $uy = y$. Every return set to a member of a countable neighborhood basis at y is an infinite member of u. Choose α by Lemma 6.2 for these sets. Consequently

$$\{y\} \times \mathbb{T}^{dM} \subseteq \overline{\{(S^n y, R_\alpha^n 0) : n \geq 1\}}. \tag{6}$$

Put $L = \beta\mathbb{N} + u$ and $K = \{v \in L : vx = y\}$. As in Section 5, K is a nonempty compact subsemigroup: $v + u = v$ on L, so every $v \in K$ also fixes y. Every member of every $v \in L$ is piecewise syndetic, because $B \in t + u$ implies $-m + B \in u$ for some integer m.

For a specified target choose (W, R, w, ψ) from Lemma 6.3, and take the image P of K in the Ellis group of the distal system $(W \times \mathbb{T}^{dM}, R \times R_\alpha)$. A compact subsemigroup of an algebraic Ellis group is a subgroup, so P is a compact subgroup. Moreover, evaluation of P at the clock basepoint covers all of $\mathbb{T}^{dM}$. Given a clock point v, (6) gives $t \in \beta\mathbb{N}$ with $ty = y$ and $t0 = v$ on the clock. Then $t + u \in K$, and u acts identically on the distal clock because its Ellis image is idempotent. Thus $(t + u)0 = v$.

Let $P_0 \leq P$ fix 0 in the lower-clock factor and let $\rho : P_0 \to \mathbb{T}^M$ be top displacement. It is a continuous homomorphism. The preceding surjectivity, applied to $(0, \ldots, 0, v)$, gives $\rho(P_0) = \mathbb{T}^M$. Thus $\tau = c_d \rho$ is also onto. Lemma 5.6 shows that $\tau((P_0)_w)$ contains a connected subtorus of codimension at most C_*. Choose $g \in (P_0)_w$ with $\tau(g) \in O_M$, and choose a preimage $v \in K$. Along this ultrafilter, the target linearization returns to w, while the lower clock tends to 0 and its top coordinate tends to $\rho(g)$. Hence

$$v\text{-}\lim_n p(n)\alpha = v\text{-}\lim_n \sum_{r=1}^{d} c_r (R_\alpha^n 0)_r = c_d \rho(g) \in O_M.$$

The host indicator tends to y with $y_0 = 1$. The polynomial has only finitely many nonpositive values on $\mathbb{N}$; v is free. It follows that the parameter set (5) belongs to $v \in L$ and is piecewise syndetic. □

6.4. **Relative palette induction and finite assembly.** We give the remaining steps to track the parameter/difference distinction. Induct simultaneously over all pairs (p, A) in Theorem 6.1, starting with an empty lower set at level $s = 0$ and $H_0 = 2$. Suppose $B \subseteq p_+(A)$ has P_s and an $(s + 1)$-color blocker of length H_s. Put $q = s + 2$, choose the gate frequencies for the finite list of

target presentations and desired length, and set

$$S = (p_+(A) \cap \{d : d\alpha \in O_M\}) \cup \left\{ d \in B : \sum_{i=1}^{M} \|d\alpha_i\| < (4\pi q H_s)^{-1} \right\}.$$

The chart coloring from Section 5 blocks S in q colors at length $L = \max(200q, H_s)$, uniformly in all host and clock data. It is applied to actual differences d; only the forcing argument is sampled in the source variable n.

For an $(s+1)$-coloring of a prescribed nilwindow, choose the symbolic minimal extension and its maximal pronilfactor as in Section 5. Denote the inverse image of the original target window by W_0, and the first-color cylinder by C_1. If the image of C_1 covers W_0, let U_1 be its restriction above W_0. Then the factor-return set equals the original target-return set. Apply [9, Theorem A] with the distinct nonconstant forms

$$p(n), 2p(n), \dots, hp(n).$$

It says that the parameters allowed by factor returns but not by U_1 returns form a non-piecewise-syndetic set of n. Proposition 6.4 supplies a piecewise syndetic set of parameters in A meeting the gate and all the target returns. One survives the exception; its positive difference $p(n)$ lies in the first part of S and gives the desired first-color progression. The forms $p, \dots, hp$ give the same progression differences as $0, p, \dots, (h-1)p$ by invertibility.

If the first-color image does not cover W_0, a nonempty finite nilfactor window has only the other s colors above it. Append a sufficiently small torus-zero window and apply P_s of B. The resulting difference belongs to the second part of S. This case is absent for $s = 0$. Thus the same finite forcing statement as in Section 5 holds for every prescribed finite family of nilpresentations, all its compact parameters, and every coloring.

Here is the host containment at assembly. If $\lambda \geq 1$, put

$$A_\lambda = A/\lambda, \qquad p_\lambda(n) = p(\lambda n)/\lambda.$$

The set A_λ is quasi-central by the division lemma of Section 5; $p_\lambda \in \mathbb{Z}[n]$ has zero constant term and positive leading coefficient. Moreover

$$\lambda(p_\lambda)_+(A_\lambda) \subseteq p_+(A). \tag{7}$$

Run the countable compact nilwindow exhaustion with mixed-radix places λ_j. At stage j use the induction hypothesis for $(p_{\lambda_j}, A_{\lambda_j})$, the first j applicable test families, and target length $j+2$. Lemma 6.3 gives a common finite dimension bound for this finite list, so Proposition 6.4 chooses one common M, α before the pointwise forcing argument. Compactness of the original translation, point, and coloring parameter spaces extracts a finite packet

$$F_j \subseteq (p_{\lambda_j})_+(A_{\lambda_j})$$

with the required finite forcing and a q, L blocker. The open cover is expressed in the original target coordinates; no continuous choice of the polynomial linearization is needed.

Choose a radix $b_j > 2L \max F_j$ with $(j+1)! \mid \lambda_j b_j$, and let $\lambda_{j+1} = \lambda_j b_j$. Then

$$D_{s+1,p,A} = \bigcup_{j \geq 1} \lambda_j F_j \subseteq p_+(A)$$

by (7). Exhaustion and the factorial divisibility give P_{s+1} exactly as in Section 5. If c_j blocks F_j and $z_j(n)$ is the digit at place λ_j, the finite sum

$$c(n) = \sum_j \left(c_j(1 + z_j(n)) - c_j(1)\right) \pmod q$$

blocks $D_{s+1,p,A}$ at length $2L$: on any $2L$ sites of step $\lambda_j f$, the lower digits are fixed and the higher prefix changes at most once, leaving L consecutive sites controlled by c_j. Therefore $H_{s+1} = 2\max(200(s+2), H_s)$ and $H_k = 400 \cdot 2^k$. This proves Theorem 6.1.

Scope. All piecewise-syndetic assertions in the polynomial gate and structural lifting concern the source parameter n, not the image $p_+(A)$. Intersectivity alone cannot replace $p(0) = 0$ uniformly over all quasi-central sources: $p(n) = n - 1$ is intersective, but it maps the central source $2\mathbb{N}$ onto the positive odd integers. Parity is a binary two-term blocker for the odd differences. The zero-constant hypothesis in Theorem 6.1 is therefore a substantive restriction.

7. A binary nil-recurrence seed in every IP_0 host

A set $A \subseteq \mathbb{N}$ is IP_0 if, for every r, it contains $\mathrm{FS}(b_1, \dots, b_r)$ for some positive integers b_i, with repetitions permitted. This is weaker than containing a single infinite finite-sums set. The next construction uses pointwise nil recurrence, not just ordinary recurrence of scalar polynomial phases and not merely the nilwindow property P_1.

Fix a pointwise nil-recurrent set S with a binary blocker at length H. Such a set with $H = 800$ is supplied by the proved Corollary 5.12 with $A = \mathbb{N}$. Products of a nilrotation and its powers let the same recurrence property impose any finite list of simultaneous returns.

Theorem 7.1. *For the pointwise recurrent seed S fixed above, every IP_0 set $A \subseteq \mathbb{N}$ contains a pointwise nil-recurrent set D with a binary blocker at length $8H$. Thus one may take $8H = 6400$ using Corollary 5.12. In particular D has the all-nil-window property P_1 and is not 2-large.*

7.1. **Symmetric Host–Kra cubes.** First let $Y = G/\Gamma$, where G is a connected simply connected s-step nilpotent Lie group and Γ is a lattice. Equip G with its lower central filtration $G_0 = G_1 = G$, $G_{s+1} = \{1\}$. All filtration groups are rational. Write $C^r(Y)$ for its compact space of Host–Kra r-cubes. We identify a vertex with a subset of $[r]$. The group cube space $C^r(G)$ is generated by face elements $[g]_I$, with $g \in G_{|I|}$, supported on the vertices containing I. In any fixed order of the subsets respecting inclusion, each group cube has a unique ordered face decomposition. In particular, two group r-cubes with equal values at every proper vertex have top values differing by right multiplication by an element of G_r. These are the standard explicit cube-group facts in [11, Proposition A.5].

We will also use the Hall–Petresco facts: sequences

$$p(n) = g_0 g_1^{\binom{n}{1}} \cdots g_s^{\binom{n}{s}}, \qquad g_j \in G_j, \tag{8}$$

are polynomial sequences, and their projections form a compact finite-dimensional nilmanifold $\mathrm{HP}(Y)$. The shift on this space is a unipotent affine nilsystem. Evaluation at zero is continuous, and the shifted sequence evaluates to $p(n)\Gamma$. These facts are proved in [15, Chapters 14–15, especially Chapter 15, Corollaries 2–3]. The polynomial-to-cube implication is also [11, Proposition A.25].

Lemma 7.2 (Lifting one corner)**.** *Suppose $c \in C^r(Y)$ and a group cube $a \in C^r(G)$ has $a(F)\Gamma = c(F)$ at every proper vertex $F \subsetneq [r]$. Then there is $h \in C^r(G)$ projecting to c at every vertex and agreeing with a at every proper vertex.*

Proof. Choose a group cube g projecting to c. On proper vertices put $b(F) = g(F)^{-1}a(F) \in \Gamma$. This is a group cube corner: its proper faces are restrictions of the group cube $g^{-1}a$. Its ordered face coefficients for proper I therefore lie in $G_{|I|}$; recursively solving for them from the vertex values also shows that they lie in Γ. Give the top face the identity coefficient. The resulting full cube b has all entries in Γ. Then $h = gb$ is the required lift. This is the explicit fibration argument in the proof of [11, Proposition A.17]. □

Lemma 7.3 (Symmetric-cube lemma)**.** *Suppose $y_0, y_1, \dots \in Y$ have the property that $F \mapsto y_{|F|}$ belongs to $C^r(Y)$ for every $r \geq 1$. Then there are $g_j \in G_j$ such that*

$$y_n = g_0 g_1^{\binom{n}{1}} \cdots g_s^{\binom{n}{s}} \Gamma \qquad (n \geq 0).$$

Proof. Choose g_0 lifting y_0. Inductively suppose $p_{j-1}(n) = g_0 \cdots g_{j-1}^{\binom{n}{j-1}}$ projects to y_n for $0 \le n < j$. Its Hamming-weight j-cube is a group cube. Apply Lemma 7.2 to that cube and the given symmetric j-cube in Y. The resulting lifted top value differs from $p_{j-1}(j)$ by right multiplication by some $g_j \in G_j$. Appending $g_j^{\binom{n}{j}}$ therefore gives agreement for $0 \le n \le j$. Continue through $j = s$.

Let p be the resulting polynomial sequence. For $n > s$, inductively compare the two Hamming-weight n-cubes given by p and y. All proper vertices agree after projection. The n-uniqueness property of an s-step nilspace forces agreement at the top vertex. For clarity, this uniqueness also follows directly from the corner lemma: after lifting the common corner, the two possible top values differ by $G_n = \{1\}$. Thus agreement holds for every $n \ge 0$. □

7.2. **Ramsey symmetrization and a uniform finite cube.** An infinite cube is a map $c : \{F \subseteq \mathbb{N} : F \text{ finite}\} \to Y$ whose restriction to every finite set of coordinates is a Host–Kra cube. The space $C^\infty(Y)$ of these maps is compact, because all its finite cube conditions are closed.

Lemma 7.4 (Cardinality recurrence on nilcubes)**.** *Let S be pointwise nil-recurrent. For every $c \in C^\infty(Y)$ and every open neighborhood V of $c(\varnothing)$, there is a nonempty finite $F \subseteq \mathbb{N}$ with $|F| \in S$ and $c(F) \in V$.*

Proof. Partition the compact metric space Y into finitely many sets of diameter at most 2^{-r} for each r. Repeated applications of the infinite Ramsey theorem give nested infinite coordinate sets X_r such that for every $1 \le j \le r$, the values $c(F)$ on $F \in [X_r]^j$ all lie in one partition cell of diameter at most 2^{-r}. For each j, these values converge uniformly as $r \to \infty$ to a point y_j. Put $y_0 = c(\varnothing)$.

For fixed q, choose q coordinates from X_r and let $r \to \infty$. The associated q-cubes converge to $F \mapsto y_{|F|}$, which is a cube because $C^q(Y)$ is closed. Lemma 7.3 identifies $(y_n)_{n\ge0}$ with a Hall–Petresco sequence. Apply the pointwise nil-recurrence of S to its shift system, with the neighborhood defined by evaluation at zero lying in V. Some $m \in S$ has $y_m \in V$. Taking r large and then any m coordinates from X_r gives $c(F) \in V$. □

Corollary 7.5 (Uniform finite version)**.** *Let $K \subset V \subseteq Y$, with K compact and V open. There is $m \ge 1$ such that every $c \in C^m(Y)$ with $c(\varnothing) \in K$ has a nonempty $F \subseteq [m]$ satisfying $|F| \in S$ and $c(F) \in V$.*

Proof. The compact subspace of $C^\infty(Y)$ with initial vertex in K is covered, by Lemma 7.4, by the open conditions $c(F) \in V$ indexed by $|F| \in S$. A finite subcover uses only coordinates in $[m]$ for some m. Every m-cube extends to an infinite cube by ignoring coordinates beyond m, so the same finite conditions cover every such m-cube. □

In particular, this bound is uniform over the translation in any fixed nilmanifold: for positive integers $b_1, \dots, b_m$, every nilrotation T and every $y \in K$, the map

$$F \longmapsto T^{\sum_{i\in F} b_i} y$$

is a Host–Kra cube. For simultaneous returns at the multiples $1, \dots, h$, use Y^h, its diagonal initial compact set, and the rotation $T \times T^2 \times \cdots \times T^h$.

7.3. **A scalar phase clock for arbitrary generators.**

Lemma 7.6 (Periodic source blocker)**.** *Suppose S has a binary blocker at length H. If $P > (2H-1)m$ is an integer, there is a P-periodic binary coloring $b : \mathbb{Z} \to \{0,1\}$ which blocks all differences in $S \cap [m]$ at length $2H$.*

Proof. Extend existence of the source blocker from $\mathbb{N}$ to $\mathbb{Z}$ by compactness of translated finite intervals. Periodically repeat its values on $[0, P-1]$. A $2H$-term progression of step at most m

crosses at most one period boundary. At least H consecutive sites have representatives forming an original arithmetic progression. □

Lemma 7.7 (Phase-clock transfer)**.** *Fix m, P as in Lemma 7.6, and choose*

$$0 < \delta < \frac{1}{Pm(4H-1)}.$$

Suppose $b_1, \dots, b_m$ are positive integers and

$$\|\alpha b_i - 1/P\| < \delta \qquad (1 \le i \le m).$$

Then the set

$$E = \left\{ \sum_{i \in F} b_i : \varnothing \neq F \subseteq [m],\ |F| \in S \right\}$$

has a binary blocker at length $4H$.

Proof. Use $C(n) = b(\lfloor P\alpha n \rfloor)$, where b is periodic as in Lemma 7.6. For $d = \sum_{i \in F} b_i$ put $s = |F|$. There are an integer ℓ and a real η with

$$\alpha d = s/P + \ell + \eta, \qquad |\eta| < m\delta.$$

Along $4H$ sites $a + td$, reduction of the integer labels modulo P gives

$$\lfloor P\alpha(a + td) \rfloor \equiv ts + \lfloor P\alpha a + tP\eta \rfloor \pmod{P}.$$

The last floor is monotone and changes at most once, since its total real variation is strictly less than one. Consequently at least $2H$ consecutive output labels form a b-progression of step $s \in S \cap [m]$, which cannot be monochromatic. □

Lemma 7.8 (Choosing the clock uniformly)**.** *For fixed m, P, δ with $0 < \delta < 1/(2P)$, every list of $r > m/(2\delta)$ positive integers has m indexed members and an $\alpha \in \mathbb{T}$ satisfying the hypothesis of Lemma 7.7. The integers need not be distinct.*

Proof. For each positive integer b, the map $\alpha \mapsto b\alpha$ preserves Haar measure on $\mathbb{T}$. Thus the expected number of indexed members whose phases lie in the interval of radius δ about $1/P$ equals $2\delta r > m$. Some α has at least m such members. No independence or equidistribution assertion is required. □

Proposition 7.9 (Finite packets in IP_0 hosts)**.** *Let A be IP_0. For any finite list of connected nilmanifolds, compact/open window pairs $K \subset V$, and a prescribed finite return order h, there is a finite $E \subseteq A$ which returns every point of every K into its V at all multiples $d, 2d, \dots, hd$ for some $d \in E$, uniformly over every translation on the specified nilmanifold. The set E has a binary blocker at length $4H$.*

Proof. Apply Corollary 7.5 to the product tests described above, and take the maximum of the finitely many resulting m values. Choose P, δ, r by Lemmas 7.6–7.8, and use the IP_0 property to obtain an r-generator finite-sums block in A. Select m of its indexed generators by Lemma 7.8. Take E to be their cardinality-S sums. Corollary 7.5 gives the returns, and Lemma 7.7 gives the single binary blocker. Enlarging m does not invalidate a smaller cube bound, because one can restrict to its initial coordinates. □

7.4. **Affine and component reductions.** Here is the standard exact reduction needed to include general finite-dimensional nilwindows. On a connected component, a return power of a niltranslation has a presentation

$$T_b(z) = bA(z) \quad \text{on } N/\Lambda,$$

where N is connected simply connected nilpotent and A is a lattice-preserving unipotent automorphism. The nilpotence of the original ambient group makes its semidirect extension nilpotent. Writing $A^t = \exp(t \log A)$, the connected nilpotent suspension

$$\widetilde{N} = N \rtimes_{A^t} \mathbb{R}, \qquad \widetilde{\Lambda} = \Lambda \rtimes_A \mathbb{Z}$$

has compact quotient. The element $(b, 1)$ preserves the embedded section $(N/\Lambda) \times \{0\}$ and acts there exactly as T_b. For a compact/open pair $K \subset V$ in that section, choose an ambient open neighborhood whose intersection with the section is contained in V. The packet proposition on the suspension therefore supplies the desired affine returns, uniformly in b.

There are countably many rational nilmanifold presentations, lattice-preserving unipotent automorphisms, component periods, and compact/open window pairs forming bases. They give an exhaustive countable list of tests. Compactness of translation-parameter sets is harmless but unnecessary in Proposition 7.9, which is uniform over all translations on a fixed suspension.

7.5. **Division and one infinite binary blocker.**

Lemma 7.10. *If A is IP_0 and $\lambda \geq 1$, then $A/\lambda = \{d \geq 1 : \lambda d \in A\}$ is IP_0.*

Proof. For a desired m generators, take a λm-generator block in A, partitioned into m disjoint successive groups of λ generators. Among the $\lambda + 1$ prefix sums of each group, including zero, two have the same residue modulo λ. Their difference is a nonempty consecutive sum divisible by λ. These m disjoint sums have all finite sums in A; dividing them by λ proves the assertion. □

Proof of Theorem 7.1. List the countable tests from the preceding section. Begin with $\lambda_1 = 1$. At stage j, retain the first j tests whose component period divides λ_j and include the one-point test. Apply Proposition 7.9 in A/λ_j, with return order j, to the corresponding powered affine transformations. It gives a finite nonempty $F_j \subseteq A/\lambda_j$ with a binary blocker c_j at length $L = 4H$. Choose an integer radix

$$q_j > 2L \max F_j, \qquad (j+1)! \mid \lambda_j q_j, \qquad \lambda_{j+1} = \lambda_j q_j,$$

and put $D = \bigcup_j \lambda_j F_j \subseteq A$.

Every prescribed nilwindow and finite return order are handled at some sufficiently late stage. Its component period then divides λ_j, and the powered test returns translate back to returns at the original differences $\lambda_j d$. Thus D is pointwise nil-recurrent at every finite order.

For the binary obstruction, let $z_j(n)$ be the mixed-radix digit at place λ_j, and define

$$c(n) = \sum_{j \geq 1} \big(c_j(1 + z_j(n)) - c_j(1)\big) \pmod 2.$$

The sum is finite. On $2L$ sites of a progression of step $\lambda_j f$, $f \in F_j$, all lower digits are fixed and the higher prefix changes at most once, because $(2L-1)f < q_j$. Some L consecutive sites have constant lower and higher digits; their jth digit advances by f. A constant output would contradict the blocker c_j. Hence c blocks D at length $2L = 8H$.

Finally, to obtain P_1, let $N_T(y, V)$ be any nilreturn window. Choose a positive a with $T^a y \in V$, possible because a nilrotation has minimal orbit closure. Apply finite-order pointwise recurrence at $T^a y$ to a neighborhood contained in V. This gives the desired progression $a, a+d, \ldots, a+(h-1)d$ in the return window. □

7.6. **A conditional colored-cardinality transfer.** For an abelian group G, a direction set $E \subseteq G \setminus \{0\}$ is k-large if every k-coloring of G has arbitrarily long monochromatic progressions with difference in E. Write $\mathbb{Z}^{(\mathbb{N})}$ for the group of finitely supported integer sequences, and 1_F for the indicator vector of a finite coordinate set F. The following statement isolates a stronger colored hypothesis than the pointwise nil recurrence used above.

Corollary 7.11 (Conditional cardinality transfer to every IP_0 host)**.** *Let $S \subseteq \mathbb{N}$ have a (q, H)-blocker, with $q \geq 2$. Suppose*

$$\mathcal{E}(S) = \{1_F \in \mathbb{Z}^{(\mathbb{N})} : \varnothing \neq F \subset \mathbb{N} \text{ finite}, \ |F| \in S\}$$

is k-large on the group $\mathbb{Z}^{(\mathbb{N})}$. Then every IP_0 host $A \subseteq \mathbb{N}$ contains a k-large subset with a $(q, 8H)$-blocker. Pointwise nil-recurrence of S is not assumed in this corollary.

Proof. All three clock lemmas preserve an arbitrary q-color palette; their proofs use only monochromaticity, not the fact that $q = 2$. For a prescribed progression length h, compactness of the coloring space gives an m such that the directions $\mathcal{E}_m(S) = \{1_F : F \subseteq [m], \ |F| \in S\}$ are (k, h)-forcing on $\mathbb{Z}^m$. Compactness once more supplies a finite vertex set $V \subset \mathbb{Z}^m$ on which the forcing is already witnessed.

Choose sufficiently many finite-sums generators in A and retain m generators $b_1, \ldots, b_m$ using the clock averaging lemma. The linear image of $\mathcal{E}_m(S)$ under $\ell(x) = \sum_i b_i x_i$ is (k, h)-forcing on $\mathbb{N}$: given a coloring of $\mathbb{N}$, choose a positive offset a with $a + \ell(V)$ positive and pull the coloring back to V. A witnessed progression maps to a genuine progression, since the image of each nonempty Boolean direction is positive. Injectivity on V is unnecessary. The clock gives this finite image a $(q, 4H)$-blocker.

Do this for increasing h in the successive IP_0 quotient hosts A/λ_j, and use the same mixed-radix assembly, now summing the digit colors in $\mathbb{Z}/q\mathbb{Z}$. The resulting union is k-large and has a $(q, 8H)$-blocker. □

Theorem 7.1 provides a binary-blockable pointwise nil-recurrent seed, not a 2-large seed. It makes no inference that every large host is IP_0 and does not transfer ordinary k-largeness to Boolean directions selected by their cardinality.

8. Transfer through dilations and power packets

A set $A \subseteq \mathbb{N}$ is *multiplicatively thick* if every finite $F \subseteq \mathbb{N}$ has a dilation $tF \subseteq A$ for some $t \in \mathbb{N}$.

Theorem 8.1 (Transfer through divisible copies)**.** *Let D be k-large and have a (q, H)-blocker. Suppose A satisfies*

$$\forall \ell \geq 1 \ \forall F \subseteq D \text{ finite} \quad \exists t \in \ell\mathbb{N} \quad tF \subseteq A. \tag{9}$$

Then A contains a k-large subset with a $(q, 2H)$-blocker. In particular this holds whenever A is multiplicatively thick.

Proof. Choose finite nonempty $(k, j+2)$-forcing packets $F_j \subseteq D$. Condition (9) supplies λ_1 with $\lambda_1 F_1 \subseteq A$. Given λ_j, choose an integer $M_j > 2H \max F_j$ and apply the condition with $\ell = \lambda_j M_j$ and $F = F_{j+1}$. It supplies λ_{j+1} divisible by $\lambda_j M_j$ with $\lambda_{j+1} F_{j+1} \subseteq A$. Thus the successive radix $b_j = \lambda_{j+1}/\lambda_j$ exceeds $2H \max F_j$. The union $B = \bigcup_j \lambda_j F_j \subseteq A$ is k-large by dilation invariance. Restrict the given blocker of D to the direction sets F_j and apply Lemma 4.3. For multiplicatively thick A, apply its definition to the finite set ℓF; the resulting scale for F is a multiple of ℓ. □

Corollary 8.2. *Every multiplicatively thick host contains a k-large but not $(k+1)$-large subset, with a blocker at length $800 \cdot 2^k$.*

Proof. Apply Theorem 8.1 to the absolute hierarchy from Corollary 5.11, with $q = k + 1$. □

Corollary 8.3 (Arbitrarily long power packets)**.** *Fix $s \geq 1$. Suppose that for arbitrarily large N there is $t \geq 1$ with*

$$t\{1^s, 2^s, \ldots, N^s\} \subseteq A.$$

Then A contains, for every $k \geq 1$, a k-large subset with a $(k + 1, 800 \cdot 2^k)$-blocker.

Proof. Theorem 6.1 with source $\mathbb{N}$ and $p(n) = n^s$ supplies $D \subseteq \{n^s : n \geq 1\}$ with P_k and a $(k+1, 400 \cdot 2^k)$-blocker. Given ℓ and a finite $F = \{m_1^s, \dots, m_r^s\} \subseteq D$, choose the displayed packet with $N \geq \ell \max_i m_i$. Then $t\ell^s F \subseteq A$ and $t\ell^s \in \ell\mathbb{N}$, verifying (9). Apply Theorem 8.1. No divisibility of the originally given packet scales t is needed; internal power scaling supplies it. □

Multiplicative thickness implies largeness: take a finite (r, h)-forcing packet in $\mathbb{N}$, supplied by van der Waerden's theorem, and dilate it into A. The converse fails, as the fixed-template obstruction in Section 15 shows. In particular the transfer theorem does not settle arbitrary large hosts by a change of scale.

9. One extra color: finite sets and linear Bohr complements

The next marker has two roles. Its short contiguous arc prevents long monochromatic progressions in the new color along old directions. A phase orbit that visits all bins rules out a monochromatic progression along a newly adjoined direction.

Theorem 9.1. *Suppose $r, H \geq 2$ and $E \subseteq \mathbb{N}$ has an (r, H)-blocker.*

(1) *For every $f \in \mathbb{N}$, the set $E \cup \{f\}$ has an $(r+1, L_1)$-blocker, where*

$$L_1 = \max\{2r,\ r(H-1)+1\}.$$

(2) *If $F \subseteq \mathbb{N}$ is finite with $s = |F| \geq 1$, then $E \cup F$ has an $(r+1, L_s)$-blocker, where*

$$L_s = \max\{r(H-1)+1,\ 4s(2r)^s + 1\}.$$

These bounds are independent of the magnitudes, divisibility, and ratios of the elements of F, and independent of the original blocking coloring. The original palette is enlarged by exactly one available color. The assertion does not give the same uniformity while retaining only r colors.

The proof uses a marker occupying a contiguous half-circle. This contiguity controls the old directions even when they are not multiples of the marker scale.

Lemma 9.2 (One arc controls the new color)**.** *Put $p = 2r$. Let $c : \mathbb{N} \to \{1, \dots, r\}$ be an (r, H)-blocker for E, and let $x : \mathbb{N} \to \mathbb{R}/p\mathbb{Z}$ be an additive rotation, so that $x(n+d) - x(n)$ depends only on d. Represent $x(n)$ in $[0, p)$ and define*

$$C(n) = \begin{cases} *, & \lfloor x(n) \rfloor = c(n), \\ c(n), & \text{otherwise.} \end{cases}$$

Then C blocks E at length $r(H-1)+1$.

Proof. A progression monochromatic in an old color i was already monochromatic in c, so H sites are impossible.

Suppose all sites $n + jd$, $0 \leq j < L$, have the new color $*$, where $d \in E$ and $L = r(H-1)+1$. Their phase representatives x_j all belong to the half-open interval $[1, r+1)$, of length $r = p/2$. Each difference $x_{j+1} - x_j$ lies in $(-r, r)$ and represents the same element $x(d)$ modulo p. There is at most one representative of a circle element in this open interval, so all these real differences equal one number θ. Consequently (x_j) is a genuine real arithmetic progression.

The labels $\lfloor x_j \rfloor = c(n + jd)$ are monotone, or constant, and take values in $\{1, \dots, r\}$. They therefore form at most r consecutive constant runs. Since $L > r(H-1)$, one run has at least H sites. It gives a monochromatic H-term progression of step d in c, a contradiction. □

Proof of Theorem 9.1(1). Use Lemma 9.2 with $x(n) = n/f \pmod p$. Along a step-f progression the integer phase labels advance by one modulo p. In p sites they visit every label. A progression monochromatic in an old color i would have original color i at every site, but its site of phase label i is marked with $*$. A progression monochromatic in $*$ cannot visit phase label zero, which is never marked. Thus p sites block the added step, and the maximum of the two bounds proves the assertion. □

Lemma 9.3 (Uniform annulus inside a finite Bohr neighborhood)**.** *Let $f_1, \dots, f_s$ be positive integers and let $p \geq 2$. There is $\beta \in \mathbb{T}$ such that*

$$\delta < \|f_i\beta\| < \frac{1}{p} \qquad (1 \leq i \leq s), \qquad \delta = \frac{1}{4sp^s}.$$

Proof. Let K be the image of $\mathbb{T}$ under $\beta \mapsto (f_1\beta, \dots, f_s\beta)$, and let μ be its normalized Haar measure. Each coordinate marginal is Haar measure on $\mathbb{T}$, because each f_i is nonzero.

Partition $\mathbb{T}^s$ into p^s half-open cubes of side $1/p$. Some cube Q has $\mu(Q) \geq p^{-s}$. Choose $z \in Q \cap K$. The translate $(Q \cap K) - z$ lies in

$$B = \{x \in K : \|x_i\| < 1/p \ \forall i\},$$

and has the same Haar measure. Hence $\mu(B) \geq p^{-s}$. The union of the coordinate strips $\|x_i\| \leq \delta$ has measure at most $2s\delta = p^{-s}/2$. Some point of B lies outside all these strips. Lift it to $\beta \in \mathbb{T}$. □

Proof of Theorem 9.1(2). Put $p = 2r$ and take β from Lemma 9.3. Use the marker of Lemma 9.2 with $x(n) = pn\beta \pmod p$. It blocks E at length $r(H-1)+1$.

For any $f \in F$, the circle-p increment has a signed real representative θ_f satisfying

$$p\delta < |\theta_f| < 1.$$

Let $J = 4sp^s = 1/\delta$. In $J+1$ consecutive sites of a step-f progression, the lifted real path has span $J|\theta_f| > p$ and consecutive gaps smaller than one. It therefore meets every unit interval $[i, i+1)$ modulo p. Explicitly, in the positive-direction case choose a translate of the left endpoint of the desired unit interval in $[x_0, x_0+p)$. The first lifted site at or beyond that endpoint lies less than one unit past it and occurs by time J. The negative-direction case follows by using the right endpoint.

Thus the phase labels visit every value $0, \dots, p-1$. As in the singleton proof, a progression constant in an old color i encounters a marked site at phase label i, and one constant in $*$ encounters an unmarked site at phase label zero. This blocks every member of F at length $J+1$. □

Corollary 9.4 (A short-arc gate without filtering the old set)**.** *Let E have an (r, H)-blocker, and suppose $B \subseteq \mathbb{N}$ satisfies*

$$\delta < \|\beta d\| < \frac{1}{2r} \quad (d \in B)$$

for some $\beta \in \mathbb{T}$ and $\delta > 0$. Then $E \cup B$ has an $(r+1, L)$-blocker with

$$L = \max\{r(H-1)+1, \ \lceil 1/\delta \rceil + 1\}.$$

There is no phase restriction on E.

Proof. Use the same marker and the same phase-orbit argument as in the proof of Theorem 9.1(2). The annulus assumption is exactly what that argument needs for each newly added direction. □

Theorem 9.5 (Adjoining a one-dimensional Bohr complement)**.** *Suppose E has an (r, H)-blocker, $r, H \geq 2$, and let $\alpha \in \mathbb{T}$ and $0 < \varepsilon \leq 1/2$. Put*

$$B = \{d \in \mathbb{N} : \|\alpha d\| \geq \varepsilon\}, \qquad L = \max\{r(H-1)+1, \ \lceil (2r)!/\varepsilon \rceil + 2r + 1\}.$$

Then $E \cup B$ has an $(r+1, L)$-blocker. The bound is uniform in the frequency α and in the original blocking coloring.

Proof. Set $p = 2r$, $M = p!$, and choose a real representative of α. Write $\beta = \alpha/M \pmod 1$. Use Lemma 9.2 with $x(n) = pn\beta \pmod p$, which already blocks E at length $r(H-1)+1$.

Fix $d \in B$ and put $\theta = d\beta \pmod 1$. Among the $p+1$ points $0, \theta, \dots, p\theta$, two lie in the same one of p half-open intervals of length $1/p$. Thus for some $1 \leq q \leq p$,

$$\eta = \|q\theta\| < 1/p.$$

Since $q \mid M$, the hypothesis on d gives

$$\varepsilon \le \|\alpha d\| = \|(M/q)(q\theta)\| \le (M/q)\eta.$$

In particular $\eta \ge q\varepsilon/M > 0$.

Along the subsequence of progression indices $0, q, 2q, \dots, Jq$, where $J = \lceil 1/\eta \rceil$, the circle phases advance by a signed increment of magnitude $\eta < 1/p$. Their real lift has span at least one, so they meet all p half-open intervals of length $1/p$. Also

$$qJ \le q/\eta + q \le M/\varepsilon + p.$$

Consequently the original progression's first $\lceil M/\varepsilon \rceil + p + 1$ sites meet every phase label $0, \dots, p-1$. The same marker argument as before rules out a constant old output color and a constant new output color. □

Corollary 9.6 (Losing one color when imposing one Bohr condition)**.** *Let $r \ge 2$ and let D be $(r+1)$-large. For every $\alpha \in \mathbb{T}$ and every $\varepsilon > 0$,*

$$D \cap \{d : \|\alpha d\| < \varepsilon\}$$

is r-large. In particular $D \cap m\mathbb{N}$ is r-large for every positive integer m.

Proof. If the displayed intersection had an (r, H)-blocker at some length H, Theorem 9.5 would block its union with the complementary Bohr set using $r+1$ colors. That union contains D, a contradiction. It is enough to consider $\varepsilon \le 1/2$; larger radii either impose no restriction or are handled by a smaller radius. For $m \ge 2$ use $\alpha = 1/m$ and $\varepsilon = 1/(2m)$; the Bohr condition then says exactly $m \mid d$. The case $m = 1$ is immediate. □

Corollary 9.7 (Bohr stability inside a hypothetical counterhost)**.** *Fix $k \ge 2$. Suppose $A \subseteq \mathbb{N}$ has the property that every k-large subset of A is $(k+1)$-large. Then every k-large $D \subseteq A$ remains k-large after intersection with any finite-dimensional Bohr neighborhood of zero.*

Proof. Write the neighborhood as the intersection of finitely many conditions $\|\alpha_i d\| < \varepsilon_i$. By the hypothesis, D is $(k+1)$-large, so imposing the first condition preserves k-largeness by Corollary 9.6. The resulting set is still a subset of A, hence is again $(k+1)$-large. Repeat for the remaining conditions. □

The additional color is part of the conclusion. A bound depending only on $r, H, |F|$ while retaining the original r colors remains unresolved here. Same-palette finite addition was proved earlier with a bound depending on the actual finite set F.

10. Piecewise-syndetic augmentation of a nilwindow seed

The preceding hierarchy supplies P_k inside quasi-central hosts. For a more general piecewise-syndetic host, the following argument increases the number of colors by one provided a suitable lower-color nilwindow seed is already available. Its conclusion is ordinary k-largeness; it does not supply P_k and hence does not automatically iterate.

Lemma 10.1 (A piecewise-syndetic part away from zero)**.** *If $A \subseteq \mathbb{N}$ is piecewise syndetic and $\alpha \in \mathbb{T}$ is irrational, there is $0 < \epsilon < 1/2$ for which*

$$B = A \cap \{d \ge 1 : \|\alpha d\| \ge \epsilon\}$$

is piecewise syndetic.

Proof. Choose a nonempty finite $F \subset \mathbb{Z}$ such that $T = \bigcup_{f \in F}(A - f)$ is thick. Choose $\epsilon > 0$ small enough that the finite union of closed arcs

$$\bigcup_{f \in F} (-f\alpha + [-\epsilon, \epsilon])$$

does not cover $\mathbb{T}$, and choose a nonempty open arc V in its complement. Irrationality makes $R = \{n \geq 1 : n\alpha \in V\}$ syndetic. Hence $T \cap R$ is piecewise syndetic. For $n \in T \cap R$, choose $f \in F$ with $n + f \in A$. The choice of V gives $\|(n+f)\alpha\| > \epsilon$, so $n + f \in B$. Thus

$$T \cap R \subseteq \bigcup_{f \in F} (B - f).$$

Partition regularity and translation invariance imply that B is piecewise syndetic. □

The structural input is the following consequence of [9, Theorem A]. If (X, T) is a minimal invertible compact metric system, $\pi : X \to Z$ is its maximal infinite-step pronilfactor, and $U \subseteq X$ is nonempty and open with $\pi(U) = Z$, then

$$\begin{gathered} \mathrm{Ret}_h(U) = \left\{ d \in \mathbb{Z} : \bigcap_{j=0}^{h-1} T^{-jd} U \neq \varnothing \right\}, \\ \mathbb{Z} \setminus \mathrm{Ret}_h(U) \text{ is not piecewise syndetic.} \end{gathered} \tag{10}$$

Here $h \geq 2$, and the distinct linear polynomials are $0, n, \ldots, (h-1)n$. The other fact used about Z is its definition as an inverse limit of finite-dimensional minimal nilsystems.

Theorem 10.2 (Piecewise-syndetic augmentation)**.** *Let $k \geq 2$. If $E \subseteq \mathbb{N}$ has P_{k-1} and $B \subseteq \mathbb{N}$ is piecewise syndetic, then $E \cup B$ is k-large.*

Proof. Fix a k-coloring c of $\mathbb{N}$ and $h \geq 2$. Extend c to a two-sided sequence, take its forward orbit limit set, and choose a minimal subsystem X of that limit set. Every finite colored pattern occurring in X consequently occurs at positive positions in c. Write $C_1, \ldots, C_k$ for the clopen color cylinders in X, and let $\pi : X \to Z$ be its maximal infinite-step pronilfactor.

If $\pi(C_1) = Z$, apply (10) to $U = C_1$. The piecewise-syndetic positive set B cannot be contained in its non-piecewise-syndetic complement. Some $d \in B \cap \mathrm{Ret}_h(C_1)$ therefore produces a monochromatic h-term pattern in X, and hence in c.

Otherwise the compact image $\pi(C_1)$ has a nonempty open complement. The inverse-limit description gives a finite-dimensional nilfactor (Y, S), a factor map $\rho : Z \to Y$, and a nonempty open $V \subseteq Y$ with

$$\rho^{-1}(V) \subseteq Z \setminus \pi(C_1).$$

Choose $x \in X$ with $y = \rho\pi(x) \in V$. All points $T^n x$ for $n \in N_S(y, V)$ have one of the remaining $k-1$ colors. Apply property P_{k-1} of E to that coloring of the return window. The resulting monochromatic h-term pattern has difference in E and again occurs in the original positive coloring. □

Theorem 10.3 (A relative separator criterion)**.** *Let $A \subseteq \mathbb{N}$ be piecewise syndetic and $k \geq 2$. Suppose A contains E with P_{k-1} and a (k, H)-blocker. Then A contains a k-large set which is not $(k+1)$-large.*

Proof. Use Lemma 10.1 to choose a piecewise-syndetic $B \subseteq A$ in a one-frequency Bohr complement, and put $D = E \cup B$. Theorem 10.2 makes D k-large. Theorem 9.5 provides a $(k+1, L)$-blocker for D. □

For a nontrivial prescribed nilwindow, the first branch would need a piecewise-syndetic intersection of B with that window's multiple-return set. Arbitrary piecewise syndeticity of B does not supply such an intersection. This explains why the theorem asserts ordinary k-largeness rather than the stronger P_k.

Corollary 10.4 (Piecewise-syndetic IP_0 hosts at two colors)**.** *Every piecewise-syndetic IP_0 host contains a 2-large subset which is not 3-large.*

Proof. Apply Theorem 10.3 to the binary-blockable P_1 seed from Theorem 7.1. □

11. Monomial Bohr complements

We now replace a linear phase by a polynomial phase. The old-direction argument uses an integer finite difference trapped in an interval of length less than one. The new-direction argument needs one external input: quantitative Weyl equidistribution. We use Green–Tao [10, Proposition 4.3 and Definition 2.7]: failure of a fixed equidistribution tolerance for a degree-s polynomial on N consecutive integers forces a bounded positive integer multiple of its top binomial coefficient to have circle norm $O(N^{-s})$. The constants depend only on s and the tolerance.

Theorem 11.1. *For every $r, H \geq 2$, $s \geq 1$, and $0 < \varepsilon \leq 1/2$, there is an integer $L = L(r, H, s, \varepsilon)$ with the following property. If $E \subseteq \mathbb{N}$ has an (r, H)-blocker and $\alpha \in \mathbb{R}$, then*

$$E \cup \{d \in \mathbb{N} : \|\alpha d^s\|_{\mathbb{T}} \geq \varepsilon\}$$

has an $(r+1, L)$-blocker. In particular, L is independent of α, E, and the original blocking coloring.

11.1. **A uniform bin-hitting lemma.**

Lemma 11.2. *For integers $s \geq 1$ and $p \geq 2$ there is a positive integer $M = M(s, p)$ such that, for each $0 < \varepsilon \leq 1/2$, there is $N = N(s, p, \varepsilon)$ with this property: if a real polynomial $P(t)$ of degree at most s has coefficient γ of t^s and $\|M\gamma\|_{\mathbb{T}} \geq \varepsilon$, then*

$$\{P(0)\}, \ldots, \{P(N-1)\}$$

meets every half-open bin $[i/p, (i+1)/p)$, $0 \leq i < p$. The same N works for every choice of the lower coefficients.

Proof. Set $\delta = 1/(16p^2)$. On each bin put the triangular function which vanishes at the endpoints, has slopes 1 and -1, and is zero off that bin. Viewed as a function on $\mathbb{T}$, it has integral $1/(4p^2)$ and Lipschitz norm at most 2, where this norm is the supremum norm plus the Lipschitz seminorm. Missing the bin therefore violates δ-equidistribution.

Quantitative Weyl equidistribution supplies constants $Q = Q(s, p) \in \mathbb{N}$ and $C = C(s, p) \geq 1$ such that any such failure on N consecutive integers implies

$$\|q\, s!\gamma\|_{\mathbb{T}} \leq CN^{-s} \qquad \text{for some } 1 \leq q \leq Q.$$

Indeed, the coefficient of $\binom{t}{s}$ is $s!\gamma$, and the $C^\infty[N]$-norm in the cited proposition bounds its circle distance after multiplication by q, with weight N^s. Translating the interval from $[N]$ to $\{0, \ldots, N-1\}$ does not change the leading coefficient.

Take

$$M = s!\,\mathrm{lcm}(1, \ldots, Q), \qquad N = \max\left\{2, \left\lfloor (MC/\varepsilon)^{1/s} \right\rfloor + 1\right\}.$$

Since $qs!$ divides M, failure would give

$$\|M\gamma\|_{\mathbb{T}} \leq \frac{M}{qs!}\,\|qs!\gamma\|_{\mathbb{T}} \leq MCN^{-s} < \varepsilon,$$

a contradiction. The hypothesis ensures $\gamma \neq 0$, so the polynomial has degree exactly s when the external result is used. □

11.2. **Preserving old directions.**

Lemma 11.3 (Polynomial arc marker)**.** *Let $p > 2^s r$ be an integer, let P be any real polynomial of degree at most s, and let $c : \mathbb{N} \to \{1, \ldots, r\}$ be an (r, H)-blocker for E. Put*

$$z(n) = \lfloor p\{P(n)\} \rfloor, \qquad C(n) = \begin{cases} *, & z(n) = c(n), \\ c(n), & z(n) \neq c(n). \end{cases}$$

Define

$$R = 2s(r-1) + 1, \qquad L_E = \max\{s+2,\ R(H-1)+1\}.$$

Then C is an $(r+1, L_E)$-blocker for E.

Proof. A progression monochromatic in an old color is monochromatic in c, so already its first H sites are impossible.

Suppose that $a + td$, $0 \le t < L_E$, is entirely colored $*$, where $d \in E$. The representatives

$$u_t = \{P(a + td)\}$$

all lie in the interval $[1/p, (r + 1)/p)$ of length r/p. For each permissible t, $\Delta^{s+1}u_t$ is an integer, because $P(a + td)$ has vanishing $(s + 1)$st finite difference. Subtracting the midpoint of this short interval in the alternating sum gives

$$|\Delta^{s+1}u_t| \le 2^s r/p < 1.$$

Thus every such finite difference is zero. Newton interpolation and its finite-difference recurrence give a real polynomial $U(t)$ of degree at most s with $U(t) = u_t$ for all these indices.

We show that the sequence $\lfloor pU(t) \rfloor \in \{1, \dots, r\}$ has at most R consecutive constant runs. A change of its value between t and $t + 1$ changes at least one of the indicators

$$\mathbf{1}_{\{pU(t) \ge i\}}, \qquad 2 \le i \le r.$$

If $pU - i$ is identically zero its indicator is constant. Otherwise it has at most s distinct real roots. Each change of the indicator gives a root in the closed interval $[t, t + 1]$; each root belongs to at most two such adjacent intervals. Consequently each indicator changes at most $2s$ times, and all the labels together change at most $2s(r - 1)$ times. This argument includes roots at sample points and tangencies.

Since $L_E > R(H - 1)$, some constant run has at least H sites. At a marked site its label equals $c(a + td)$, so this run gives an H-term monochromatic progression of step d in c, contrary to the original blocker. □

Proof of Theorem 11.1. Choose $p = 2^{s+1}r + 1$ and let M, N be supplied by Lemma 11.2. Apply Lemma 11.3 with

$$P(n) = \frac{\alpha}{M} n^s, \qquad L = \max\{L_E, N\}.$$

It remains to block a difference d with $\|\alpha d^s\|_{\mathbb{T}} \ge \varepsilon$. Along an arbitrary progression $a + td$ the phase polynomial $P(a + td)$ has leading coefficient $\gamma = \alpha d^s/M$. Lemma 11.2 says that in the first N sites its phase visits every bin, uniformly in a.

If the progression were monochromatic in an old color i, every original color on it would equal i. Its site of phase label i would therefore be marked $*$, a contradiction. If it were entirely colored $*$, its site of phase label 0 would be a contradiction, since a site with this label is never marked. This proves the stated length bound. □

Corollary 11.4 (One-color localization)**.** *If D is $(r+1)$-large, then for every real α, integer $s \ge 1$, and $\varepsilon > 0$,*

$$D \cap \{d : \|\alpha d^s\|_{\mathbb{T}} < \varepsilon\}$$

is r-large.

Proof. It is enough to consider $\varepsilon \le 1/2$. If the intersection had an (r, H)-blocker, Theorem 11.1 would give an $(r + 1, L)$-blocker for a set containing D, a contradiction. □

Corollary 11.5 (Polynomial Bohr stability in a collapsed host)**.** *Suppose $k \ge 2$ and $A \subseteq \mathbb{N}$ has the property that every k-large subset of A is $(k + 1)$-large. If $D \subseteq A$ is k-large, $P_1, \dots, P_m \in \mathbb{R}[x]$ have zero constant term, and $\varepsilon > 0$, then*

$$D \cap \bigcap_{j=1}^{m} \{d : \|P_j(d)\|_{\mathbb{T}} < \varepsilon\}$$

is k-large.

Proof. Write all the nonzero monomial terms in the polynomials as $\alpha_\ell x^{s_\ell}$, $1 \le \ell \le t$, allowing repetitions. If there are none the assertion is immediate. Starting with D, successively intersect with

$$\{d : \|\alpha_\ell d^{s_\ell}\|_{\mathbb{T}} < \varepsilon/t\}.$$

At each stage the current set is k-large, hence $(k+1)$-large by the host hypothesis; Corollary 11.4 makes the next intersection k-large. The resulting set lies in every required polynomial neighborhood by the triangle inequality on $\mathbb{T}$. Supersets of k-large sets are k-large, proving the claim. □

The implication for ordinary polynomial neighborhoods comes from successively imposing monomial conditions inside the counterhost. It is not an assertion that a single extra color adjoins the complement of every ordinary polynomial neighborhood in one step.

12. Rounded linear products and nil-Bohr stability

For this section write $\langle x \rangle = \lfloor x + 1/2 \rfloor$. Thus ties are rounded upward. This fixed convention matters in the exact carry identities below. The two polynomial marker facts needed in the proof are precisely Lemmas 11.2 and 11.3. In the first, translation of the argument preserves the leading coefficient, so the bin-hitting conclusion holds on any N consecutive integers.

Theorem 12.1 (Conditional one-color addition)**.** *Fix $r, H \ge 2$, $m \ge 1$, integers $e_0 \ge 0$ and $e_i \ge 1$ for $1 \le i \le m$, and $0 < \varepsilon \le 1/2$. There exist $L \in \mathbb{N}$ and $\delta > 0$, depending only on these parameters, with the following property for all real $\alpha, \beta_1, \dots, \beta_m$. Define*

$$F(n) = \alpha n^{e_0} \prod_{i=1}^{m} \langle \beta_i n \rangle^{e_i}, \qquad G = \{d \in \mathbb{N} : \|\beta_i d\|_{\mathbb{T}} < \delta \ (1 \le i \le m)\}.$$

If $E \subseteq G$ has an (r, H)-blocker, then

$$E \cup \{d \in G : \|F(d)\|_{\mathbb{T}} \ge \varepsilon\}$$

has an $(r+1, L)$-blocker. The parameters L, δ are independent of the real coefficients and the original blocker.

Proof. Put $s = e_0 + \cdots + e_m$, choose $p = 2^{s+1}r + 1$, and take M, N, L_E from Lemmas 11.2 and 11.3. Set

$$K = m + 1, \quad T = \max\{L_E, N\}, \quad L = K(T-1) + 1, \quad \delta = \frac{1}{2L}.$$

Let c be the given blocker and use the coloring

$$C(n) = \begin{cases} *, & \lfloor p\{F(n)/M\} \rfloor = c(n), \\ c(n), & \text{otherwise.} \end{cases}$$

We first describe the phase along an arbitrary L-site progression $a + td$ with $d \in G$. Put $q_i = \langle \beta_i d \rangle$ and $\eta_i = \beta_i d - q_i$, so $|\eta_i| < \delta$. For $0 \le t < L$ we have the exact identity

$$\langle \beta_i (a + td) \rangle = t q_i + \left\lfloor \beta_i a + \tfrac{1}{2} + t\eta_i \right\rfloor.$$

The expression inside the last floor is monotone and varies over an interval of length less than $1/2$. Its floor therefore changes at most once. Taking the union of these at most m change positions partitions $\{0, \dots, L-1\}$ into at most K consecutive intervals. On each interval every rounded term equals $b_i + t q_i$ with b_i constant. Thus $F(a+td)/M$ is a real polynomial of degree at most s on that interval, and its coefficient of t^s is always

$$\gamma = \frac{\alpha}{M} d^{e_0} \prod_{i=1}^{m} q_i^{e_i} = \frac{F(d)}{M},$$

independent of the interval and of a. In particular, one interval has at least T sites.

If $d \in E$, a monochromatic old-color progression contradicts c already on its first H sites. An all-$*$ progression contains an interval of at least L_E sites on which the phase is polynomial; Lemma 11.3 gives a contradiction there.

If $d \in G$ and $\|F(d)\|_{\mathbb{T}} \geq \varepsilon$, the interval of length at least N meets every phase bin by Lemma 11.2. A progression monochromatic in an old color i encounters a site of label i, which would be marked. An all-$*$ progression encounters label 0, which is never marked. This excludes all monochromatic progressions claimed in the theorem. □

Corollary 12.2 (Lossless rounded-linear localization in a counterhost)**.** *Suppose $k \geq 2$ and every k-large subset of $A \subseteq \mathbb{N}$ is $(k+1)$-large. Let $D \subseteq A$ be k-large and let F have the form in Theorem 12.1. Then, for every $\varepsilon > 0$,*

$$D \cap \{d : \|F(d)\|_{\mathbb{T}} < \varepsilon\}$$

is k-large.

Proof. Theorem 11.1 implies that every k-large subset of this host stays k-large after any finite family of ordinary Bohr restrictions: apply its one-color-loss corollary and restore the lost color by the host hypothesis at each stage.

Suppose the displayed set has a (k, H)-blocker; it suffices to treat $\varepsilon \leq 1/2$. Choose L, δ, G from Theorem 12.1, allowing these choices to depend on this H. The set $D \cap G$ is k-large by the preceding Bohr stability, hence $(k+1)$-large by the host hypothesis. Its subset

$$E = D \cap G \cap \{d : \|F(d)\|_{\mathbb{T}} < \varepsilon\}$$

inherits the supposed (k, H)-blocker. The conditional theorem now blocks a set containing $D \cap G$ with $k+1$ colors, a contradiction. □

Finite families of these conditions may be imposed successively. Finite real linear combinations are also covered: impose each monomial summand with a smaller radius, then use the triangle inequality. Ordinary monomials can be included using Theorem 11.1. Thus the conclusion includes finite families of expressions such as

$$ad + bd^2 + cd\langle \beta d \rangle + e\langle \beta d \rangle \langle \gamma d \rangle.$$

Lemma 12.3 (Removing an outer linear rounding)**.** *Under the same host hypothesis, let $Q(n)$ be an ordinary monomial or a product of the form in Theorem 12.1. Then every k-large $D \subseteq A$ remains k-large after imposing*

$$\|\alpha d \langle Q(d) \rangle\|_{\mathbb{T}} < \varepsilon.$$

Either nearest-integer convention may be used for this outer rounding, even at ties.

Proof. For any nearest integers $[p]$ and $[q]$, direct expansion gives

$$p[q] \equiv pq - q[p] - (p - [p])(q - [q]) \pmod 1.$$

Take $p = \alpha d$ and $q = Q(d)$. Both $\alpha d Q(d)$ and $Q(d)\langle \alpha d \rangle$ are covered products or ordinary monomials. Impose their circle norms to be less than $\varepsilon/3$, and also impose $\|\alpha d\|_{\mathbb{T}} < \min\{1/4, \varepsilon/3\}$. All three restrictions preserve k-largeness. Since $|q - [q]| \leq 1/2$, the displayed identity makes the desired norm less than ε. Inner linear rounding ties, if one wishes to change their convention, can first be excluded by the corresponding ordinary Bohr restrictions. □

Definition 12.4. A *three-step nil-Bohr neighborhood of zero* in $\mathbb{N}$ is a set containing $\{n \geq 1 : T^n x \in U\}$ for some minimal nilsystem (X, T) of step at most three and some open neighborhood U of x. A nilsystem is a compact nilmanifold G/Γ with translation by a fixed element of the nilpotent Lie group G.

We use the following precise external characterization. By Huang–Shao–Ye [17, Theorem B, Theorem 4.2.11, and Definition 4.1.2], the filter of three-step nil-Bohr neighborhoods is generated by finite intersections of $\{n : \|Q(n)\|_{\mathbb{T}} < \varepsilon\}$ for these seven types:

$$\begin{gathered} an, \qquad an^2, \qquad an\langle bn\rangle, \qquad an^3, \\ an^2\langle bn\rangle, \qquad an\langle bn^2\rangle, \qquad an\langle bn\langle cn\rangle\rangle. \end{gathered} \tag{11}$$

Coefficients are arbitrary real numbers. Their nearest-integer notation rounds a tie downward. The following proof reconciles that convention rather than assuming that ties are absent.

Corollary 12.5 (Three-step stability in a counterhost)**.** *Assume* (C_k). *If $D \subseteq A$ is k-large and B is any three-step nil-Bohr neighborhood of zero, then $D \cap B$ is k-large.*

Proof. Choose finitely many special conditions from (11) whose intersection lies in B. For every linear expression βn appearing directly inside a rounding symbol, also impose $\|\beta d\|_{\mathbb{T}} < 1/4$. These finitely many Bohr restrictions preserve k-largeness by Corollary 9.7; they exclude ties for those linear expressions, so either convention has the same value.

The first five types in (11) are covered by the ordinary monomial theorem and Corollary 12.2. The sixth and seventh follow from Lemma 12.3, applied respectively to $Q(n) = bn^2$ and $Q(n) = bn\langle cn\rangle$. That lemma explicitly permits either convention at the outer tie. Impose the finitely many conditions successively. The remaining set is k-large and is contained in $D \cap B$. □

This proves stability of the *allowed differences*. It does not assert the existence of colored progressions whose *sites* lie inside arbitrary three-step nilsystem windows. The distinction will matter in Section 13. Nor has this argument treated the degree-four special expression $an^2\langle bn^2\rangle$. Section 15.3 gives an explicit obstruction to its most direct scalar-marker extension.

12.1. **One extra color at every finite nilpotency order.** The preceding rounded-product arguments give direct low-degree proofs. For the full nil-Bohr filter we use additional information from the construction of the Alweiss–Farhangi coloring. An orbit segment missing a color has a long subprogression on which the marker changes value only a bounded number of times. Rescaling the original blocker controls the bounded multiple of the difference on that subprogression.

For special generalized polynomials we use the nearest-integer convention and the class in Alweiss–Farhangi [4, Definition 1.6 and formula (1.1)]; explicitly the exponents w_j are positive and the formula is

$$P(n) = \mathcal{L}(\alpha_1 n^{w_1}, \dots, \alpha_s n^{w_s}), \quad \mathcal{L}(x) = x, \quad \mathcal{L}(x_1, \dots, x_s) = x_1 \lfloor \mathcal{L}(x_2, \dots, x_s) + 1/2 \rfloor.$$

Theorem 12.6. *Let $k, H \geq 2$. Fix finitely many special generalized polynomials $P_1, \dots, P_m$ and $\varepsilon > 0$, and put*

$$B = \{d \in \mathbb{N} : \|P_i(d)\| < \varepsilon \text{ for every } i\}.$$

There is a length K such that for every (k, H)-blockable set E, the set $E \cup (\mathbb{N} \setminus B)$ is $(k+1, K)$-blockable. For degree bound D one may take $K = K(k, H, m, D, \varepsilon)$. The length may be chosen independently of E, of its witness coloring, and of the real coefficients when the degree bounds and number of polynomials are fixed. The same holds for every superset of such a basic nil-Bohr neighborhood, by monotonicity of blockability.

The proof uses the shell coloring and the bounded subprogression in the *proof* of [4, Proposition 3.7], in addition to their polychromatic theorem. Invoking the polychromatic theorem alone would not establish the result: one has to treat a possible long star-colored progression in a direction from $E \cap B$.

Lemma 12.7 (Rescaling a blocker)**.** *Fix $W \geq 1$, put $L = \operatorname{lcm}(1, \dots, W)$, and let $c : \mathbb{N} \to \{1, \dots, k\}$ block (E, H). Define*

$$c^*(n) = c(1 + \lfloor n/L \rfloor).$$

Then c^* blocks every direction qd, $d \in E$, $1 \le q \le W$, at the common length $H^* = 1 + (H-1)L$.

Proof. If an H^*-term progression of step qd were monochromatic, sample its indices $0, L/q, \dots, (H-1)L/q$. The arguments of c are $1 + \lfloor n/L \rfloor + td$ for $0 \le t < H$, a contradiction. □

Lemma 12.8 (The additional bounded-run consequence of the AF coloring)**.** *Fix $r \ge 2$ and a finite family of special generalized polynomials of degrees at most D. There are a coloring $b : \mathbb{N} \to \{0, \dots, r-1\}$ and constants $W, N_0, T \ge 1$, $C > 0$, with the following two properties for all $a, d \ge 1$ and $N \ge N_0$.*

1. *If the N samples $b(a + jd)$ omit a color, then $\|P_i(d)\| \le CN^{-\deg P_i}$ for every i.*
2. *In the same event there exist $0 \le u \le W$, $1 \le q \le W$ such that the sequence*

$$b(a + (u + qt)d), \qquad 0 \le t \le J = \lfloor (N - 1 - u)/q \rfloor,$$

is a concatenation of at most T constant runs. All bounds here are independent of the real coefficients for fixed degree and number of polynomials.

Proof. Use the primary shell coloring in [4, Theorem 1.9], in the notation of its proof. For a finite product G of real unitriangular matrix groups with their integer lattices Γ, let

$$F(A) = \min_{\gamma \in \Gamma} \sum_{i<j} (A\gamma)_{ij}^2, \qquad \kappa(A) = \lfloor F(A)/h_0 \rfloor \pmod r,$$

with the sum over the off-diagonal entries of all blocks and $h_0 > 0$ the fixed shell width of [4, (3.18)]. The coloring of the integers is $b(n) = \kappa(g(n/L_{\mathrm{AF}}))$ for their polynomial matrix g and scale L_{AF}; this auxiliary scale is distinct from L in Lemma 12.7. The first assertion is exactly their Theorem 1.9.

Here are the further details from their proof needed for (2). The auxiliary bound W is fixed in [4, (3.23)–(3.24)] before a, d, N and the real coefficients. Increase N_0 to be at least $4W + 2$ and at least the coloring's theorem threshold. With $p(t) = g((a + dt)/L_{\mathrm{AF}})$, the proof of Proposition 3.7, equations (3.26)–(3.28), gives u, q in the displayed bounds and polynomial matrices $C_0(t), Q(t)$ such that

$$p(u + qt)C_0(t) = Q(t), \quad C_0(\mathbb{Z}) \subseteq \Gamma, \quad \deg Q_{ij} \le D, \quad |Q(t)|_\infty \le 3/4 \ \ (0 \le t \le J).$$

The norm here is the maximum over off-diagonal coordinates. The last estimate holds throughout the *real* interval: their (3.28) bounds the displacement by $1/4$ from the canonical initial point whose coordinates have magnitude at most $1/2$. Right-lattice invariance gives $b(a + (u + qt)d) = \kappa(Q(t))$ at all the indicated integers, whether or not $Q(t)$ remains canonical.

Let $\mathcal{K} \subset G$ be the fixed compact box of off-diagonal radius $3/4$. By [4, Lemma 3.3], only a finite set $\Gamma_0 \subset \Gamma$ can minimize F on this box. One can also see this directly: a bounded candidate for the minimum bounds the entries of $A\gamma$, while A^{-1} is uniformly bounded on $\mathcal{K}$, hence bounds the integer entries of $\gamma = A^{-1}(A\gamma)$. Thus on $[0, J]$,

$$F(Q(t)) = \min_{\gamma \in \Gamma_0} p_\gamma(t), \qquad p_\gamma(t) = \sum_{i<j} (Q(t)\gamma)_{ij}^2, \qquad \deg p_\gamma \le 2D.$$

The range of F is in $[0, \nu/4]$, where ν is the number of off-diagonal coordinates. Consider all equations $p_\gamma(t) = \ell h_0$ for integers $0 \le \ell \le 1 + \lceil \nu/(4h_0) \rceil$. Discard the equations that hold identically. The remaining union of real solution sets has at most

$$R = 2D\,|\Gamma_0|\,(2 + \lceil \nu/(4h_0) \rceil)$$

points. On any complementary interval every p_γ lies on a fixed side of each shell boundary or is identically equal to it; these comparisons determine the shell of their minimum, including the floor convention at equality. Thus $\kappa(Q(t))$ is constant there. Each exceptional point may contribute its own single label. On integer samples there are at most $T = 2R + 1$ constant runs. The finitely many possible weight patterns for fixed D, m allow maxima of all constants, as in [4]. □

Proof of Theorem 12.6. Use Lemma 12.8 with $r = k+1$ and fix its b, W, N_0, T, C. Choose c^* and H^* from Lemma 12.7. Choose K so large that $K \geq N_0$, $C/K < \varepsilon$, and

$$1 + \lfloor (K - 1 - W)/W \rfloor > T(H^* - 1).$$

Use $k+1$ colors $1, \ldots, k, \star$, assigning

$$C^*(n) = \begin{cases} \star, & b(n) = c^*(n), \\ c^*(n), & b(n) \neq c^*(n). \end{cases}$$

On a progression monochromatic in an ordinary label l, one has $c^* = l$ and $b \neq l$ throughout. On a star-monochromatic progression one has $b = c^* \in \{1, \ldots, k\}$ throughout, so b omits 0. In either case a monochromatic progression of length K forces $d \in B$ by Lemma 12.8(1). If also $d \in E$, an ordinary monochromatic label contradicts Lemma 12.7 with $q = 1$. For a star label, Lemma 12.8(2) and the size of K give H^* consecutive equal b-values on one of the subprogressions of step qd, $q \leq W$. On that subprogression $c^* = b$, again contradicting Lemma 12.7. This blocks precisely as required. □

Corollary 12.9 (Full nil-Bohr localization of differences)**.** *Let $k \geq 2$ and let B be a nil-Bohr neighborhood of zero of any finite order. If D is $(k+1)$-large, then $D \cap B$ is k-large. In particular, if A satisfies* (C_k)*, then $D \cap B$ is k-large for every k-large $D \subseteq A$.*

Proof. The nil-Bohr filter characterization [17, Theorem B and Theorem 4.2.11], also stated in [4, Definition 1.6], gives a finite basic intersection inside B. For precision about rounding, take its special expressions in the convention of Theorem 12.6 and also impose $\|R(d)\| < 1/4$ for each proper nested tail R that occurs under a rounding symbol. Those tails are themselves special expressions. Induction from the innermost rounding shows that the upward and downward tie conventions agree on this refined basic intersection. Taking one common smaller radius gives a basic set $B_0 \subseteq B$ in the required convention.

If $E = D \cap B$ had a (k, H)-blocker, Theorem 12.6 would give a $(k+1, K)$-blocker for $E \cup (\mathbb{N} \setminus B_0)$. This union contains D, contradicting $(k+1)$-largeness. Under (C_k) each k-large $D \subseteq A$ is already $(k+1)$-large, giving the second conclusion. □

This theorem restricts the allowed differences inside D. For progressions whose sites are required to belong to a prescribed window, Section 13 proves a separate rotation-window result. Theorem 13.6 combines both restrictions in a single coloring for all finite nilpotency orders.

13. Bohr localization and nilreturn windows of positions

For $\alpha \in \mathbb{T}^m$ and $\varepsilon > 0$, set

$$B(\alpha, \varepsilon) = \{d \in \mathbb{N} : \|d\alpha\|_\infty < \varepsilon\}.$$

The zero-dimensional case means $B = \mathbb{N}$.

Theorem 13.1. *For $r \geq 1$ and $D \subseteq \mathbb{N}$, the following are equivalent.*

1. *For every m, α, ε, the set $D \cap B(\alpha, \varepsilon)$ is r-large.*
2. *For every rotation $Tz = z + \alpha$ of $\mathbb{T}^m$, every $y \in \mathbb{T}^m$, every open neighborhood V of y, and every r-coloring of*
$$N(y, V) = \{n \in \mathbb{N} : y + n\alpha \in V\},$$
there are arbitrarily long monochromatic progressions contained in $N(y, V)$ and having difference in D.

The second assertion is equally valid for any open V meeting the orbit closure of y, whether or not $y \in V$.

Proof. Assume (1), and put $Y = \overline{\{y + n\alpha : n \in \mathbb{Z}\}}$. Rotation on Y is minimal. Choose finitely many nonnegative integers $j_1, \dots, j_t$ so that the sets $T^{-j_i}V$ cover Y. Choose an integer M large enough that, for every half-open grid box Q of mesh $1/M$ in $\mathbb{T}^m$ meeting Y, its closure in Y is contained in some $T^{-j_i}V$. This follows from a Lebesgue number for the finite cover of Y. Assign one such j_i to each box, writing $j(Q)$.

Given the window coloring c, define an r-coloring of all positive integers by

$$b(n) = c(n + j(Q(y + n\alpha))).$$

The argument of c belongs to $N(y, V)$ by construction. Fix a desired length $h \geq 2$ and put

$$L = (m + 1)(h - 1) + 1, \qquad \varepsilon < \frac{1}{ML}.$$

By (1), b has a monochromatic L-term progression $a, a + d, \dots, a + (L - 1)d$ with $d \in D \cap B(\alpha, \varepsilon)$. Choose the centered real lift θ of $d\alpha$. The lifted phase path is $z + t\theta$, $0 \leq t < L$. Each coordinate moves through an interval of length less than $1/M$, and hence crosses at most one hyperplane of the periodic grid of mesh $1/M$. The list of grid boxes can therefore change at most m times. It divides the L sites into at most $m + 1$ consecutive runs; one has at least h sites. On that run $j(Q)$ is constant, say j. Adding this same j to the corresponding integer sites gives the required monochromatic h-progression inside $N(y, V)$, still with difference $d \in D$.

Conversely assume (2). Given an arbitrary r-coloring of $\mathbb{N}$, restrict it to visits of 0 under $+\alpha$ to a sufficiently small open box $V \ni 0$ with $V - V \subseteq \{z : \|z\|_\infty < \varepsilon\}$. An h-term progression from (2), with $h \geq 2$, has difference $d \in D$, and the phases of its first two sites show $d \in B(\alpha, \varepsilon)$. The case $m = 0$ includes r-largeness of D itself. □

Corollary 13.2 (Consequence of local consecutive-color collapse)**.** *Fix $k \geq 2$. Suppose $A \subseteq \mathbb{N}$ satisfies*

$$\forall E \subseteq A \qquad [E \text{ is } k\text{-large} \Longrightarrow E \text{ is } (k + 1)\text{-large}]. \tag{C}$$

Using Theorem 9.5, every k-large $D \subseteq A$ has the rotation-window property in Theorem 13.1 with both $r = k$ and $r = k + 1$.

Proof. Corollary 9.6 states that if E is $(r + 1)$-large, then $E \cap \{d : \|d\beta\| < \delta\}$ is r-large. Begin with a k-large $D \subseteq A$. By (C) it is $(k + 1)$-large, so the first one-frequency slice is k-large. This slice remains a subset of A, and hence (C) promotes it to $(k+1)$-large. Repeat for each coordinate. Every resulting finite-dimensional Bohr slice is both k-large and $(k+1)$-large. Apply Theorem 13.1 for the two palettes. □

Lemma 13.3 (A uniform finite packet over compact phase parameters)**.** *Let P be a compact space, and for each $d \in D$ let $U_d \subseteq P$ be open. Suppose that*

$$D_p = \{d \in D : p \in U_d\}$$

is r-large for every $p \in P$. For every $h \geq 2$ there exist a finite $F \subseteq D$ and $N \in \mathbb{N}$ such that, for every $p \in P$ and every r-coloring of $[1, N]$, there is a monochromatic h-term progression contained in $[1, N]$ with difference in $F \cap D_p$.

Proof. For each p, compactness of the coloring space gives a finite (r, h)-forcing packet $F_p \subseteq D_p$ and a finite interval $[1, N_p]$ witnessing its forcing. The open set $V_p = \bigcap_{d \in F_p} U_d$ contains p, and the same packet works for every parameter in V_p. Take a finite subcover $V_{p_1}, \dots, V_{p_t}$ of P, set $F = \bigcup_i F_{p_i}$, and take $N = \max_i N_{p_i}$. □

In particular, ordinary polynomial-Bohr stability in a counterhost, proved in Section 11, has the following uniform consequence. Fix h, m, s and $\varepsilon > 0$. Every k-large D in that host admits a finite $F \subseteq D$ and N such that, simultaneously for every coefficient array $(\alpha_{jt}) \in \mathbb{T}^{ms}$ and every coloring

of $[1, N]$ in k colors (or, separately, in $k+1$ colors), there is a monochromatic h-term progression with difference $d \in F$ satisfying

$$\left\| \sum_{t=1}^{s} \alpha_{jt} d^t \right\|_{\mathbb{T}} < \varepsilon \qquad (1 \le j \le m).$$

The finite packet and interval depend on the fixed structural parameters, but not on the real coefficients or the coloring. This compactness uniformity does not resolve the dependence of a nilwindow's lower coefficients on the unknown progression start.

Why the same folding proof does not cover a skew shift. For

$$T(x, z) = (x + \alpha, z + x) \quad \text{on } \mathbb{T}^2,$$

the d-step action is

$$T^d(x, z) = (x + d\alpha, z + dx + \tbinom{d}{2}\alpha).$$

Even if $d\alpha$ and $\binom{d}{2}\alpha$ are very small modulo one, the term dx is uncontrolled when the progression starts at an arbitrary orbit point. Along the orbit of $(0, 0)$ starting at time a, it contributes $ad\alpha$ modulo one. A fixed small bound on $\|d\alpha\|$ gives no bound on $\|ad\alpha\|$ uniform in a. Consequently the bounded-grid-crossing argument above cannot simply be repeated for nilrotations or polynomial visit sequences. This is a limitation of this proof, not a counterexample to a stronger window theorem.

13.1. **One additional color for arbitrary nilreturn windows of positions.** We now pass from rotations to finite-dimensional nilrotations. Write $P_k(D)$ for the property defined in Section 5. The geometric part of the shell argument allows a bounded correction which puts an entire progression of the original difference back in a prescribed nilreturn window.

Theorem 13.4. *For every $k \ge 1$ and $D \subseteq \mathbb{N}$, $(k+1)$-largeness of D implies $P_k(D)$. The trivial system gives the converse bound $P_k(D) \Rightarrow k$-largeness.*

We prove a more concrete assertion in which R may be any set containing a syndetic basic neighborhood

$$B = \{m \in \mathbb{N} : \|P_i(m)\| < \epsilon,\ 1 \le i \le v\}. \tag{12}$$

Here the P_i are nested special generalized polynomials as in Alweiss–Farhangi (AF), with the nearest-integer convention $\langle x \rangle = \lfloor x + 1/2 \rfloor$, and $\epsilon > 0$. The nil-Bohr neighborhood characterization of Huang–Shao–Ye, in the formulation of [4, Theorem 4.1], supplies such a B in any nilreturn window. For clarity, if the convention at half-integers differs, add all proper nested tails that are rounded in the given expressions to the constraints, and require distances less than $\min(\epsilon, 1/4)$. Induction from the innermost tail gives agreement of the two rounding rules on this refined set. The refined set is itself syndetic: the reverse neighborhood characterization gives a return window inside it (use the same tail refinement in the reverse convention if needed), and nilrotation orbit closures are minimal. Equivalently one can obtain syndeticity by linearizing the finitely many polynomial matrix orbits below and returning close to their common identity coset, where all last-column residuals are small.

Reading the basic neighborhood from matrices. For an expression $P = \mathcal{L}(\alpha_1 t^{a_1}, \dots, \alpha_s t^{a_s})$, let $g_P(t) \in \mathrm{UT}_{s+1}(\mathbb{R})$ be the bidiagonal unipotent matrix with first superdiagonal

$$(-\alpha_1 t^{a_1}, \dots, -\alpha_{s-1} t^{a_{s-1}}, \alpha_s t^{a_s}).$$

Write $w_{ij} = a_i + \cdots + a_{j-1}$. By AF, Lemma 3.8, the modulus of the top right entry of the canonical representative of $g_P(m)\,\mathrm{UT}_{s+1}(\mathbb{Z})$ is exactly $\|P(m)\|$. Canonical means off-diagonal entries in $[-1/2, 1/2)$. Successive column reductions, or the triangular recursion of AF, give a unique such representative. On any compact input box this operation has finitely many semialgebraic pieces: right integer corrections are bounded by boundedness of the input inverse and the canonical output.

A fixed matrix space for all residues. Fix F so that for each $m \geq 0$ one of $m+1, \dots, m+F$ lies in B. For each pair (P_i, f), $1 \leq f \leq F$, use a separate copy of the following real group J. Its elements are unitriangular matrices $M(X)$ of polynomials with $\deg M_{ij} \leq w_{ij}$ for $i < j$. Addition of weights under matrix multiplication shows that this is a group; the coefficient-integral matrices constitute a subgroup Λ. The quotient J/Λ has a bounded set of representatives. Indeed right multiplication by $I + p(X)E_{ij}$ adds p to entry (i,j) and affects otherwise only entries (i',j) with $i' < i$. Reduce the coefficients of each entry into $[-1/2, 1/2)$ in order of increasing $j-i$, choosing $p \in \mathbb{Z}[X]$ within the entry's degree bound. These operations preserve all degree bounds. In particular the closed box with coefficients in $[-1/2, 1/2]$ contains a representative of every coset.

Put $D_0 = \sum_j a_j$. Embed J into a real upper unitriangular matrix group by replacing an entry $p(X)$ by the matrix of multiplication by p on $\mathbb{R}[X]/(X^{D_0+1})$, using the basis $1, X, \dots, X^{D_0}$. The diagonal blocks are identities and all other nonzero blocks are strictly above them. Products in the polynomial matrix group satisfy the degree bounds, so that truncation changes none of their entries. The representation is faithful (read the first column of each block), and

$$\rho(J) \cap \mathrm{UT}(\mathbb{Z}) = \rho(\Lambda). \tag{13}$$

The image is a closed affine subspace of matrices cut out by linear coefficient constraints. Insert D_0 unused identity coordinates between successive diagonal blocks. Now every possible nonzero off-diagonal coordinate of $\rho(M)$ has column index minus row index at least D_0. Take the block diagonal product of these padded embeddings for all (i,f), denoting the product groups again by J, Λ and the embedding by $\rho : J \to \mathrm{UT}_b(\mathbb{R})$. In particular b is fixed before any scaling parameter is chosen.

The shell and its polynomial subprogression. For $r = k+1$ use the shell coloring κ and parameters W, M of AF, Sections 3.5–3.8, on $\mathrm{UT}_b(\mathbb{R})$ with weights $w'_{ij} = j - i$. The scale $L = \mathrm{lcm}(1, \dots, W)$ below is our scale; we use the geometric part of the proof of AF, Proposition 3.7, which applies directly to polynomial matrix curves of coordinate degrees at most w'_{ij} and does not require the coefficient-decoding scale of AF. Form the product jet

$$G(n)_{i,f}(X) = g_{P_i}\big((n+X)/L + f\big), \qquad n \in \mathbb{R}, \qquad b(n) = \kappa(\rho(G(n))). \tag{14}$$

The polynomial in X in each entry of $G(n)$ has the allowed degree, and its coefficients have degree at most D_0 in n in the corresponding block. The padding guarantees the AF coordinate bounds. Consequently, if $b(a + jd)$ omits one shell color for $0 \leq j < N$, $N \geq 4W + 2$, AF equations (3.26)–(3.28) give $0 \leq u \leq W$, $1 \leq q \leq W$, and a real polynomial matrix $Q(t)$ on $0 \leq t \leq S = \lfloor (N - 1 - u)/q \rfloor$ with bounded coordinate degrees (at most b), such that

$$\|Q(t)\|_{\mathrm{off},\infty} \leq 3/4, \qquad Q(t)\,\mathrm{UT}_b(\mathbb{Z}) = \rho(G(a + (u+qt)d))\,\mathrm{UT}_b(\mathbb{Z}) \quad (t \in [0,S] \cap \mathbb{Z}). \tag{15}$$

The norm estimate holds on the entire real interval.

Lemma 13.5. *Fix $0 \leq r_0 < L$. On any integer arithmetic subprogression of the parameter t in (15) on which $n = a + (u+qt)d \equiv r_0 \pmod{L}$, the following two labels together have a bounded number of consecutive constant runs:*

$$b(n), \qquad \min\{1 \leq f \leq F : \lfloor n/L \rfloor + f \in B\}. \tag{16}$$

The bound is independent of a, d, N and of the chosen subprogression.

Proof. Here all spaces, boxes, and residue choices are fixed; constants need not be uniform over the original window. On the box of (15) the shell label $\kappa(Q)$ is determined by finitely many polynomial sign tests: the size function is the minimum of the squared off-diagonal norms of $Q\gamma$ over a finite list of integral γ, by compactness and boundedness of Q^{-1}, and its range meets only finitely many shell thresholds.

Let K be the image under ρ of the closed coefficient representative box from above. At every integer sample of (15), some $\gamma \in \mathrm{UT}_b(\mathbb{Z})$ has $Q(t)\gamma \in K$; only finitely many γ can occur, as

$\gamma = Q(t)^{-1}(Q(t)\gamma)$. Select the first one from a fixed finite list, using any default when none exists at an unsampled real point. This selection uses only fixed polynomial sign tests since K is a compact polytope. At valid samples write the result $Q(t)\gamma = \rho(M_t)$. By (13) and (15), $M_t = G(n)\lambda_t$ for some $\lambda_t \in \Lambda$. Reading the blocks of M_t and evaluating them at $X = -r_0$ are fixed linear operations. Since $-r_0$ is an integer, each $\lambda_t(-r_0)$ is an integral unipotent matrix. Thus the evaluated (i, f) block represents precisely the right coset of $g_{P_i}(\lfloor n/L \rfloor + f)$. Evaluation of K at each of the finitely many residues is bounded, so the canonical reductions and the subsequent tests in (12) again use only finitely many polynomial sign tests. This detects the least correction in (16), with an arbitrary default at irrelevant real points.

After substituting $Q(t)$ each of the finitely many tests is a polynomial sign test of bounded degree in one real variable. Discard identically zero polynomials, retaining their equality outcomes. The number of remaining roots is bounded by the sum of the degree bounds. The intervals and singleton roots give a uniform bound on the number of label runs, valid also on any increasing subsequence of the specified residue. □

Proof of Theorem 13.4. Given a k-coloring c of R, with labels $1, \dots, k$, let $f(m)$ be the least correction putting $m + f(m)$ in B, and put $c_0(n) = c(\lfloor n/L \rfloor + f(\lfloor n/L \rfloor))$. Give n the new label $\star$ if $b(n) = c_0(n)$ and give it $c_0(n)$ otherwise. The shell labels are $0, \dots, k$. On a progression monochromatic in an ordinary color i, the shell omits i, and $c_0 = i$ throughout. On an all-star progression the shell omits zero and $c_0 = b$ throughout.

By $(k+1)$-largeness find such progressions $a+jd$, $d \in D$, of arbitrarily large length N. Apply (15) and sample $t = 0, L/q, 2L/q, \dots$ up to S. The resulting n have one fixed residue r_0 mod L, and their floors $m = \lfloor n/L \rfloor$ advance by exactly d. The number of samples tends to infinity uniformly in u, q as $N \to \infty$ because L, W are fixed. By Lemma 13.5 they include an arbitrarily long consecutive run with both b and $f(m)$ constant. On this run c_0 is constant in either monochromatic case. The positions $m + f(m) \in B \subseteq R$ therefore form the required progression in c, with the original difference $d \in D$. □

Theorem 13.6 (Simultaneous localization of differences and positions)**.** *If D is $(k+1)$-large, then $P_k(D \cap B_{\text{diff}})$ holds for every nil-Bohr neighborhood B_{diff} of zero, at any fixed finite order.*

Proof. Choose a finite basic difference test contained in B_{diff}, say $\|P_l'(d)\| < \epsilon'$ for all l. In (14) take the product of the jet ambient group with one separate native bidiagonal unitriangular factor per P_l', retaining its native weights. Choose the AF shell on this product, then its W, and set $L = \operatorname{lcm}(1, \dots, W)$ and $L_{\text{AF}} = D'!\, L$, where D' is the maximum weight in any ambient factor. Keep the position jet at its scale L; on each difference factor use the ordinary homogeneous block $g_{P_l'}(n/L_{\text{AF}})$. The whole curve is in the AF matrix-curve class.

An omitted color still yields (15) on the jet factor. To read the shell label on the selected subprogression, use the bounded lifts of *all* factors from AF (3.28); their product gives the same finite sign-test proof for the shell. In addition Proposition 3.7 gives $\delta_{L_{\text{AF}}}\tau(p)\gamma = E$ with errors of weight w bounded by $C_w N^{-w}$. Here $p(j)$ is the full curve on $a + jd$ and τ takes the top coefficient of the assigned weight in each entry. On the native difference factors homogeneity gives $\delta_{L_{\text{AF}}}\tau(p)_l = g_{P_l'}(d)$, regardless of the jet factors. The lattice is a product. Thus AF, Lemma 3.8, gives $\|P_l'(d)\| \le C_l N^{-\deg P_l'}$ for large N. For all sufficiently long input progressions d is therefore in B_{diff}. The proof already given supplies the prescribed colored position progression with this very d. □

In particular on any counterhost (one where all k-large subsets are $(k+1)$-large), every k-large subset has this simultaneous localization at the same level k. Conversely, to separate in a given host it would suffice to find a k-large subset for which the conclusion fails for one pair of windows. This is a sufficient test, not a necessary condition on a separating set: the positive quasi-central-host constructions can retain strong nilwindow properties despite an extra-color blocker.

14. Independent thinning cannot produce separation

Theorem 14.1. *Let A be large and let $0 < p \le 1$. Retain each element of A independently with probability p, obtaining A_p. Then A_p is large almost surely.*

Proof. We recall the elementary properties needed for the probability argument. Nonlarge sets form an ideal: product colorings combine two finite-color blockers, with forbidden length the larger of their two forbidden lengths. Thus a finite partition of a large set has a large cell. Every finite set F is nonlarge: if $M = \max F$, the binary coloring $n \mapsto \lfloor (n-1)/M \rfloor \bmod 2$ blocks all differences in F at length $2M+1$. (For $F = \varnothing$ there is nothing to prove.) It follows from the ideal property that finite deletion preserves largeness.

The event that A_p is large is measurable. For each r, h, compactness of $\{1, \dots, r\}^{\mathbb{N}}$ shows that a forcing set has a finite forcing subset. Consequently (r,h)-forcing is an open property of the selected subset, and largeness is a countable intersection of such open properties. Finite-change invariance makes this a tail event, whose probability is zero or one.

First take $p = 1/m$. Independently give the elements of A one of m labels, uniformly. Each of the m cells has the same distribution as $A_{1/m}$. At least one cell is large for every outcome, by finite partition regularity. Therefore the probability that a specified cell is large is positive (indeed at least $1/m$), and the tail zero-one law makes it one. For arbitrary $p > 0$, choose m with $1/m \le p$ and use the usual monotone coupling of the two Bernoulli subsets. Largeness is upward closed, so the conclusion follows. □

Corollary 14.2 (Finite probabilistic witnesses)**.** *For every large A, every r, h, every $0 < p \le 1$, and every $\varepsilon > 0$, there is a finite $F \subseteq A$ such that the independent p-thinning of F is (r,h)-forcing with probability at least $1 - \varepsilon$.*

Proof. Enumerate A and let F_N be its first N elements. Under a single coupled thinning, the events that $(F_N)_p$ is (r,h)-forcing increase with N. Their union is the event that A_p is (r,h)-forcing, because forcing has finite witnesses. The theorem and continuity from below of probability give the claim. □

The result also holds for independent retention probabilities bounded below by a positive constant, by coupling. It does not apply to the dependent torus selections used in the color hierarchy construction. Indeed, an independently thinned large host remains large and so cannot itself supply the requested k-large but not $(k+1)$-large separation.

15. Obstructions to several proposed routes

The results here rule out specific mechanisms. None is a counterexample to Conjecture 1.2.

15.1. Critical packets do not supply a fixed forbidden length.

Lemma 15.1 (One additional marker color)**.** *Suppose $E \subseteq \mathbb{N}$ has a k-color blocker of length h, and let $p > k$ be a prime. Then $E \cup (\mathbb{N} \setminus p\mathbb{N})$ has a $(k+1)$-color blocker of length $\max(h, p)$.*

Proof. Label the original colors $1, \dots, k$ and write the original coloring as c. Recolor n with a new color $*$ when $n \equiv c(n) \pmod p$, leaving its color unchanged otherwise.

For a difference not divisible by p, a progression of p sites visits every residue. It cannot be monochromatic of an old color i, because the site of residue i would have been recolored. It cannot be monochromatic of color $*$, because the site of residue zero is never recolored.

For a difference $d \in E \cap p\mathbb{N}$, a progression of length h monochromatic in an old color was already monochromatic in c. If it is monochromatic of color $*$, all its residues are the same, and the marker rule forces all its original colors to equal that residue. Either case contradicts the original blocker. □

Corollary 15.2 (Critical finite packets)**.** *Let F be a finite inclusion-minimal (k,h)-forcing set of differences, where $k,h \geq 2$. For every prime $p > k$, the set F has a $(k+1)$-color blocker of length* $\max(h,p)$.

Proof. Divide by $g = \gcd(F)$, which preserves all fixed forcing and blocking properties. The normalized set has an element d_0 not divisible by p. By minimality $F \setminus \{d_0\}$ has a k-color blocker of length h. The preceding lemma blocks this set together with every nonmultiple of p, in particular d_0. Finally restore the dilation by g. □

Every forcing host contains a finite inclusion-minimal forcing packet, by Lemma 2.1. Thus the critical-packet corollary always gives finite lower and upper witnesses. Its upper length $\max(h,p)$ still grows with h. It therefore does not establish $T_H(A)$ for any fixed H.

15.2. **No finite list of prescribed dilation templates is universal.** A set is *syndetic* if it has bounded gaps. An *IP set* is a set containing $\mathrm{FS}(b_i)$ for some infinite sequence of positive integers.

Theorem 15.3. *Given finitely many ordered pairs (a_i, b_i) of distinct positive integers, there is a syndetic large set $A \subseteq \mathbb{N}$ that contains an IP set and satisfies*

$$\forall i \ \forall m \geq 1 \qquad \{a_i m, b_i m\} \not\subseteq A.$$

Proof. Exchange the entries when necessary so $a_i < b_i$, and put $r_i = b_i/a_i > 1$. Choose constants

$$1 < c < \min_i r_i, \qquad \beta > c \max_i r_i.$$

For a sufficiently large J, the open subset of the circle

$$U = \bigcup_{j \geq J} (\beta^{-j}, c\beta^{-j})$$

lies in $(0, \varepsilon)$, where $\varepsilon \max_i(a_i + b_i) < 1$. For each i, the interval $r_i(\beta^{-j}, c\beta^{-j})$ lies strictly between the jth and $(j-1)$st intervals defining U. Hence $U \cap r_i U = \varnothing$, considering these as subsets of the positive real line.

Fix an irrational α and let $A = \{n \in \mathbb{N} : \{n\alpha\} \in U\}$. This is syndetic, since an irrational rotation is minimal and U is nonempty and open. Suppose $a_i m, b_i m \in A$. Put $u = \{a_i m \alpha\}$ and $v = \{b_i m \alpha\}$. Then $b_i u - a_i v$ is an integer, and its absolute value is less than $(a_i + b_i)\varepsilon < 1$. It is therefore zero, so $v = r_i u \in U \cap r_i U$, a contradiction.

We verify that A contains an IP set, rather than merely invoking a boundary-return assertion. Recursively choose increasing positive integers $n_1, n_2, \ldots$ such that the fractional parts of all nonempty finite sums of the selected n_j belong to U. At a finite stage those fractional parts form a finite subset of the open real set $U \subset (0, \varepsilon)$. There is $\delta > 0$ such that adding any number in $(0, \delta)$ keeps all of them in U, without crossing 1. Since $0 \in \overline{U}$, the open set $U \cap (0, \delta)$ is nonempty. The irrational orbit visits it at arbitrarily large positive times, so choose the next n_j there. Its own fractional part and every newly created finite sum then belong to U. This completes the induction.

Finally every IP set is large by Hales–Jewett: for an r-coloring and a target length h, color words in $\{0, \ldots, h-1\}^M$ by the color of $1 + \sum_i w_i n_i$. A monochromatic combinatorial line gives an h-term progression with difference a nonempty finite sum of the n_i. □

Corollary 15.4. *For any finite collection $F_1, \ldots, F_t$ of finite subsets of $\mathbb{N}$, each of size at least two, there is a syndetic large host A containing no dilated copy mF_i for any $m \geq 1$ and any i. Consequently no finite list of fixed direction templates, considered up to dilation, is a universal source of finite forcing packets inside all large hosts, even at one fixed nontrivial color/length target.*

Proof. Select two distinct entries from each F_i and apply Theorem 15.3. For at least two colors and target length at least two, a singleton direction set cannot be forcing: the binary coloring $n \mapsto \lfloor (n-1)/d \rfloor \bmod 2$ blocks its only difference already at length two. □

The preceding construction leaves room for finite packets whose arithmetic labels depend on the host. It excludes a shortcut in which one selects from a finite catalogue of fixed direction sets and then rescales one of them into every large host.

15.3. **A direct quartic scalar marker fails uniformly.**

Proposition 15.5. *Fix an odd integer $p > 2$ and any positive scaling constant M. For every $L \in \mathbb{N}$ and $\eta > 0$ there exist $0 < \alpha, \beta < \eta$ and $a \in 4\mathbb{N}$ such that the following happens. Put*

$$F(n) = \alpha n^2 \left\lfloor \beta n^2 + \tfrac{1}{2} \right\rfloor, \qquad c(n) = \begin{cases} 1, & n \text{ even}, \\ 2, & n \text{ odd}, \end{cases}$$

and define the direct scalar marker

$$C(n) = \begin{cases} *, & \lfloor p\{F(n)/M\} \rfloor = c(n), \\ c(n), & \text{otherwise}. \end{cases}$$

Then c is a $(2,2)$-blocker for $E = \{1\}$, but $C(a), \dots, C(a+L-1)$ all equal $*$. In particular, this marker gives no coefficient-uniform finite blocking length for the old direction 1, even when both α and β are arbitrarily close to zero.

Proof. Set

$$b = \frac{5}{4p}, \qquad \delta_A = \frac{2}{p}\left(1 + \frac{2b}{A}\right), \qquad \beta_A = \frac{p}{4A}, \qquad \alpha_A = \frac{M\delta_A}{A^2},$$

where A tends to infinity through multiples of 4. For fixed $0 \le t < L$, write $e_t = 0$ if t is even and $e_t = 1$ if t is odd. We have

$$\beta_A(A+t)^2 = \frac{pA}{4} + \frac{pt}{2} + \frac{pt^2}{4A}.$$

The first term is an integer. The second is an integer when t is even and an integer plus $1/2$ when t is odd. For sufficiently large A, the last term is in $[0, 1/2)$ for all these finitely many t. Consequently the stipulated nearest-integer convention gives the exact identity

$$\left\lfloor \beta_A(A+t)^2 + \tfrac{1}{2} \right\rfloor = \frac{pA}{4} + \frac{pt}{2} + \frac{e_t}{2}.$$

Multiplication now gives

$$\begin{aligned} \frac{F_A(A+t)}{M} &= \frac{\delta_A}{A^2}(A+t)^2 \left(\frac{pA}{4} + \frac{pt}{2} + \frac{e_t}{2}\right) \\ &= \frac{A}{2} + 2t + b + \frac{e_t}{p} + O_{p,L}(A^{-1}), \end{aligned}$$

uniformly for $0 \le t < L$. One can see the error directly from

$$\frac{(A+t)^2}{A^2}\left(\frac{pA}{4} + \frac{pt}{2} + \frac{e_t}{2}\right) = \frac{pA}{4} + pt + \frac{e_t}{2} + \frac{5pt^2}{4A} + \frac{e_t t}{A} + \frac{pt^3 + e_t t^2}{2A^2}.$$

All the error terms after multiplying by δ_A are nonnegative and converge uniformly to zero on this finite interval.

Since $A/2 + 2t$ is an integer,

$$p\{F_A(A+t)/M\} \longrightarrow \frac{5}{4} + e_t.$$

Both limiting values lie strictly inside their respective phase bins and are less than p (including when $p = 3$). Thus, for all sufficiently large A,

$$\lfloor p\{F_A(A+t)/M\} \rfloor = 1 + e_t = c(A+t) \qquad (0 \le t < L).$$

Every one of these sites is therefore marked $*$. Finally $\alpha_A, \beta_A \to 0$, so they can also be made less than the prescribed η. The original coloring alternates on step 1, proving its $(2,2)$-blocking property. □

At the direction $d = 1$, conditions such as $\|\alpha d^2\| < \eta$ and $\|\beta d^2\| < \eta$ therefore do not repair this specific marker. For large A one even has $F_A(1) = 0$. The failure comes from the starting point $a = A$: its mixed linear coefficient creates alternating rounding carries despite the small direction coefficients. The construction concerns the direct scalar marker with this particular phase; other colorings may still prove the desired addition or localization theorem.

15.4. **Winding constraints for proper colorings.** For directions E in an abelian group, a coloring is *proper* when $c(x+d) \neq c(x)$ for every x and every $d \in E$. The source obstruction length $H = 2$ is unusually rigid. The following elementary character arguments explain why it cannot supply the $k = 2$, $q = 3$ approximate extractor. They also give periodization on arbitrarily large finite tori. The ternary winding characterization is attributed to the unpublished working note [8]; the finite-generator argument is included for the scope used here.

Lemma 15.6 (A half-turn character)**.** *Let $E \subseteq \mathbb{Z}^m \setminus \{0\}$ be finite. If there is a binary coloring $c : \mathbb{Z}^m \to \{0,1\}$ satisfying $c(x+d) \neq c(x)$ for all $x \in \mathbb{Z}^m$ and $d \in E$, there exists $\omega \in \mathbb{Q}^m$ such that*

$$\omega \cdot d = \tfrac{1}{2} \pmod 1 \qquad (d \in E).$$

Proof. Let Λ be the subgroup generated by E. Every integer relation $\sum_{d\in E} n_d d = 0$ has $\sum_d n_d$ even: the relation gives a closed walk of length $\sum_d |n_d|$ in the Cayley graph, and a binary proper coloring forces that length to be even. The two sums have the same parity. Consequently

$$\chi\Big(\sum_d n_d d\Big) = \tfrac{1}{2}\sum_d n_d \pmod 1$$

is a well-defined homomorphism $\Lambda \to \{0, 1/2\} \subset \mathbb{T}$. Take a $\mathbb{Z}$-basis $v_1, \dots, v_s$ of Λ, choose representatives $\beta_j \in \{0, 1/2\}$ of $\chi(v_j)$, and solve the rational linear equations $\omega \cdot v_j = \beta_j$. The basis vectors are linearly independent over $\mathbb{Q}$, so a rational solution exists. It has the required property. □

Lemma 15.7 (Ternary winding character)**.** *If a finite $E = \{v_1, \dots, v_s\} \subseteq \mathbb{Z}^m \setminus \{0\}$ admits a proper ternary coloring, there is a character $\xi : \mathbb{Z}^m \to \mathbb{T}$ with $\xi(v_i) \in [1/3, 2/3]$ for all i.*

Proof. Pull the coloring back by $\pi : \mathbb{Z}^s \to \mathbb{Z}^m$, $e_i \mapsto v_i$. Lift positive edge differences to $\sigma_i(x) \in \{-1, 1\}$. The two sums around a commuting square belong to $\{-2, 0, 2\}$ and agree modulo 3, hence agree. Integrating gives an integer height u with these gradients and

$$u(x) \equiv c(\pi x) - c(0) \pmod 3, \qquad u(x) \equiv \sum_i x_i \pmod 2.$$

For $z \in \ker \pi$, gradient periodicity makes $u(x+z) - u(x) = w(z)$ independent of x. Thus $w(z) \in 3\mathbb{Z}$ and $w(z) \equiv \sum_i z_i \pmod 2$. Subsequential box averages give $f_i \in [-1, 1]$; telescoping and translation invariance yield $\sum_i z_i f_i = w(z)$. Set $\eta_i = 1/2 + f_i/6$. Then

$$\eta_i \in [1/3, 2/3], \qquad \sum_i z_i \eta_i = \frac{\sum_i z_i + w(z)/3}{2} \in \mathbb{Z}.$$

So the character with generator values η_i factors through $\pi(\mathbb{Z}^s)$. Smith normal form extends it to $\mathbb{Z}^m$ by taking roots on the diagonal multiples of an ambient basis. □

Lemma 15.8 (Robust half-turn triples)**.** *If $\theta \in \mathbb{T}$ has a representative in $(1/4, 3/4)$, the three points $u, u+\theta, u+2\theta$ cannot all belong to either of the half-open semicircles $[0, 1/2)$ and $[1/2, 1)$.*

Proof. For $\theta \in (1/4, 1/2)$, the circular gaps between the three points are $\theta, \theta, 1-2\theta$, all strictly smaller than $1/2$. A set contained in a semicircle would have a complementary circular gap of at least $1/2$. For $\theta \in (1/2, 3/4)$ reverse the order and apply the same argument to $1-\theta$. At $\theta = 1/2$, two of the points are antipodal and belong to different half-open semicircles. □

Theorem 15.9 (Uniform triple blockers on finite tori)**.** *Let $E \subseteq \mathbb{Z}^m \setminus \{0\}$ be finite and put $K = \max_{v\in E} \|v\|_1$. If E has a binary proper blocker, then for every $L > 2K$ its image in $(\mathbb{Z}/L\mathbb{Z})^m$ has a binary three-term blocker. If it has a ternary proper blocker, the same conclusion holds for every $L > 6K$.*

Proof. In the binary case choose the character from the half-turn lemma; in the ternary case use Lemma 15.7. Write either character as $x \mapsto \sum_i \alpha_i x_i \pmod 1$. Choose integers a_i with $|a_i/L - \alpha_i| \le 1/(2L)$. The character

$$\xi_L(x) = \frac{\sum_i a_i x_i}{L} \pmod 1$$

factors through the finite torus and changes its value on every $v \in E$ by distance at most $K/(2L)$. In the binary case this is less than $1/4$ from the exact half turn; in the ternary case it is less than $1/12$ from $[1/3, 2/3]$. Thus every direction phase belongs to $(1/4, 3/4)$. Partition the circle into the two half-open semicircles and apply the preceding triple lemma. □

For Boolean directions $E \subseteq \{0,1\}^m \setminus \{0\}$, one may use $K \le m$. In particular arbitrarily large moduli coprime to any prescribed integer are allowed. Before approximation, the same half-circle coloring is already a binary triple blocker on the original lattice. By compactness this holds for countable direction sets in $\mathbb{Z}$. Consequently a 2-large subset of $\mathbb{N}$ cannot have a ternary two-term blocker: the property $T_2(A)$ is impossible for $k = 2$, in every host A.

For a ternary blocker of triples, adjacent colors may coincide. The lifted edge increments then lie in $\{-1, 0, 1\}$, and square path sums may differ by 3. The zero-curl proof above no longer applies. The next source length $H = 3$ is not settled by it.

15.5. **Blocker selection need not be equivariant.** For $r, H \ge 2$ and $D \subseteq \mathbb{N}$, write $X_{r,H}(D)$ for the compact shift-invariant space of (r, H)-blockers on $\mathbb{Z}$. For $D = \{1\}$ every subset is binary two-term blockable, but there is no shift-equivariant map from $X_{3,2}(D)$ to $X_{2,2}(D)$. Indeed $x(n) = n \pmod 3$ is fixed by the third power of the shift, whereas every binary proper coloring is alternating and is not fixed by that power. An equivariant map would preserve the fixed point condition. Thus abstract existence of compressed blockers cannot be upgraded to such a dynamical selection without a new argument. This example is a warning about the proposed method; its finite host is not a counterhost.

15.6. **A failed structural classification.** We write

$$d^*(B) = \limsup_{L\to\infty} \sup_{a\ge 0} \frac{|B \cap [a+1, a+L]|}{L}$$

for upper Banach density.

Theorem 15.10. *There is a large set $A \subseteq \{n^3 : n \ge 1\}$ such that*

$$d^*(\{n \ge 1 : p(n) \in A\}) = 0 \qquad \text{for every nonconstant } p \in \mathbb{Q}[n]. \tag{17}$$

Consequently A contains no finite-sums set with two positive generators, and contains no polynomial image of a quasi-central source. More strongly, it contains no $p(B)$ with p a nonconstant real polynomial and $B \subseteq \mathbb{N}$ of positive upper Banach density.

Thus the proposed reduction "every large host contains an IP_0 set or a polynomial image of a quasi-central source" is false. This does not furnish a counterexample to the hereditary color-hierarchy conjecture: it only excludes that particular classification strategy.

Lemma 15.11 (Separated polynomial preimages of a dilated packet)**.** *Fix $N, J \geq 1$, finitely many nonconstant rational polynomials $p_1, \ldots, p_t$ of positive leading coefficient, and a bound $B \geq 0$. For all sufficiently large positive integers λ, each set*

$$E_i(\lambda) = \{n \geq 1 : p_i(n) \in \lambda^3\{1^3, 2^3, \ldots, N^3\}\}$$

lies above B and any two distinct members have distance greater than J.

Proof. For a fixed $p = p_i$ of degree d, choose a threshold beyond which p is positive and strictly increasing. Since the packet's minimum λ^3 tends to infinity, all its positive integer preimages eventually lie beyond both this threshold and B. There are constants C, C' depending on p, N such that every such preimage satisfies $n \leq C\lambda^{3/d}$ and

$$\sup_{0 \leq x \leq C\lambda^{3/d}+J} |p'(x)| \leq C'\lambda^{3(d-1)/d}$$

for all large λ (the degree-one case is included). If $n < n'$ are distinct preimages with $n' - n \leq J$, the mean value theorem gives

$$0 < p(n') - p(n) \leq JC'\lambda^{3(d-1)/d} < \lambda^3$$

for large λ. But distinct packet values differ by at least λ^3, a contradiction. Take the maximum of the finitely many thresholds for $i = 1, \ldots, t$. □

Proof of Theorem 15.10. The polynomial van der Waerden theorem [5] implies that the positive cubes form a large set. By finite coloring compactness, for each $j \geq 2$ there is N_j such that

$$F_j = \{1^3, 2^3, \ldots, N_j^3\}$$

forces a monochromatic j-term progression in every j-coloring of $\mathbb{N}$. To spell out compactness, if no finite initial cube set worked, the compact space of j-colorings would contain a coloring avoiding all such progressions with any cube difference, contrary to polynomial van der Waerden. Enlarge them so $N_j \geq j$.

Enumerate all nonconstant rational polynomials of positive leading coefficient as $p_1, p_2, \ldots$. Choose positive integers λ_j recursively and put $A_j = \lambda_j^3 F_j$. At stage j, apply Lemma 15.11 with $J = j$ to the first j polynomials. Choose its bound B greater than j plus every integer preimage under these polynomials of the finite set $\bigcup_{\ell<j} A_\ell$. Such a bound exists because a nonconstant polynomial takes each value only finitely often. Also require $\min A_j > \max \bigcup_{\ell<j} A_\ell$. Then, for every $i \leq j$, the set $p_i^{-1}(A_j) \cap \mathbb{N}$ is j-separated and lies at distance greater than j above all earlier preimages for the same polynomial. Set

$$A = \bigcup_{j \geq 2} A_j.$$

The set A is large. Given an r-coloring and desired length h, choose $j \geq \max(r, h, 2)$. Read the coloring at the multiples of λ_j^3. The forcing property of F_j gives a monochromatic j-term progression; rescaling produces one of difference in $\lambda_j^3 F_j \subseteq A$.

Fix p_i and an integer $R \geq 1$. From stage $j \geq \max(i, R, 2)$ onward, all its preimages of the packets form an R-separated set, including the gaps between packets. The earlier preimages are finite. An R-separated set has upper Banach density at most $1/R$, and adding finitely many points does not change that density. Since R is arbitrary, (17) follows for p_i. A rational polynomial of negative leading coefficient has only finitely many positive values on $\mathbb{N}$, so its preimage of $A \subseteq \mathbb{N}$ is finite; this handles all remaining nonconstant rational polynomials.

Every element of A is a positive cube. If $a, b, a + b \in A$, there would be positive integers x, y, z with $x^3 + y^3 = z^3$, contrary to the classical exponent-three case of Fermat's theorem. Thus A is not even IP_2.

Finally, suppose $p(B) \subseteq A$ for a nonconstant real polynomial p and an infinite $B \subseteq \mathbb{N}$. If $d = \deg p$, choose $d+1$ distinct points of B. Their arguments and values are integers, so interpolation through the rational Vandermonde system shows that every coefficient of p is rational. Therefore $B \subseteq p^{-1}(A) \cap \mathbb{N}$ and (17) gives $d^*(B) = 0$. In particular B cannot be quasi-central, since every quasi-central set is piecewise syndetic and hence has positive upper Banach density. □

Nearby simpler examples. The positive cubes alone already show that largeness does not imply IP_0. Even piecewise-syndetic largeness does not imply IP_0: the set consisting of the odd integers together with the positive cubes is syndetic and large, but not IP_4. Indeed, split four proposed finite-sums generators into two pairs. Each pair contains a nonempty subblock of even sum. The two disjoint even sums and their sum would all be in the even part of the host, hence would all be positive cubes. Theorem 15.10 adds the simultaneous exclusion of all positive-density polynomial source images.

The host just constructed nevertheless has the full hierarchy. Indeed, it contains arbitrarily long dilated initial cube packets, so Corollary 8.3 applies with $s = 3$. The construction therefore disproves a suggested classification of large hosts; it is not a new unresolved host for the original conjecture. It also illustrates why the power-packet transfer is strictly useful beyond the quasi-central-source and IP_0 classes.

16. IP-host obstructions to full-factor lifting

The next construction isolates a failure of one possible extension of the quasi-central proof. It gives no counterexample to the hereditary color hierarchy. Even a color cylinder projecting onto the entire maximal pronilfactor need not lift recurrence along a torus-gated IP host. The gate retains a finite-tail covering property, so that property cannot replace piecewise syndeticity in the structural step. Theorem 16.6 strengthens the obstruction to all color cylinders simultaneously, using the same cocycle. Set $b_1 = 16$, $b_{j+1} = b_j^{j+4}$, and

$$\omega = \sum_{j \geq 1} \frac{1}{b_j}, \qquad A = \mathrm{FS}(b_j).$$

The integer sequence and hence the IP host A will remain fixed. This A is large: color the words in $\{0, \dots, h-1\}^n$ by the color of $1 + \sum_{j \leq n} w_j b_j$. The Hales–Jewett theorem gives a monochromatic line, whose associated integers form an h-term progression with difference in A. On the other hand A is not piecewise syndetic. Modulo b_j it occupies at most 2^{j-1} residue classes. For any fixed finite set of translates, their union therefore misses a residue class modulo b_j for all sufficiently large j, and cannot be thick. For $M \geq 1$ and $0 < a < 1/10$, let $O \subset \mathbb{T}^M$ be a nonempty open set such that every $\theta \in O$ has some coordinate with

$$\|\theta_l\| > a/2. \tag{18}$$

In particular this includes the root window O_M at distinguished velocity $-a$, since its error parameter is much smaller than $a/2$.

Theorem 16.1. *For almost every $\beta \in \mathbb{T}^M$ there is a strictly ergodic distal system (X, T_β), an open nonempty $U \subset X$, and its maximal infinite-step pronilfactor map $\pi : X \to \mathbb{T}$ such that*

$$\pi(U) = \mathbb{T}, \qquad A \cap \{d : d\beta \in O\} \cap \mathrm{Ret}_2(U) = \varnothing. \tag{19}$$

Here $\mathrm{Ret}_2(U) = \{d \geq 1 : U \cap T_\beta^{-d} U \neq \varnothing\}$. Consequently (19) also holds with $\mathrm{Ret}_h(U)$ for every $h \geq 2$. The sequence $(b_j\beta)$ is dense in $\mathbb{T}^M$, with every tail dense. Moreover, for every R there is a finite set $I \subset \{R+1, R+2, \dots\}$ such that, for every nonempty finite F with $\min F > \max I$, some $i \in I$ satisfies

$$b_{F \cup \{i\}}\beta \in O, \qquad b_F = \sum_{j \in F} b_j. \tag{20}$$

Thus the gated set has a finite-translate covering property on every tail of the partial semigroup of finite subsets.

16.1. **An elementary lacunary cocycle.** For each j let δ_j be the fractional part of $b_j\omega$. Divisibility of the sequence gives

$$0 < \delta_j = \sum_{t>j} \frac{b_j}{b_t} < \frac{2b_j}{b_{j+1}} \le \frac{2}{b_j^3}. \tag{21}$$

These inequalities also prove that ω is irrational: if it had a fixed positive denominator, its nonzero fractional parts in (21) could not tend to zero.

Partition $\mathbb{N}$ into its M residue classes $J_1, \dots, J_M$. For a positive constant τ, to be fixed later, define in real $L^2(\mathbb{T})$

$$f_l(x) = \tau \sum_{j\in J_l} \frac{\cos(2\pi b_j x)}{j}, \qquad 1 \le l \le M. \tag{22}$$

The series converge in L^2 by orthogonality. Put

$$\phi_l(x) = f_l(x+\omega) - f_l(x). \tag{23}$$

The series obtained by subtracting the terms in (22) converges uniformly, since its jth term has uniform norm at most $2\pi\tau\delta_j/j$. Thus (23) has a continuous real representative, of integral zero. For $n \ge 1$ put

$$\phi_{l,n}(x) = \sum_{t=0}^{n-1} \phi_l(x + t\omega).$$

Lemma 16.2. *For every l,*

$$\sum_{i\ge 1} \|\phi_{l,b_i}\|_\infty < \tau. \tag{24}$$

In particular $\|\phi_{l,d}\|_\infty < \tau$ for every $d \in A$.

Proof. The telescoped Fourier series converges uniformly for every fixed n, and gives

$$\|\phi_{l,b_i}\|_\infty \le 2\pi\tau \sum_{j\ge 1} \frac{\|b_i b_j \omega\|}{j}.$$

For $j \le i$ use $\|b_i b_j \omega\| \le b_j\delta_i$ and $\sum_{j\le i} b_j < 2b_i$, obtaining a contribution below $4/b_i^2$. For $j > i$ use $\|b_i b_j\omega\| \le b_i\delta_j$. The growth of the b_j and (21) give $\sum_{j>i}\delta_j < 3/b_{i+1}^3$, so this contribution is below $3b_i/b_{i+1}^3 < 1/b_i^2$. Consequently

$$\|\phi_{l,b_i}\|_\infty < 10\pi\tau/b_i^2.$$

Since $b_1 = 16$ and $b_{i+1} \ge 2b_i$, $\sum_i b_i^{-2} \le 1/192$, and $10\pi/192 < 1$. This proves (24). The cocycle identity gives $\|\phi_{l,n+m}\|_\infty \le \|\phi_{l,n}\|_\infty + \|\phi_{l,m}\|_\infty$; apply it to a finite sum of distinct b_i. □

Lemma 16.3. *Every nonzero integer linear combination $\sum_l m_l f_l$ is essentially unbounded. In particular it has no continuous representative.*

Proof. Its Fourier series has the form $f = \sum_j a_j \cos(2\pi b_j x)$ with $a_j = \tau m_l/j$ for $j \in J_l$. For $N \ge 1$ put

$$R_N(x) = \prod_{j\le N} (1 + \sigma_j \cos(2\pi b_j x)),$$

where $\sigma_j = \operatorname{sgn}(a_j)$ if $a_j \ne 0$ and $\sigma_j = 1$ otherwise. The superlacunary inequality $b_j > 3\sum_{i<j} b_i$ makes every signed frequency representation needed in this product unique, including the coefficients at $\pm b_j$. Thus $R_N \ge 0$, $\int R_N = 1$, and

$$\int f R_N = \frac{1}{2} \sum_{j\le N} |a_j|.$$

No later Fourier frequency occurs in R_N, because its maximal frequency is below b_{N+1}. At least one residue class has $m_l \neq 0$, so the right side tends to infinity. Essential boundedness of f would bound the left side by $\|f\|_\infty$, a contradiction. □

16.2. **A hidden measurable rotation.** Choose β so that $1, \omega, \beta_1, \dots, \beta_M$ are rationally independent, and set

$$X = \mathbb{T}^{M+1}, \quad T_\beta(x, y) = \big(x + \omega,\ y + \beta + \phi(x)\big), \quad \phi = (\phi_l)_{l=1}^{M}. \tag{25}$$

This is a homeomorphism and a compact abelian group extension of a rotation, hence is distal. On an invariant conull set of base points,

$$(x, y) \longmapsto (x, y - f(x)) \tag{26}$$

conjugates (25) measurably to the torus translation by (ω, β). The latter is uniquely ergodic by rational independence. Every T_β-invariant probability projects to Haar measure on the base. Thus (26) is defined almost everywhere for every such probability, and carries it to the unique invariant probability of that torus translation. Fiberwise translation preserves Haar measure, so Haar measure on X is the only invariant probability for (25). It has full support, which also implies minimality: a proper nonempty closed invariant subset would support a different invariant probability. The system is therefore strictly ergodic and has pure discrete measure-theoretic spectrum.

Lemma 16.4. *Every continuous eigenfunction of T_β depends only on x. Consequently the maximal equicontinuous factor is the base circle.*

Proof. Let $h \circ T_\beta = \lambda h$ be a nonzero continuous eigenfunction. For $m \in \mathbb{Z}^M$ take its continuous vertical Fourier coefficient $h_m(x) = \int_{\mathbb{T}^M} h(x, y) e(-m \cdot y)\, dy$. If $h_m \neq 0$, its absolute value is invariant under the irrational base rotation and hence is a nonzero constant. Write

$$h_m(x) = c\, e(\ell x + g(x)),$$

where $\ell \in \mathbb{Z}$, $c \neq 0$, and g is a continuous real periodic function. This is the usual degree decomposition of a nonvanishing circle-valued function. The eigenfunction equation gives

$$e\big(m \cdot \beta + m \cdot \phi(x) + \ell\omega + g(x + \omega) - g(x)\big) = \lambda.$$

The real expression inside e is continuous and constant modulo one, so it is constant. Its integral equals $m \cdot \beta + \ell\omega$; therefore

$$m \cdot \phi(x) + g(x + \omega) - g(x) = 0.$$

By (23), $m \cdot f + g$ is invariant in L^2 under the irrational rotation and thus is constant almost everywhere. If $m \neq 0$, this contradicts Lemma 16.3. All nonzero vertical Fourier coefficients vanish, proving that h depends only on x. Characters of an equicontinuous factor pull back to continuous eigenfunctions and separate its points; hence every such factor factors through x. The base circle itself is an equicontinuous factor, proving the last assertion. □

Lemma 16.5. *The maximal infinite-step pronilfactor of* (25) *is the base circle.*

Proof. We use the standard spectral fact that the Kronecker factor of an ergodic nilsystem is its continuous abelian quotient; the eigenfunction assertion is recorded in [2, Proposition 2.3]. In particular a minimal nilsystem with pure discrete Haar spectrum is equicontinuous. Indeed, pure discrete spectrum makes the abelian quotient generate the full measurable sigma algebra; a proper homogeneous quotient has nontrivial Haar fibers and cannot do so.

Every nilsystem factor of (25) has pure discrete spectrum, as a factor of the measurable torus rotation (26). It is therefore equicontinuous, and Lemma 16.4 makes it factor through the base circle. The same holds for an inverse limit of nilsystem factors. Since the base circle is itself a nilsystem factor, it is exactly the maximal infinite-step pronilfactor. □

16.3. **The failed full-fiber lifting.** Take $\tau = a/100$ in (22) and put

$$U = \mathbb{T} \times (-a/10, a/10)^M.$$

Plainly $\pi(U) = \mathbb{T}$. If $d \in A$ and $d\beta \in O$, choose l as in (18). Formula (25) and Lemma 16.2 show that the lth fiber displacement at time d has circle norm at least

$$\|d\beta_l\| - \|\phi_{l,d}\|_\infty > a/2 - a/100 > 2a/10.$$

Two points in $(-a/10, a/10)$ cannot differ by that displacement. Thus $U \cap T_\beta^{-d} U = \varnothing$, proving (19).

For almost every β, $(b_j\beta)$ is equidistributed in $\mathbb{T}^M$. For completeness, for each nonzero $v \in \mathbb{Z}^M$, orthogonality gives

$$\int_{\mathbb{T}^M} \left| \frac{1}{N} \sum_{j \le N} e(b_j v \cdot \beta) \right|^2 d\beta = \frac{1}{N}.$$

Borel–Cantelli on square values of N, followed by interpolation between successive squares, proves Weyl's criterion almost everywhere. The required rational independence excludes only countably many null subtori. Every generator tail is therefore dense. For any R, the translates $O - b_i\beta$, $i > R$, cover $\mathbb{T}^M$. Choose a finite subcover indexed by I. Apply that cover to $b_F\beta$ for $\min F > \max I$ to obtain (20). This finishes Theorem 16.1.

16.4. **Every color cylinder can have full factor image.** The same hidden cocycle gives a stronger failure: all color cylinders can project onto the entire pronilfactor, while every root-gated direction visits every color within one fixed progression length. The host A and the irrational number ω remain those fixed at the beginning of this section. Fix $q \ge 2$, $M \ge 1$, and put

$$Q = 100q, \quad a = 1/Q, \quad \eta = (10^6 Q^2)^{-1}, \quad H = 2Q.$$

Let $O_M \subset \mathbb{T}^M$ be the root window: for some coordinate i, its centered representatives satisfy

$$|\theta_i + a| < \eta, \qquad |\theta_l| < \eta \ (l \ne i), \qquad \sum_{l \ne i} |\theta_l| < 2a.$$

Theorem 16.6. *For almost every $\beta \in \mathbb{T}^M$, there is a minimal q-symbol subshift (Y, σ) with maximal pronilfactor map $\rho : Y \to Z$ such that every color cylinder C_r, $r \in \mathbb{Z}/q\mathbb{Z}$, satisfies*

$$\rho(C_r) = Z, \qquad (A \cap \{d : d\beta \in O_M\}) \cap \bigcup_{r \in \mathbb{Z}/q\mathbb{Z}} \mathrm{Ret}_H(C_r) = \varnothing.$$

In fact every H-term progression along a gated direction visits all q colors, for every point of Y. The host A is IP and hence large, and every tail of $(b_j\beta)$ is dense in $\mathbb{T}^M$.

The proof reuses Lemmas 16.2, 16.4, and 16.5, valid for every $\tau > 0$. The almost-everywhere frequency assertion was proved following Theorem 16.1. We now choose a smaller τ in the same construction (22)–(25). Choose

$$0 < \tau < \min\left\{ a/100, \frac{1}{1000\pi M Q} \right\}.$$

Define the analytic function on the vertical torus

$$F(y) = \sum_{l=1}^{M} (\cos(2\pi y_l) - 1).$$

Away from its finitely many level sets $qF(y) \in \mathbb{Z}$, give (x, y) the color $\lfloor qF(y) \rfloor \pmod q$. Boundary values may be assigned by either adjacent integer label. Each color has a nonempty open vertical region, over every base point x.

16.5. **Uniform polychromaticity on the IP gate.**

Lemma 16.7. *For every $y \in \mathbb{T}^M$ and $\theta \in O_M$, the values $F(y + t\theta)$, $0 \le t < 2Q$, have range greater than* 1.95*. Their neighboring differences have absolute value at most $2\pi(3a + \eta)$.*

Proof. The neighboring bound follows from $\sum_l \|\theta_l\| < 3a+\eta$. Let i be the distinguished coordinate. Choose $0 \le t < Q$ so that $\|y_i - t/Q\| \le 1/(2Q)$, and put $m = Q/2$. Then

$$\|y_i + t\theta_i\| \le 1/(2Q) + Q\eta, \qquad |m\theta_i + 1/2| < Q\eta/2.$$

The distinguished contribution to the second difference at $t, t+m, t+2m$ is

$$\mathrm{Re}\left(\mathrm{e}(y_i + t\theta_i)(\mathrm{e}(m\theta_i) - 1)^2\right), \qquad \mathrm{e}(u) = \exp(2\pi i u),$$

and is greater than $4 - 4\pi/Q - 12\pi Q\eta$. The absolute sum of all other contributions is bounded by

$$\pi^2 Q^2 \sum_{l \ne i} \theta_l^2 < 2\pi^2 Q\eta.$$

Since $Q \ge 200$ and $\eta = (10^6 Q^2)^{-1}$, the total second difference is greater than 3.9. A second difference has absolute value at most twice the range, proving the claim. All three selected indices lie in $[0, 2Q-1]$. □

Fix $d \in A$ with $d\beta \in O_M$, and start at any $(x, y) \in X$. The vertical coordinate at time td equals

$$y + td\beta + \phi_{td}(x).$$

The cocycle identity and Lemma 16.2 give $|\phi_{l,td}(x)| \le t\tau$. Consequently its F value differs from the ideal linear orbit value by at most

$$2\pi M t\tau < 4\pi MQ\tau < 0.004.$$

The actual second difference is therefore greater than $3.9 - 16\pi MQ\tau > 3.884$, so the actual range is greater than $1.942 > 1.94$. Writing F_t for the actual value at time td, consecutive values satisfy

$$q|F_{t+1} - F_t| \le 2\pi q(3a + \eta + M\tau) < 1. \tag{27}$$

Choose any allowed integer labels J_t with $qF_t \in [J_t, J_t + 1]$. Inequality (27) implies $|J_{t+1} - J_t| \le 1$. Their range is greater than $1.94q - 2 > q - 1$, so the integer walk (J_t) visits every residue modulo q. This proves the asserted polychromaticity, including all boundary choices compatible with one-sided limits of the vertical partition.

16.6. **Minimal symbolic realization and full factor images.** The partition boundary is a finite union of level sets of a nonconstant real-analytic function, hence is closed with empty interior. Choose an orbit avoiding all its translates and code it by the vertical partition. Take a minimal subsystem X^* of the joined orbit closure of the coding and its point in X. The projection $p : X^* \to X$ is onto and almost one-to-one: every point whose full orbit avoids the boundary has a uniquely prescribed coding. These points form a dense invariant G_δ, by the Baire category theorem. Such a minimal extension is proximal: two points above the same base point can be moved arbitrarily close by moving that point toward a singleton fiber and using upper semicontinuity of fibers. Every distal factor identifies a proximal pair. Thus every distal, and in particular every pronil, factor of X^* factors through X. Its maximal pronilfactor is therefore the same base circle.

Let Y be the minimal symbolic factor of X^*, with factor map $q_0 : X^* \to Y$. Its maximal pronilfactor map $\rho : Y \to Z$ satisfies

$$\rho q_0 = r\pi p$$

for a surjection $r : \mathbb{T} \to Z$. For each color r_0, a nonempty open vertical region of that color lies over every base point. Every lift of a point in such a region has symbol r_0 at zero. Hence the r_0 cylinder of X^* projects onto the full base circle, and the cylinder C_{r_0} of Y satisfies $\rho(C_{r_0}) = Z$.

The polychromaticity proved above passes to the symbolic orbit closure, since omission of a color on a specified finite progression is a clopen cylinder condition. It holds on Y for every gated d, proving Theorem 16.6.

Exact implication. The integer piecewise-syndetic intersection in the structural lifting argument cannot be replaced by a torus-gated IP intersection merely by requiring every color cylinder to cover the pronilfactor. This failure persists for generic frequencies and dense generator tails. It does not rule out an argument using additional seed directions or a different relative extraction method. Nor does it contradict the quasi-central theorem: the sparse IP host here is not piecewise syndetic, and its finite-tail covering property (20) is a different assertion.

Quadratic difference-set supplement. An independently audited supplement [20] strengthens the obstruction in another direction. For the fixed sparse host A of Section 16, every $q \geq 2$ and every irrational $\theta \in \mathbb{T}$ admit a finite $H \geq 2$ and an open arc $J \ni \theta$ with the following property. For almost every $\alpha \in \mathbb{T}$, $B = A \cap \{d : \alpha d^2 \in J\}$ contains $\Delta(E) = \{u - v : u, v \in E,\ u > v\}$ for some infinite $E \subseteq \mathbb{N}$. There is a minimal q-symbol subshift in which every color cylinder maps onto the full maximal pronilfactor and every H-term progression of difference in B sees every symbol. In particular B is van der Corput. Indeed, vanishing on $\Delta(E)$ would make the characters indexed by E orthonormal in the test measure, and Bessel's inequality at its atom at zero forces that mass to vanish. The supplement gives the full quadratic cocycle and prefix-difference construction. Its factor argument uses measure-to-topology rigidity for factor maps between ergodic nilsystems [16, Appendix, Theorem A.1].

17. Changing-degree power packets: a transfer criterion and a stress host

Fixed-degree power packets were covered by Corollary 8.3. Allowing the exponent to grow removes the fixed finite patterns that made that argument work. We give an exact finite-size condition under which the mixed-radix transfer still works, and then construct a large host outside the positive host mechanisms proved here. The new host is an unresolved test case, not a counterexample to the conjecture.

Write $[N]^s = \{1^s, \dots, N^s\}$. A finite set of differences is (r, h)-forcing if every r-coloring of $\mathbb{N}$ has a monochromatic h-term progression with one of those differences. It is (r, h)-blocked if some r-coloring has no such progression. Theorem 6.1, with source $\mathbb{N}$ and $p(n) = n^s$, gives, for every $s, k \geq 1$, a k-large subset of the positive sth powers with a $(k + 1, H_k)$ blocker, where

$$H_k = 400 \cdot 2^k.$$

The numerical value of H_k is independent of s. The finite extraction size below need not be uniform in s.

17.1. **The precise finite comparison.** For fixed $k, h, s \geq 1$, with $h \geq 2$, let $M_{k,h}(s)$ be the least M such that $[M]^s$ contains a (k, h)-forcing subset admitting a $(k + 1, H_k)$ blocker. This integer exists by compactness applied to the polynomial hierarchy. Also let

$$W_s(r, h) = \min\{N : [N]^s \text{ is } (r, h)\text{-forcing}\}.$$

The polynomial van der Waerden theorem [5] makes $W_s(r, h)$ finite.

Proposition 17.1. *Suppose A contains $t_j[N_j]^{s_j}$ for $j \geq 1$, where all three parameters are positive integers. Fix k. If*

$$\forall h \geq 2\ \forall \ell \geq 1\ \exists j \geq 1 \qquad N_j \geq \ell M_{k,h}(s_j), \tag{28}$$

then A contains a k-large subset with a $(k + 1, 2H_k)$ blocker.

Proof. At stage i request $h = i + 2$. Let ℓ be a prescribed multiple of the previous mixed-radix place, large enough to make the next radix exceed $2H_k$ times the largest previous unscaled difference.

Choose j by (28) and a finite (k,h)-forcing, $(k+1,H_k)$-blocked set $F_i \subseteq [M_{k,h}(s_j)]^{s_j}$. Then

$$\lambda_i F_i \subseteq t_j[N_j]^{s_j} \subseteq A, \qquad \lambda_i = t_j \ell^{s_j} \in \ell\mathbb{N}.$$

Thus the scales can be chosen recursively with $\lambda_{i+1}/\lambda_i > 2H_k \max F_i$. Their union $B = \bigcup_i \lambda_i F_i$ is k-large by dilation invariance.

Let c_i be an individual $(k+1,H_k)$ blocker for F_i, with palette identified with $\mathbb{Z}/(k+1)\mathbb{Z}$. If $z_i(n)$ is the digit at place λ_i, then

$$C(n) = \sum_i \big(c_i(1+z_i(n)) - c_i(1)\big) \pmod{k+1}$$

is well defined. On $2H_k$ sites with difference $\lambda_i f$, $f \in F_i$, the lower digits are fixed and the higher prefix changes at most once. Hence H_k consecutive sites have every digit except the ith fixed. Their ith digits advance by f, contradicting the blocker c_i if all output colors agree. □

In particular, the following finite comparison would suffice to solve every changing-degree host whose jth packet is (j,j)-forcing:

$$\boxed{\forall k \ge 1\ \forall h \ge 2\ \forall \ell \ge 1\ \exists J \ge 2\ \forall s \ge 1 \qquad W_s(J,J) \ge \ell M_{k,h}(s).} \tag{29}$$

Indeed a packet of index $j \ge J$ has $N_j \ge W_{s_j}(J,J)$, so (28) follows. The existing polynomial hierarchy proves only that each $M_{k,h}(s)$ is finite. Even though H_k is uniform, it does not supply (29): the required index J could depend on s. No assertion that (29) is false is made here. It is a concrete sufficient comparison left unresolved by this route.

17.2. **Uniform comparison under division of the root index.** Write $P_s(X) = \{n^s : n \in X\}$ and $X/\ell = \{m : \ell m \in X\}$. We first isolate a divisibility fact valid in every degree. It removes one quantifier from the outstanding comparison (29). A sharper version then uses one additional color and applies to primes that may grow with the degree.

Theorem 17.2 (Root division at the same length)**.** *For every $s,\ell,r \ge 1$, $h \ge 2$, and $X \subseteq \mathbb{N}$,*

$$P_s(X) \text{ is } (\ell r,h)\text{-forcing} \quad\Longrightarrow\quad P_s(X/\ell) \text{ is } (r,h)\text{-forcing.}$$

In particular $W_s(\ell r,h) \ge \ell W_s(r,h)$.

Proof. For a prime p, and any coloring $c : \mathbb{N} \to [r]$, color x by

$$(x \bmod p,\ c(1+\lfloor x/p^s \rfloor)).$$

A monochromatic progression of step n^s, $n \in X$, has $p \mid n^s$, so $n = pm$ and $p^s \mid n^s$. Its quotient sites form a monochromatic c-progression of step m^s, with $m \in X/p$. This proves the prime case. Iterating through the prime factors of ℓ, with repetitions, proves the general case. Take $X = [W_s(\ell r,h)]$ to get the numerical inequality. □

Use $H_* = H_k$ in the definition of $M_{k,h}(s)$ above.

Corollary 17.3 (The dilation margin in the comparison is redundant)**.** *For each fixed k,h, the following are equivalent:*

$$\exists J \ge 2\ \forall s \ge 1 \qquad W_s(J,J) \ge M_{k,h}(s);$$
$$\forall \ell \ge 1\ \exists J_\ell \ge 2\ \forall s \ge 1 \qquad W_s(J_\ell,J_\ell) \ge \ell M_{k,h}(s).$$

Proof. Given J in the first line, take $J_\ell = \ell J$ and use

$$W_s(\ell J,\ell J) \ \ge\ W_s(\ell J, J) \ \ge\ \ell W_s(J,J).$$

The converse takes $\ell = 1$. □

For large degree the palette cost of imposing *any* fixed root divisibility condition can be reduced to just one color, at a bounded initial cost in progression length. The key is that the marker is exactly periodic on the retained differences.

Lemma 17.4 (A polychromatic torus test)**.** *For integers $m \geq 1$, $q \geq 2$, there exist an integer $L \geq 1$ and a function $\kappa : \mathbb{T}^m \to \{0, \dots, q-1\}$ with this property: for every $\varepsilon > 0$ there exists K such that, whenever $v \in \mathbb{T}^m$ satisfies $\max_i \|Lv_i\| \geq \varepsilon$, every list*

$$\kappa(x + jv), \qquad 0 \leq j < K,$$

contains all q values. Here L is chosen independently of ε.

Proof. This is the torus shell method of Alweiss–Farhangi [4, Sections 2, 3.6]; we record the dependence of L. Let $F(x) = \sum_i \|x_i\|^2$. For $x \in \mathbb{T}^m$ and $u \in [-1,1]^m$ with $\|u\|_\infty = 1$, the function $F(x+tu)$ on $[0,1]$ is nonconstant: away from finitely many crossings of moving coordinates through half-integers it is a quadratic with strictly positive quadratic coefficient $\sum_i u_i^2$. Its oscillation has a positive minimum η over the compact family (x,u), by continuity. Put $w = \eta/(4q)$ and $\kappa(x) = \lfloor F(x)/w \rfloor \bmod q$. Every such path crosses the centers of q consecutive bands. Uniform continuity gives $0 < \delta < 1/2$ so that samples of parameter mesh at most δ, including time zero and reaching within δ of time one, meet the interiors of those bands: choose δ so that parameters at distance at most δ have F-values differing by less than $w/3$, uniformly in (x,u).

Partition $\mathbb{T}^m$ into B half-open cubes of side less than δ. Among $0, v, \dots, Bv$ two lie in one cube. Thus some $1 \leq a \leq B$ satisfies $\|av\|_\infty < \delta$. Set $L = B!$. Under the lemma's hypothesis, the centered representative v' of av has $\sigma = \|v'\|_\infty \geq a\varepsilon/L > 0$, since $a \mid L$. Along $x + t(v'/\sigma)$ the samples of mesh σ are $x + jav$, $0 \leq j \leq \lfloor 1/\sigma \rfloor$. They see all colors, and their original indices are at most $a/\sigma \leq L/\varepsilon$. Taking $K = 1 + \lceil L/\varepsilon \rceil$ suffices. □

Theorem 17.5 (One extra color for large degrees)**.** *Fix $r \geq 2$ and $\ell \geq 2$. There are s_0 and K, depending only on r, ℓ, such that for every $s \geq s_0$, $X \subseteq \mathbb{N}$ and $h \geq 2$,*

$$P_s(X) \text{ is } (r+1, \max\{h, K\})\text{-forcing} \quad \Longrightarrow \quad P_s(X/\ell) \text{ is } (r,h)\text{-forcing.}$$

In particular, for all $s \geq s_0$ and $h \geq K$,

$$W_s(r+1, h) \;\geq\; \ell W_s(r,h).$$

Proof. Write $\ell = \prod_{p \mid \ell} p^{a_p}$ and $m = \sum_{p \mid \ell} a_p$. Choose L, κ from Lemma 17.4 with $q = r+1$ and this m. Set $b_p = 1 + v_p(L)$ and $s_0 = \max_p b_p$. For $s \geq s_0$ define

$$\beta_{p,t} = p^{-(st + b_p)}, \qquad p \mid \ell, \quad 0 \leq t < a_p.$$

All these rational frequencies have denominator dividing ℓ^s, since $st + b_p \leq sa_p$. If $\ell \nmid n$, choose a deficient prime p and put $t = v_p(n) < a_p$. Then $L\beta_{p,t} n^s$, in lowest terms modulo one, has denominator exactly p. Hence its distance from an integer is at least $1/p$. Applying the lemma with $\varepsilon = 1/\max_{p \mid \ell} p$ gives K independent of s. The periodic marker $b(x) = \kappa(x\beta)$ is constant on every progression of step divisible by ℓ^s, and visits all $r+1$ labels in K sites on any progression of step n^s with $\ell \nmid n$.

Suppose $c : \mathbb{N} \to \{1, \dots, r\}$ blocks $P_s(X/\ell)$ at length h. Read $c_0(x) = c(1 + \lfloor x/\ell^s \rfloor)$ and recolor a site with the new color $*$ exactly when $b(x) = c_0(x)$, leaving it $c_0(x)$ otherwise. On a progression of step n^s with $\ell \mid n$, the marker is constant. A constant output then implies a constant original color: this is immediate for an old output, and an all-$*$ run has original color equal to the one fixed marker. The quotient sites advance by $(n/\ell)^s$, so h such sites are impossible for $n \in X$. For $\ell \nmid n$ any run of K sites meets every marker label. An all-$*$ run cannot meet the unmarked label zero. A run of old color i has original color i throughout and would be recolored at a site with marker label i. Thus the new coloring blocks $P_s(X)$ at length $\max(h, K)$. □

Theorem 17.6 (A growing prime at a fixed cost)**.** *Let $q \geq 2$, let $p \geq q^2$ be prime, and suppose $p-1 \mid s$. For all $X \subseteq \mathbb{N}$, if $P_s(X)$ is (q,h)-forcing and $h \geq q+1$, then $P_s(X/p)$ is $(q-1,h)$-forcing. Thus*

$$W_s(q,h) \geq pW_s(q-1,h) \qquad (h \geq q+1).$$

For $j \geq 3$, $p \geq j^2$ and $p-1 \mid s$, this gives $W_s(j,j) \geq p^{j-2}$.

Proof. Take $a = \lfloor p/q \rfloor$. On steps $d = n^s$ with $p \nmid n$, the marker phase $ax/p \bmod 1$ advances by $\theta = a/p$, since $n^s \equiv 1 \bmod p$. We have

$$0 < (q-1)/q^2 \;\leq\; 1/q - 1/p \;<\; \theta \;<\; 1/q.$$

The $q+1$ points $0, \theta, \ldots, q\theta$ in cyclic order have gaps θ or $1-q\theta$, all at most $1/q$. Every translate of them therefore meets each of the q half-open equal bins of width $1/q$. On steps with $p \mid n$, the phase is constant. Number the bins $0, \ldots, q-1$ and use them as the marker in the proof of Theorem 17.5, now reading the original $(q-1)$-coloring at $1 + \lfloor x/p^s \rfloor$. This construction also works with one original color; then a blocked difference set must be empty. The claimed implication follows because $h \geq q+1$. To obtain the last estimate, iterate the inequality at length j for $q = 2, \ldots, j-1$, use $W_s(1,j) = 1$ and use monotonicity in the color number. □

Corollary 17.7 (Several growing primes)**.** *Suppose $p_1, \ldots, p_m$ are distinct primes with $p_i - 1 \mid s$ and $p_i \geq 2qm$, where $q \geq 2$. Set $P = \prod_i p_i$. Then*

$$W_s(q,h) \geq PW_s(q-1,h) \qquad (h \geq 2qm+1).$$

Proof. Use the phase $\alpha = \sum_i \lfloor p_i/(qm) \rfloor / p_i$. Each summand lies in $[1/(2qm), 1/(qm))$. On a power step n^s its phase increment is the sum of just the terms with $p_i \nmid n$, read modulo one. It vanishes on the core $P \mid n$ and otherwise belongs to $[1/(2qm), 1/q)$. In $2qm+1$ points a lifted path with such a nonzero increment has span at least one and step less than $1/q$, so it meets all q equal bins. Apply the same marker construction, with original quotient read at $1 + \lfloor x/P^s \rfloor$. □

These inequalities compare two ordinary forcing thresholds. The remaining comparison asks whether, for each fixed k, h, there is one $J \geq 2$ such that $W_s(J,J) \geq M_{k,h}(s)$ in every degree.

17.3. **A necessary growth estimate.**

Lemma 17.8. *If positive integers $d_1 < \cdots < d_v$ satisfy $d_{i+1} \geq 3d_i$, then $\{d_1, \ldots, d_v\}$ has a binary blocker at length three.*

Proof. Construct nested closed real intervals I_i of length $1/(2d_i)$ such that $\alpha d_i \bmod 1 \in [1/4, 3/4]$ for $\alpha \in I_i$. Every interval of length at least $3/(2d_{i+1})$ contains a full interval of the form $[(a + 1/4)/d_{i+1}, (a+3/4)/d_{i+1}]$. Hence the ratio assumption permits the induction. Choose $\alpha \in I_v$ and color n by the half circle containing $n\alpha \bmod 1$.

For a step with phase $\theta \in [1/4, 3/4]$, three successive sites cannot lie in the same half-open semicircle. If $\theta = 1/2$, already the first two have different colors. Otherwise three lifts to the same interval of length $1/2$ have consecutive differences all equal to θ or all equal to $\theta - 1$, since only one of these numbers has absolute value below $1/2$. Their total span is $2\min(\theta, 1-\theta) \geq 1/2$, a contradiction. □

Proposition 17.9. *If $[N]^s$ is (j,j)-forcing for $j \geq 3$, then*

$$\frac{N}{s} > \frac{\log_2 j - 1}{\log 3}. \tag{30}$$

Consequently changing-degree forcing packets necessarily satisfy $N_j/s_j \longrightarrow \infty$.

Proof. Set $m = \min\{N, \lceil (N/s)\log 3\rceil\}$. Partition the indices $1,\dots,N$ into their residue classes modulo m. If $m = N$, each class is a singleton. Otherwise two successive indices $n, n+m$ in a class satisfy

$$\frac{(n+m)^s}{n^s} \geq \left(\frac{N}{N-m}\right)^s > \exp(sm/N) \geq 3.$$

Apply Lemma 17.8 to each class and take the product of their binary colorings. This gives a 2^m-color blocker at length three for all of $[N]^s$. Therefore (j,j)-forcing implies $2^m > j$, so $\lceil (N/s)\log 3\rceil > \log_2 j$, which gives (30). □

The growth estimate is much weaker than (29). It controls the ratio N/s, while the known compactness argument supplies no corresponding bound on $M_{k,h}(s)$.

17.4. **A host outside the present transfer mechanisms.** For $j \geq 2$, the following construction uses the minimal lengths $N_j = W_{s_j}(j,j)$; its scales impose no extra growth on N_j. For $B \subseteq \mathbb{N}$ write

$$d^*(B) = \limsup_{L\to\infty} \sup_{a\geq 0} \frac{|B \cap (a, a+L]|}{L}.$$

Theorem 17.10. *There are integers $s_j \to \infty$ divisible by three, lengths $N_j = W_{s_j}(j,j)$, and positive integers λ_j such that*

$$A = \bigcup_{j\geq 2} \lambda_j^3 [N_j]^{s_j}$$

is large and has all of the following properties:

1. *A is contained in the positive cubes, so contains no finite-sums set with two positive generators.*
2. *For every distinct $a, b \in \mathbb{N}$, the set $\{t : at, bt \in A\}$ is finite.*
3. *For every nonconstant $p \in \mathbb{Q}[n]$, $d^*(\{n \in \mathbb{N} : p(n) \in A\}) = 0$.*
4. *No $a, b, c \in A$ satisfy $ab = c$.*

Thus A has no IP_0 subset, no polynomial image of a positive-upper-Banach-density source, and no subset to which the divisible-dilation transfer applies with a direction set containing two distinct elements. The hereditary color-hierarchy question for this particular host is not settled by these facts.

Proof. Enumerate the nonconstant positive-leading rational polynomials as $p_1, p_2, \dots$ and the rationals greater than one as $r_1, r_2, \dots$. Start with $s_1 = 0$. Suppose the earlier packets have been chosen and let B be their largest value, or $B = 1$ initially. Choose s_j divisible by three, larger than s_{j-1}, and with $2^{s_j} > B$; then put $N_j = W_{s_j}(j,j)$. Take λ_j to be a prime larger than N_j and B, sufficiently large to impose the following two conditions.

First, for every $i \leq j$, the preimage $p_i^{-1}(\lambda_j^3[N_j]^{s_j}) \cap \mathbb{N}$ is j-separated and lies at distance greater than j above every preimage under p_i of an earlier packet. To see that this is possible, fix a polynomial of degree d and the finite packet $[N_j]^{s_j}$. All relevant large preimages are at most $C\lambda_j^{3/d}$, and on that range $|p_i'| \leq C'\lambda_j^{3(d-1)/d}$. Two arguments at distance at most j therefore have values at distance at most $jC'\lambda_j^{3(d-1)/d} < \lambda_j^3$ for large λ_j. Distinct packet values are separated by at least λ_j^3. All positive preimages tend to infinity with λ_j, giving the condition relative to earlier packets as well.

Second, every new packet value divided by every earlier packet value is greater than $\max_{i\leq j} r_i$. This also follows by taking λ_j large. These are only finitely many conditions at each stage.

Each packet is (j,j)-forcing after dilation, so their union is large. Every value is a cube, and the exponent-three case of Fermat's theorem gives the first assertion. For fixed $r_i > 1$, cross-packet pairs with ratio r_i can involve only finitely many packets, by the second condition. Within a single packet, such a ratio would equal $(n/m)^{s_j}$. Take any prime with nonzero valuation in r_i. That

valuation would have to be divisible by s_j, which is impossible for all sufficiently large j. Thus there are only finitely many pairs with that fixed ratio, proving the second assertion.

For the fourth assertion, every new packet value has λ_j-adic valuation exactly three. Products of two new values cannot be new values, since their valuation is six. Products of two earlier values are at most $B^2 < \lambda_j^3$, so cannot be new. If an earlier value a times a new value were another new value, then $a = (n'/n)^{s_j}$ for positive integers $n, n' \le N_j$. A rational number whose positive integer power is an integer is itself an integer. Since $a > 1$, this forces $a \ge 2^{s_j} > B$, a contradiction. Thus induction makes the whole union product-free.

For a fixed polynomial p_i and any R, the union of all its packet preimages after stage $\max(i, R)$ is R-separated. The earlier preimages are finite. Hence their upper Banach density is at most $1/R$ for every R, proving the third assertion for positive-leading polynomials. Negative-leading polynomials have only finitely many positive preimages and cause no difficulty.

Finally, a nonconstant real polynomial taking integer values at infinitely many distinct integers has rational coefficients, by Vandermonde interpolation at $d+1$ such arguments. Thus assertion three excludes images of all positive-density sources under any nonconstant real polynomial with integer values on that source. If the divisible-dilation transfer held for two distinct directions a, b, it would produce t divisible by every prescribed integer with $at, bt \in A$. A finite set of possible t cannot do that, contradicting assertion two. □

There is a corresponding obstruction to either elementary semigroup method. No additive idempotent ultrafilter can contain A, since an idempotent member contains a two-generator finite-sums configuration. No multiplicative idempotent can contain A, since an idempotent member contains a, b, ab (and $1 \notin A$). Moreover, for every free ultrafilter p, each of the sets

$$\{n : A - n \in p\}, \qquad \{n : A/n \in p\}$$

has at most one element. For the multiplicative statement use assertion two. For the additive statement, distinct translates of the positive cubes have finite intersection: a fixed nonzero difference of two cubes has finitely many positive solutions, by factorization. Membership of two such translates in a free ultrafilter is therefore impossible.

This theorem strengthens the classification obstruction in Section 15: its fixed-degree cube packets were resolved by Corollary 8.3. Here the ratio-finiteness assertion rules out that transfer entirely. It remains possible that a different argument solves this host, or that the comparison (29) itself holds. The construction is not a counterexample to the original conjecture.

18. Van der Corput transfer through interval cardinalities

In the Boolean direction problem from Section 7, fixing the requested progression length at three permits a spectral argument. We prove the finite transfer uniformly over all positive integer weights and apply it inside every IP_0 host.

Definitions. Write $\mathrm{e}(t) = \exp(2\pi i t)$ and $\widehat{\sigma}(d) = \int_{\mathbb{T}} \mathrm{e}(dt)\, d\sigma(t)$. We use the one-dimensional spectral characterization discussed in [6, Section 1.2.2]. Namely, $S \subseteq \mathbb{N}$ is a van der Corput set if every positive probability σ on $\mathbb{T}$ with $\widehat{\sigma}(d) = 0$ for all $d \in S$ satisfies $\sigma(\{0\}) = 0$. For $0 < \delta \le 1$, say a finite packet $E \subseteq \mathbb{N}$ detects atoms of size δ if for every probability σ with $\sigma(\{0\}) \ge \delta$, some $e \in E$ has $\widehat{\sigma}(e) \ne 0$. All intervals of coordinates below are finite, nonempty, and consecutive in the specified list.

Lemma 18.1 (Uniform finite Gram test)**.** *If S is van der Corput and $0 < \delta \le 1$, some m has the following property: any unit vectors $v_0, \dots, v_m$ in a complex Hilbert space with a common component of squared norm δ have $\langle v_i, v_j \rangle \ne 0$ for some $0 \le i < j \le m$ with $j - i \in S$. A common component means $v_i = w + y_i$ with $\|w\|^2 = \delta$ and $y_i \perp w$ for all i.*

Proof. Otherwise compactness of bounded Gram matrices, or a diagonal subsequence of their entries, gives a positive semidefinite kernel K on $\mathbb{N}_0$ with $K(i,i) = 1$, $K(i,j) = 0$ whenever $|i-j| \in S$,

and $K-\delta\mathbf{1}\mathbf{1}^*$ positive semidefinite. For every integer $d \geq 0$ take simultaneous subsequential limits of $N^{-1}\sum_{n=1}^{N} K(n, n+d)$, calling the result $a(d)$; put $a(-d) = \overline{a(d)}$. The kernels $(a(j-i))_{i,j\in\mathbb{Z}}$ and $(a(j-i)-\delta)_{i,j\in\mathbb{Z}}$ are positive semidefinite: on each fixed finite index list they are limits of translates averaged over a common interval, with endpoint errors tending to zero. Herglotz therefore represents a by a probability σ and $a-\delta$ by a positive measure. By uniqueness, $\sigma \geq \delta\,\delta_0$. But $a(d) = 0$ on S, a contradiction. □

Theorem 18.2 (Arbitrary positive weights)**.** *For van der Corput S and any $0 < \delta \leq 1$, choose m as in Lemma 18.1. For* every *list of positive integers $b_1, \ldots, b_m$, the sums of intervals of lengths in S detect atoms of size δ. In particular, for any infinite positive sequence of weights, the corresponding interval sums of lengths in S form a van der Corput set.*

Proof. Set $p_n = \sum_{i=1}^{n} b_i$ with $p_0 = 0$. For a test σ, the functions $v_n(t) = \mathrm{e}(p_n t)$ are unit vectors in $L^2(\sigma)$, and their Gram entries are $\widehat{\sigma}(p_j - p_i)$ up to complex conjugation. They have the common component supported at $t = 0$ of squared norm $\sigma(\{0\})$. To apply the lemma with exactly δ when the atom is larger, split off a separate mass δ at zero, retaining its excess in the residual measure. The lemma gives the required nonzero entry. □

Theorem 18.3 (Binary triple certificate)**.** *Let $S \subseteq \mathbb{N}$. If some binary coloring of $\mathbb{Z}^{(\mathbb{N})}$ blocks all three-term progressions in directions 1_I with I an interval of size in S, then there is a symmetric probability σ on $\mathbb{T}$ with*

$$\sigma(\{0\}) \geq \tfrac{1}{4}, \qquad \widehat{\sigma}(d) = 0 \quad (d \in S).$$

Consequently every van der Corput S forces a monochromatic binary triple already with an interval direction of cardinality in S. The same conclusion holds along intervals of arbitrary positive integer weights on $\mathbb{N}$. In both cases a single finite list length $m = m(S)$ suffices for all binary colorings and all positive weight lists of that length.

Proof. Take a shift-invariant probability on the orbit closure of the blocking coloring (the countable abelian shift group is amenable), and let u be its coordinate at the origin, taking values ± 1. Every coloring in this hull still avoids the forbidden triples. Write U^v for the shift operators, and $r(v) = \langle u, U^v u\rangle$. For $v = 1_I$, $|I| \in S$, use the exact identity

$$1_{a=b=c} = (1 + ab + ac + bc)/4 \quad (a, b, c \in \{-1, 1\})$$

and invariance to get $1 + 2r(v) + r(2v) = 0$. Put $p_n = \sum_{i=1}^{n} e_i$, $p_0 = 0$. In the orthogonal direct sum $\mathbb{R} \oplus L^2 \oplus L^2$, the unit vectors

$$w_n = \tfrac{1}{2}(1, \sqrt{2}\,U^{p_n} u, U^{2p_n} u)$$

have a common component of squared norm $1/4$. Their Gram entries are zero whenever $|j-i| \in S$, because $\pm(p_j - p_i)$ is such an interval vector. The stationarization in Lemma 18.1 gives the asserted measure, symmetric since the entries are real. For weights one can substitute $p_n = \sum_{i=1}^{n} b_i$ in this argument on $\mathbb{Z}$; a blocker on $\mathbb{N}$ extends to $\mathbb{Z}$ by limits of translations. Equivalently, apply the preceding assertion to the pullback of this two-sided coloring under the weight map. For finite uniformity, take the m of Lemma 18.1 at $\delta = 1/4$. The same proof on $\mathbb{Z}^m$ uses just $w_0, \ldots, w_m$, and a blocker would make their Gram entries vanish at every permitted interval length. The lemma forbids this. The weight map $e_i \mapsto b_i$ gives the finite weighted assertion. □

An explicit binary-blockable van der Corput source. Fix irrational $\alpha, \theta \in \mathbb{T}$. There are an open interval $J \ni \theta$ and an $H \geq 2$ such that

$$S = \{d \in \mathbb{N} : \alpha d^2 \bmod 1 \in J\}$$

has a binary H-blocker and is van der Corput. Indeed, Weyl equidistribution of $x + ty + t^2\theta$ for every $x, y \in \mathbb{T}$ shows that each such orbit enters the interiors of both fixed semicircles among its initial sites. Openness and compactness of $\mathbb{T}^2$ give a uniform H and an interval J of leading coefficients

with the same property (take J with length in $(0,1)$). Color n by the half containing αn^2; expansion along an arithmetic progression proves blocking. For every $\beta \neq 0$ in $\mathbb{T}$, approximation of 1_J from above and below by trigonometric polynomials and Weyl's criterion give

$$\lim_{N\to\infty} |S \cap [1,N]|^{-1} \sum_{d\in S\cap[1,N]} \mathrm{e}(d\beta) = 0.$$

Indeed each nonconstant Fourier mode in the αd^2 variable has irrational quadratic coefficient; the constant mode against $\mathrm{e}(d\beta)$ has mean zero. The unweighted density of S equals $|J|$. Dominated convergence against any spectral test σ therefore gives the limit $\sigma(\{0\})$ for its averaged Fourier coefficients on S, proving the van der Corput assertion.

Theorem 18.4 (A van der Corput seed in every IP_0 host)**.** *Suppose S is van der Corput and has a binary blocker at length H. Each IP_0 host A contains a van der Corput subset with a binary blocker at length $8H$. One may also require simultaneous pointwise recurrence for every finite-dimensional nilsystem, with a binary blocker at length $8\max(H,800)$ using Theorem 7.1.*

Proof. At stage j take m from Lemma 18.1 for $\delta = 1/j$. The divided host A/λ_j is IP_0 by the division argument in Section 7. Lemmas 7.6, 7.7, and 7.8 supply m selected generators from a sufficiently long finite-sums configuration in that host. The periodic phase clock gives a binary blocker at length $4H$ for all subset sums of those selected generators whose cardinalities lie in S. In particular it blocks the finite set E_j of their interval sums of lengths in S. Apply Theorem 18.2 to the selected generators in their inherited order. The packet E_j detects every atom of size $1/j$. The dilated packet $\lambda_j E_j$ has the same detection property. Indeed, pushing a test measure forward under $t \mapsto \lambda_j t$ preserves at least its original mass at zero, and its Fourier coefficient at e is the original coefficient at $\lambda_j e$.

Choose successive scales with $\lambda_{j+1}/\lambda_j \in \mathbb{N}$ and $\lambda_{j+1}/\lambda_j > 8H \max E_j$. The union $D = \bigcup_j \lambda_j E_j \subseteq A$ detects every positive mass at zero. Lemma 4.3 supplies its binary blocker at length $8H$. For the simultaneous assertion, interleave these stages with the pointwise nilsystem test packets in the proof of Theorem 7.1. At each nilsystem stage impose the required factorial divisibility on the next scale and use the powered tests at that scale. Their local binary blockers have length $4 \cdot 800$. Both stage types now admit length $4\max(H,800)$, so the same assembly gives a single binary blocker at length $8\max(H,800)$. □

The binary identity in Theorem 18.3 detects three sites through pair correlations. Its length restriction is substantive. For a source S with a binary H-blocker c, extend c to $\mathbb{Z}$ by compactness and pull it back under $x \mapsto \sum_i x_i$ on $\mathbb{Z}^{(\mathbb{N})}$. An increment 1_F then advances the sum by $|F|$. Thus this coloring blocks $\mathcal{E}(S)$ at length H, even if S has the van der Corput property and hence forces binary triples in interval directions. The conditional $k = 2$ transfer in Corollary 7.11 allows a ternary-blockable source and requires binary forcing at every length in its Boolean lift. The arbitrary-IP_0 hierarchy at $k = 2$ remains unresolved here.

19. A separate intersective-polynomial input

The following older project theorem has a different scope from Theorem 6.1. It allows general intersective polynomials but requires a thick parameter set. Its longer proof is not reproduced here, and none of the central finite reductions or marker lemmas depends on it.

Separate input 19.1 (Intersective polynomial images)**.** Let $p \in \mathbb{Z}[n]$ be nonconstant, have positive leading coefficient, and be *intersective*: for every $m \geq 1$ some $a \in \mathbb{Z}$ satisfies $p(a) \equiv 0 \pmod m$. If $S \subseteq \mathbb{N}$ is thick, then

$$\{p(n) : n \in S,\ p(n) > 0\}$$

contains, for each $k \geq 1$, a set with P_k and a $(k+1, 400 \cdot 2^k)$-blocker. This is the separately proved and internally audited assertion of [19], not an externally refereed or proof-assistant-certified input.

Theorem 6.1 treats arbitrary quasi-central parameter sets when $p(0) = 0$. The displayed input treats arbitrary thick parameter sets and allows intersective p with nonzero constant term. Neither statement covers the image of every prescribed large parameter set. The boundary nilsystem result formerly recorded as a separate input is now subsumed by the proved, more general boundary minimal-return Lemma 5.10 and Theorem 5.1.

20. What remains to be proved

The positive quasi-central and polynomial-image theorems do not settle the arbitrary-large-host problem. Its finite target has a concise exact form. For a fixed $k \geq 2$, prove that every large host has some $H \geq 2$ such that finite $(k+1, H)$-blockable subsets have unbounded k-forcing length. If this assertion holds for every large host, it also holds for every large subhost, and Theorem 4.4 proves Conjecture 1.2.

Conversely, a counterhost can be reduced to a large subhost with the simultaneous compression bounds of Theorem 4.5. All its k-large subsets are stable under finite ordinary polynomial restrictions and nil-Bohr restrictions of every finite order by Corollary 12.9, and have both the k- and $(k+1)$-color rotation-window properties. These are necessary structural conditions. More strongly, Theorem 13.6 gives $P_k(D \cap B)$ for every k-large subset D of a counterhost and every nil-Bohr neighborhood B of zero. An incompatibility between these conditions and largeness remains to be proved.

The current marker methods do not bridge this gap. Critical packets give an upper obstruction length growing with the requested lower forcing length. Independent thinning retains full largeness. Fixed arithmetic templates can be avoided by large hosts. The direct quartic scalar marker can create arbitrarily long runs in the new color, despite arbitrarily small coefficients. The proper-color winding argument settles a rigid source-length-two case and stops before the relevant source-length-three problem.

In particular, the basic sparse benchmark

$$\mathrm{FS}(1, 3, 3^2, \ldots) = \{n \geq 1 : \text{every ternary digit of } n \text{ is 0 or 1}\}$$

is large by Hales–Jewett, but the present work does not establish hereditary separation in it for $k = 2$. Theorem 7.1 does supply a pointwise nil-recurrent, binary-blockable seed inside this host; it does not make that seed 2-large. Resolving this benchmark, or proving a host-dependent uniform finite extractor, would be a substantive advance beyond the results reported here.

For IP_0 hosts, Corollary 7.11 gives another precise sufficient input: a $(k+1, H)$-blockable set S whose Boolean direction set $\mathcal{E}(S)$ is k-large in $\mathbb{Z}^{(\mathbb{N})}$. No implication from ordinary k-largeness of S to this stronger colored-cardinality property is proved here. Theorem 18.3 proves the binary interval-direction conclusion at length three for every van der Corput source.

The changing-degree host in Theorem 17.10 gives a second explicit benchmark. It has no IP_0 subset, no polynomial image of a positive-density source, and only finitely many simultaneous dilations of any two distinct directions. Its strict hierarchy remains unresolved here. The degree-uniform finite comparison (29) would settle it. By the root-division comparison it suffices to treat $\ell = 1$. Uniform control over the degree s remains open. The simultaneous full-factor obstruction in Theorem 16.6 likewise blocks a proposed lifting step, rather than the conjecture itself.

Provenance and verification. The source is a standalone document: it has no input dependency on the project working notes. The central finite and marker proofs have been consolidated from notes developed and independently checked within this research session. External theorem statements are identified at the points of use. The quasi-central and zero-constant polynomial-image proofs are included in full; the older general intersective-polynomial theorem is explicitly classified as a separate unpublished input. No novelty priority, external peer review, or Lean verification is claimed. The joint position and difference theorem, all-order nil-Bohr marker, palette, changing-degree, root-division, and finite van der Corput arguments were checked separately. The quadratic

difference-set obstruction is recorded as a separately proved supplement. This checkpoint is 2026-11-04 23:10 UTC. The arbitrary-large-host conjecture remains unresolved.

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